



General Aptitude (GA)

Q.1 – Q.5 Multiple Choice Question (MCQ), carry ONE mark each (for each wrong answer: – 1/3).

Q.1	The current population of a city is 11,02,500. If it has been increasing at the rate of 5% per annum, what was its population 2 years ago?
(A)	9,92,500
(B)	9,95,006
(C)	10,00,000
(D)	12,51,506

Q.2	p and q are positive integers and $\frac{p}{q} + \frac{q}{p} = 3$, then, $\frac{p^2}{q^2} + \frac{q^2}{p^2} =$
(A)	3
(B)	7
(C)	9
(D)	11



Q.3	The least number of squares that must be added so that the line P-Q becomes the line of symmetry is _____
(A)	4
(B)	3
(C)	6
(D)	7

Q.4	<i>Nostalgia</i> is to <i>anticipation</i> as _____ is to _____ Which one of the following options maintains a similar logical relation in the above sentence?
(A)	Present, past
(B)	Future, past
(C)	Past, future
(D)	Future, present



Q.5	Consider the following sentences: (i) I woke up from sleep. (ii) I woked up from sleep. (iii) I was woken up from sleep. (iv) I was wokened up from sleep. Which of the above sentences are grammatically CORRECT?
(A)	(i) and (ii)
(B)	(i) and (iii)
(C)	(ii) and (iii)
(D)	(i) and (iv)



Electronics and Communication Engineering (EC)

Q. 6 – Q. 10 Multiple Choice Question (MCQ), carry TWO marks each (for each wrong answer: – 2/3).

Q.6	Given below are two statements and two conclusions. Statement 1: All purple are green. Statement 2: All black are green. Conclusion I: Some black are purple. Conclusion II: No black is purple. Based on the above statements and conclusions, which one of the following options is logically CORRECT?
(A)	Only conclusion I is correct.
(B)	Only conclusion II is correct.
(C)	Either conclusion I or II is correct.
(D)	Both conclusion I and II are correct.

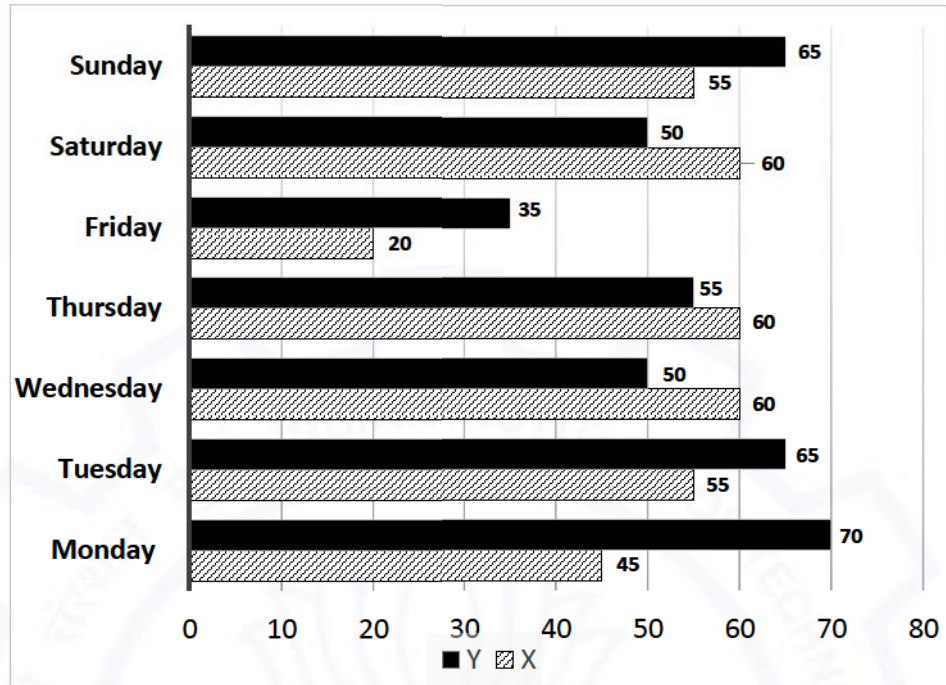


Q.7	Computers are ubiquitous. They are used to improve efficiency in almost all fields from agriculture to space exploration. Artificial intelligence (AI) is currently a hot topic. AI enables computers to learn, given enough training data. For humans, sitting in front of a computer for long hours can lead to health issues. Which of the following can be deduced from the above passage? (i) Nowadays, computers are present in almost all places. (ii) Computers cannot be used for solving problems in engineering. (iii) For humans, there are both positive and negative effects of using computers. (iv) Artificial intelligence can be done without data.
(A)	(ii) and (iii)
(B)	(ii) and (iv)
(C)	(i), (iii) and (iv)
(D)	(i) and (iii)

Q.8	Consider a square sheet of side 1 unit. In the first step, it is cut along the main diagonal to get two triangles. In the next step, one of the cut triangles is revolved about its short edge to form a solid cone. The volume of the resulting cone, in cubic units, is _____
(A)	$\frac{\pi}{3}$
(B)	$\frac{2\pi}{3}$
(C)	$\frac{3\pi}{2}$
(D)	3π



Q.9



The number of minutes spent by two students, X and Y, exercising every day in a given week are shown in the bar chart above.

The number of days in the given week in which one of the students spent a minimum of 10% more than the other student, on a given day, is

(A) 4

(B) 5

(C) 6

(D) 7



Q.10	Corners are cut from an equilateral triangle to produce a regular convex hexagon as shown in the figure above. The ratio of the area of the regular convex hexagon to the area of the original equilateral triangle is
(A)	2 : 3
(B)	3 : 4
(C)	4 : 5
(D)	5 : 6



Electronics and Communication Engineering (EC)

Q.1 – Q.15 Multiple Choice Question (MCQ), carry ONE mark each (for each wrong answer: – 1/3).

Q.1	The vector function $F(\mathbf{r}) = -x\hat{i} + y\hat{j}$ is defined over a circular arc C shown in the figure. The line integral of $\int_C F(\mathbf{r}) \cdot d\mathbf{r}$ is
(A)	$\frac{1}{2}$
(B)	$\frac{1}{4}$
(C)	$\frac{1}{6}$
(D)	$\frac{1}{3}$



Q.2	Consider the differential equation given below. $\frac{dy}{dx} + \frac{x}{1-x^2}y = x\sqrt{y}$ The integrating factor of the differential equation is
(A)	$(1-x^2)^{-3/4}$
(B)	$(1-x^2)^{-1/4}$
(C)	$(1-x^2)^{-3/2}$
(D)	$(1-x^2)^{-1/2}$

Q.3	Two continuous random variables X and Y are related as $Y = 2X + 3$ Let σ_X^2 and σ_Y^2 denote the variances of X and Y, respectively. The variances are related as
(A)	$\sigma_Y^2 = 2 \sigma_X^2$
(B)	$\sigma_Y^2 = 4 \sigma_X^2$
(C)	$\sigma_Y^2 = 5 \sigma_X^2$
(D)	$\sigma_Y^2 = 25 \sigma_X^2$

Q.4	Consider a real-valued base-band signal $x(t)$, band limited to 10 kHz. The Nyquist rate for the signal $y(t) = x(t) x\left(1 + \frac{t}{2}\right)$ is
(A)	15 kHz
(B)	30 kHz
(C)	60 kHz
(D)	20 kHz



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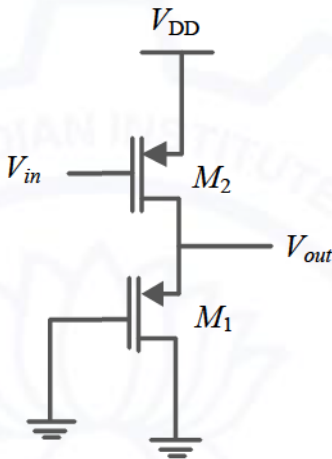
Q.5	Consider two 16-point sequences $x[n]$ and $h[n]$. Let the linear convolution of $x[n]$ and $h[n]$ be denoted by $y[n]$, while $z[n]$ denotes the 16-point inverse discrete Fourier transform (IDFT) of the product of the 16-point DFTs of $x[n]$ and $h[n]$. The value(s) of k for which $z[k] = y[k]$ is/are
(A)	$k = 0, 1, 2, \dots, 15$
(B)	$k = 0$
(C)	$k = 15$
(D)	$k = 0$ and $k = 15$

Q.6	A bar of silicon is doped with boron concentration of 10^{16} cm^{-3} and assumed to be fully ionized. It is exposed to light such that electron-hole pairs are generated throughout the volume of the bar at the rate of $10^{20} \text{ cm}^{-3}\text{s}^{-1}$. If the recombination lifetime is $100 \text{ }\mu\text{s}$, intrinsic carrier concentration of silicon is 10^{10} cm^{-3} and assuming 100% ionization of boron, then the approximate product of steady-state electron and hole concentrations due to this light exposure is
(A)	10^{20} cm^{-6}
(B)	$2 \times 10^{20} \text{ cm}^{-6}$
(C)	10^{32} cm^{-6}
(D)	$2 \times 10^{32} \text{ cm}^{-6}$



Q.7	The energy band diagram of a p-type semiconductor bar of length L under equilibrium condition (<i>i.e.</i>, the Fermi energy level E_F is constant) is shown in the figure. The valance band E_V is sloped since doping is non-uniform along the bar. The difference between the energy levels of the valance band at the two edges of the bar is Δ. If the charge of an electron is q, then the magnitude of the electric field developed inside this semiconductor bar is
(A)	$\frac{\Delta}{qL}$
(B)	$\frac{2\Delta}{qL}$
(C)	$\frac{\Delta}{2qL}$
(D)	$\frac{3\Delta}{2qL}$

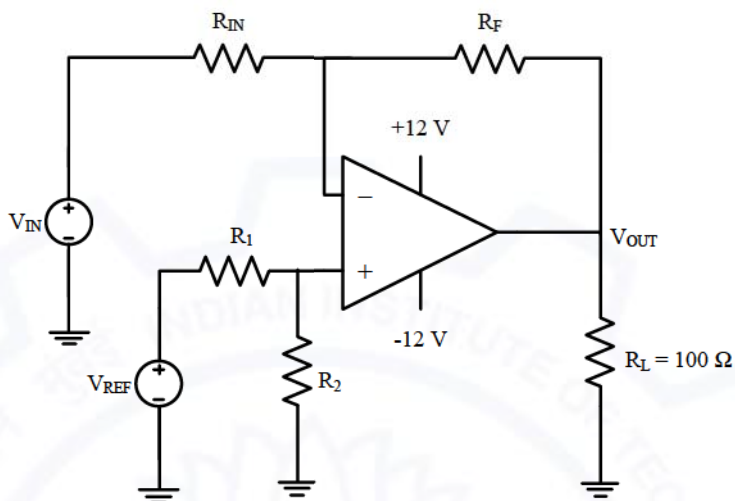


Q.8	In the circuit shown in the figure, the transistors M_1 and M_2 are operating in saturation. The channel length modulation coefficients of both the transistors are non-zero. The transconductance of the MOSFETs M_1 and M_2 are g_{m1} and g_{m2}, respectively, and the internal resistance of the MOSFETs M_1 and M_2 are r_{o1} and r_{o2}, respectively.  Ignoring the body effect, the ac small signal voltage gain ($\partial V_{out}/\partial V_{in}$) of the circuit is
(A)	$-g_{m2} (r_{o1} r_{o2})$
(B)	$-g_{m2} \left(\frac{1}{g_{m1}} r_{o2} \right)$
(C)	$-g_{m1} \left(\frac{1}{g_{m2}} r_{o1} r_{o2} \right)$
(D)	$-g_{m2} \left(\frac{1}{g_{m1}} r_{o1} r_{o2} \right)$



Q.9

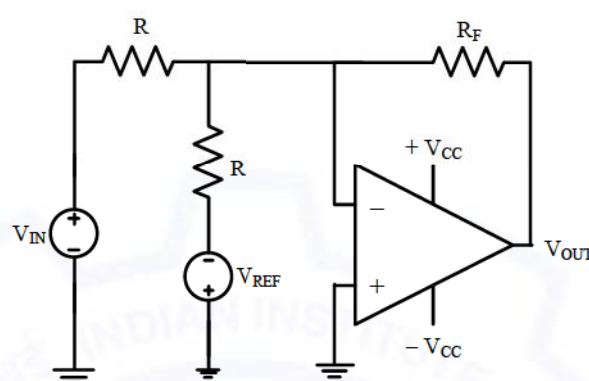
For the circuit with an ideal OPAMP shown in the figure, V_{REF} is fixed.



If $V_{OUT} = 1$ volt for $V_{IN} = 0.1$ volt and $V_{OUT} = 6$ volt for $V_{IN} = 1$ volt, where V_{OUT} is measured across R_L connected at the output of this OPAMP, the value of R_F/R_{IN} is

- (A) 3.285
(B) 2.860
(C) 3.825
(D) 5.555



Q.10	Consider the circuit with an ideal OPAMP shown in the figure.  Assuming $V_{IN} \ll V_{CC}$ and $V_{REF} \ll V_{CC}$, the condition at which V_{OUT} equals to zero is
(A)	$V_{IN} = V_{REF}$
(B)	$V_{IN} = 0.5 V_{REF}$
(C)	$V_{IN} = 2 V_{REF}$
(D)	$V_{IN} = 2 + V_{REF}$

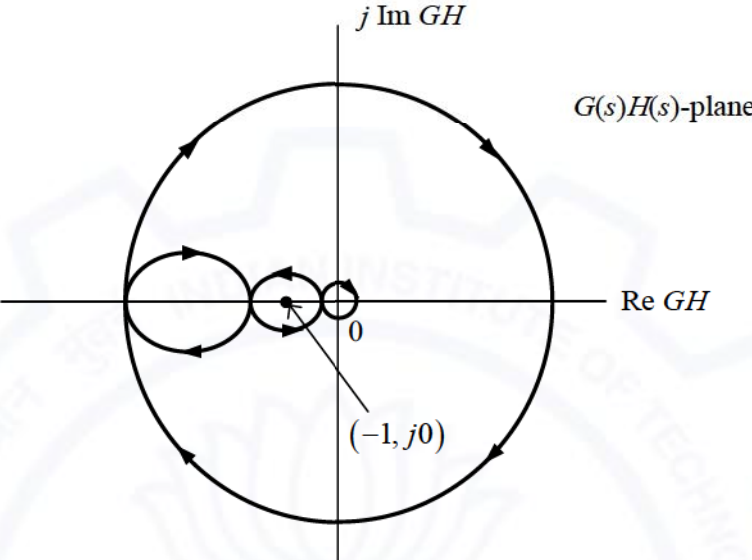
Q.11	If $(1235)_x = (3033)_y$, where x and y indicate the bases of the corresponding numbers, then
(A)	$x = 7$ and $y = 5$
(B)	$x = 8$ and $y = 6$
(C)	$x = 6$ and $y = 4$
(D)	$x = 9$ and $y = 7$



Q.12	Addressing of a $32K \times 16$ memory is realized using a single decoder. The minimum number of AND gates required for the decoder is
(A)	2^8
(B)	2^{32}
(C)	2^{15}
(D)	2^{19}

Q.13	The block diagram of a feedback control system is shown in the figure. The transfer function $\frac{Y(s)}{X(s)}$ of the system is
(A)	$\frac{G_1 + G_2 + G_1 G_2 H}{1 + G_1 H}$
(B)	$\frac{G_1 + G_2}{1 + G_1 H + G_2 H}$
(C)	$\frac{G_1 + G_2}{1 + G_1 H}$
(D)	$\frac{G_1 + G_2 + G_1 G_2 H}{1 + G_1 H + G_2 H}$



Q.14	The complete Nyquist plot of the open-loop transfer function $G(s)H(s)$ of a feedback control system is shown in the figure.  If $G(s)H(s)$ has one zero in the right-half of the s-plane, the number of poles that the closed-loop system will have in the right-half of the s-plane is
(A)	0
(B)	1
(C)	4
(D)	3



Electronics and Communication Engineering (EC)

Q.15	Consider a rectangular coordinate system (x, y, z) with unit vectors $\mathbf{a}_x$, $\mathbf{a}_y$, and $\mathbf{a}_z$. A plane wave traveling in the region $z \geq 0$ with electric field vector $\mathbf{E} = 10 \cos(2 \times 10^8 t + \beta z) \mathbf{a}_y$ is incident normally on the plane at $z = 0$, where β is the phase constant. The region $z \geq 0$ is in free space and the region $z < 0$ is filled with a lossless medium (permittivity $\epsilon = \epsilon_0$, permeability $\mu = 4\mu_0$, where $\epsilon_0 = 8.85 \times 10^{-12}$ F/m and $\mu_0 = 4\pi \times 10^{-7}$ H/m). The value of the reflection coefficient is
(A)	$\frac{1}{3}$
(B)	$\frac{3}{5}$
(C)	$\frac{2}{5}$
(D)	$\frac{2}{3}$



Q.16 – Q.25 Numerical Answer Type (NAT), carry ONE mark each (no negative marks).

Q.16	If the vectors $(1.0, -1.0, 2.0)$, $(7.0, 3.0, x)$ and $(2.0, 3.0, 1.0)$ in $\mathbb{R}^3$ are linearly dependent, the value of x is _____
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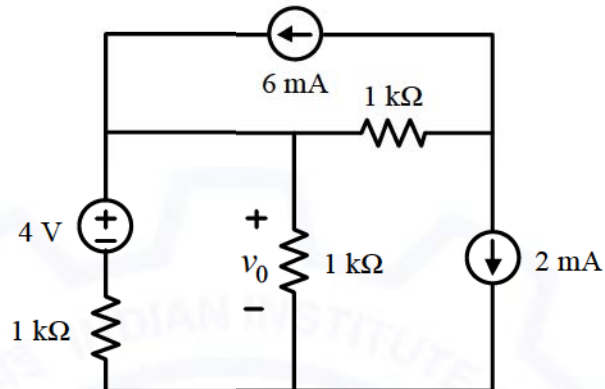
Q.17	Consider the vector field $\mathbf{F} = a_x(4y - c_1z) + a_y(4x + 2z) + a_z(2y + z)$ in a rectangular coordinate system (x, y, z) with unit vectors a_x , a_y , and a_z . If the field $\mathbf{F}$ is irrotational (conservative), then the constant c_1 (in integer) is _____
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Q.18	Consider the circuit shown in the figure. The current I flowing through the $7\ \Omega$ resistor between P and Q (rounded off to one decimal place) is _____ A.
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Q.19

Consider the circuit shown in the figure.

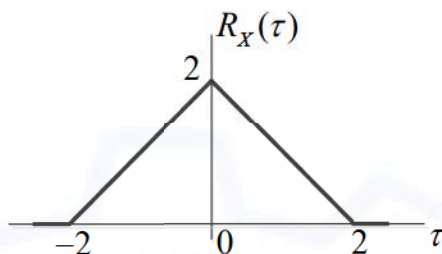


The value of v_0 (rounded off to one decimal place) is _____ V.

Q.20

An 8-bit unipolar (all analog output values are positive) digital-to-analog converter (DAC) has a full-scale voltage range from 0 V to 7.68 V. If the digital input code is 10010110 (the leftmost bit is MSB), then the analog output voltage of the DAC (rounded off to one decimal place) is _____ V.



Q.21	The autocorrelation function $R_X(\tau)$ of a wide-sense stationary random process $X(t)$ is shown in the figure.  The average power of $X(t)$ is _____
Q.22	Consider a carrier signal which is amplitude modulated by a single-tone sinusoidal message signal with a modulation index of 50%. If the carrier and one of the sidebands are suppressed in the modulated signal, the percentage of power saved (<i>rounded off to one decimal place</i>) is _____
Q.23	A speech signal, band limited to 4 kHz, is sampled at 1.25 times the Nyquist rate. The speech samples, assumed to be statistically independent and uniformly distributed in the range -5 V to $+5$ V, are subsequently quantized in an 8-bit uniform quantizer and then transmitted over a voice-grade AWGN telephone channel. If the ratio of transmitted signal power to channel noise power is 26 dB, the minimum channel bandwidth required to ensure reliable transmission of the signal with arbitrarily small probability of transmission error (<i>rounded off to two decimal places</i>) is _____ kHz.
Q.24	A 4 kHz sinusoidal message signal having amplitude 4 V is fed to a delta modulator (DM) operating at a sampling rate of 32 kHz. The minimum step size required to avoid slope overload noise in the DM (<i>rounded off to two decimal places</i>) is _____ V.
Q.25	The refractive indices of the core and cladding of an optical fiber are 1.50 and 1.48, respectively. The critical propagation angle, which is defined as the maximum angle that the light beam makes with the axis of the optical fiber to achieve the total internal reflection, (<i>rounded off to two decimal places</i>) is _____ degree.

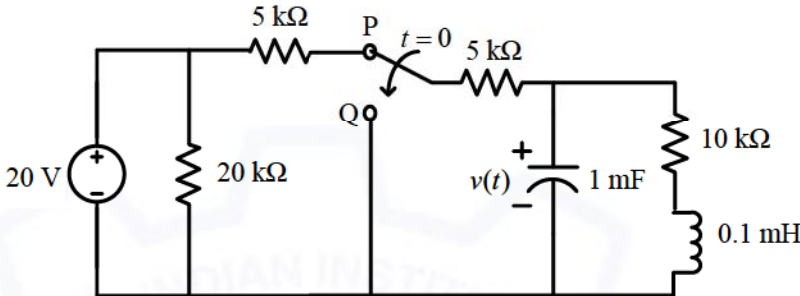


Q. 26 – Q. 35 Multiple Choice Question (MCQ), carry TWO marks each (for each wrong answer: – 2/3).

Q.26	Consider the integral $\oint_C \frac{\sin(x)}{x^2(x^2 + 4)} dx$ where C is a counter-clockwise oriented circle defined as $x - i = 2$. The value of the integral is
(A)	$-\frac{\pi}{8} \sin(2i)$
(B)	$\frac{\pi}{8} \sin(2i)$
(C)	$-\frac{\pi}{4} \sin(2i)$
(D)	$\frac{\pi}{4} \sin(2i)$

Q.27	A box contains the following three coins. I. A fair coin with head on one face and tail on the other face. II. A coin with heads on both the faces. III. A coin with tails on both the faces. A coin is picked randomly from the box and tossed. Out of the two remaining coins in the box, one coin is then picked randomly and tossed. If the first toss results in a head, the probability of getting a head in the second toss is
(A)	$\frac{2}{5}$
(B)	$\frac{1}{3}$
(C)	$\frac{1}{2}$
(D)	$\frac{2}{3}$



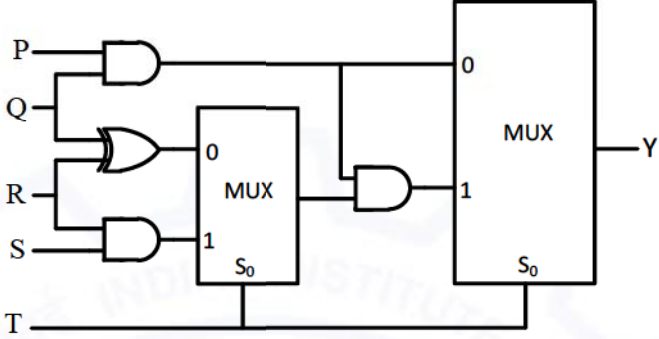
Q.28	The switch in the circuit in the figure is in position P for a long time and then moved to position Q at time $t = 0$.  The value of $\frac{dv(t)}{dt}$ at $t = 0^+$ is
(A)	0 V/s
(B)	3 V/s
(C)	−3 V/s
(D)	−5 V/s



Q.29	Consider the two-port network shown in the figure. The admittance parameters, in siemens, are
(A)	$y_{11} = 2, y_{12} = -4, y_{21} = -4, y_{22} = 2$
(B)	$y_{11} = 1, y_{12} = -2, y_{21} = -1, y_{22} = 3$
(C)	$y_{11} = 2, y_{12} = -4, y_{21} = -1, y_{22} = 2$
(D)	$y_{11} = 2, y_{12} = -4, y_{21} = -4, y_{22} = 3$

Q.30	For an n-channel silicon MOSFET with 10 nm gate oxide thickness, the substrate sensitivity ($\partial V_T / \partial V_{BS}$) is found to be 50 mV/V at a substrate voltage $V_{BS} = 2\text{ V}$, where V_T is the threshold voltage of the MOSFET. Assume that, $V_{BS} \gg 2\Phi_B$, where $q\Phi_B$ is the separation between the Fermi energy level E_F and the intrinsic level E_i in the bulk. Parameters given are Electron charge (q) = $1.6 \times 10^{-19}\text{ C}$ Vacuum permittivity (ϵ_0) = $8.85 \times 10^{-12}\text{ F/m}$ Relative permittivity of silicon (ϵ_{Si}) = 12 Relative permittivity of oxide (ϵ_{ox}) = 4 The doping concentration of the substrate is
(A)	$7.37 \times 10^{15}\text{ cm}^{-3}$
(B)	$4.37 \times 10^{15}\text{ cm}^{-3}$
(C)	$2.37 \times 10^{15}\text{ cm}^{-3}$
(D)	$9.37 \times 10^{15}\text{ cm}^{-3}$



Q.31	The propagation delays of the XOR gate, AND gate and multiplexer (MUX) in the circuit shown in the figure are 4 ns, 2 ns and 1 ns, respectively.  If all the inputs P, Q, R, S and T are applied simultaneously and held constant, the maximum propagation delay of the circuit is
(A)	3 ns
(B)	5 ns
(C)	6 ns
(D)	7 ns



Q.32	The content of the registers are $R_1 = 25H$, $R_2 = 30H$ and $R_3 = 40H$. The following machine instructions are executed. PUSH{R₁} PUSH{R₂} PUSH{R₃} POP{R₁} POP{R₂} POP{R₃} After execution, the content of registers R_1, R_2, R_3 are
(A)	$R_1 = 40H$, $R_2 = 30H$, $R_3 = 25H$
(B)	$R_1 = 25H$, $R_2 = 30H$, $R_3 = 40H$
(C)	$R_1 = 30H$, $R_2 = 40H$, $R_3 = 25H$
(D)	$R_1 = 40H$, $R_2 = 25H$, $R_3 = 30H$



Q.33	The electrical system shown in the figure converts input source current $i_s(t)$ to output voltage $v_o(t)$. Current $i_L(t)$ in the inductor and voltage $v_c(t)$ across the capacitor are taken as the state variables, both assumed to be initially equal to zero, i.e., $i_L(0) = 0$ and $v_c(0) = 0$. The system is
(A)	completely state controllable as well as completely observable
(B)	completely state controllable but not observable
(C)	completely observable but not state controllable
(D)	neither state controllable nor observable



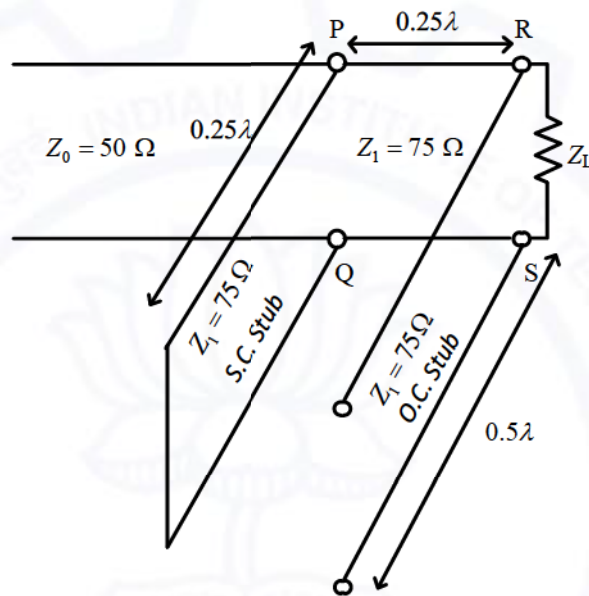
Electronics and Communication Engineering (EC)

Q.34	A digital transmission system uses a (7, 4) systematic linear Hamming code for transmitting data over a noisy channel. If three of the message-codeword pairs in this code ($m_i; c_i$), where c_i is the codeword corresponding to the i^{th} message m_i , are known to be (1 1 0 0; 0 1 0 1 1 0 0), (1 1 1 0; 0 0 1 1 1 1 0) and (0 1 1 0; 1 0 0 0 1 1 0), then which of the following is a valid codeword in this code?
(A)	1 1 0 1 0 0 1
(B)	1 0 1 1 0 1 0
(C)	0 0 0 1 0 1 1
(D)	0 1 1 0 1 0 0



Q.35

The impedance matching network shown in the figure is to match a lossless line having characteristic impedance $Z_0 = 50 \Omega$ with a load impedance Z_L . A quarter-wave line having a characteristic impedance $Z_1 = 75 \Omega$ is connected to Z_L . Two stubs having characteristic impedance of 75Ω each are connected to this quarter-wave line. One is a short-circuited (S.C.) stub of length 0.25λ connected across PQ and the other one is an open-circuited (O.C.) stub of length 0.5λ connected across RS.



The impedance matching is achieved when the real part of Z_L is

- (A) 112.5Ω
(B) 75.0Ω
(C) 50.0Ω
(D) 33.3Ω



Q.36 – Q.55 Numerical Answer Type (NAT), carry TWO mark each (no negative marks).

Q.36	A real 2×2 non-singular matrix A with repeated eigenvalue is given as $A = \begin{bmatrix} x & -3.0 \\ 3.0 & 4.0 \end{bmatrix}$ where x is a real positive number. The value of x (rounded off to one decimal place) is _____
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Q.37	For a vector field $D = \rho \cos^2 \phi \mathbf{a}_\rho + z^2 \sin^2 \phi \mathbf{a}_\phi$ in a cylindrical coordinate system (ρ, ϕ, z) with unit vectors $\mathbf{a}_\rho, \mathbf{a}_\phi$ and $\mathbf{a}_z$, the net flux of D leaving the closed surface of the cylinder ($\rho = 3, 0 \leq z \leq 2$) (rounded off to two decimal places) is _____
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Q.38	In the circuit shown in the figure, the switch is closed at time $t = 0$, while the capacitor is initially charged to -5 V (i.e., $v_c(0) = -5$ V). The time after which the voltage across the capacitor becomes zero (rounded off to three decimal places) is _____ ms.
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Q.39	The exponential Fourier series representation of a continuous-time periodic signal $x(t)$ is defined as $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$ where ω_0 is the fundamental angular frequency of $x(t)$ and the coefficients of the series are a_k. The following information is given about $x(t)$ and a_k. I. $x(t)$ is real and even, having a fundamental period of 6 II. The average value of $x(t)$ is 2 III. $a_k = \begin{cases} k, & 1 \leq k \leq 3 \\ 0, & k > 3 \end{cases}$ The average power of the signal $x(t)$ (rounded off to one decimal place) is _____
Q.40	For a unit step input $u[n]$, a discrete-time LTI system produces an output signal $(2\delta[n+1] + \delta[n] + \delta[n-1])$. Let $y[n]$ be the output of the system for an input $\left(\left(\frac{1}{2}\right)^n u[n]\right)$. The value of $y[0]$ is _____
Q.41	Consider the signals $x[n] = 2^{n-1} u[-n+2]$ and $y[n] = 2^{-n+2} u[n+1]$, where $u[n]$ is the unit step sequence. Let $X(e^{j\omega})$ and $Y(e^{j\omega})$ be the discrete-time Fourier transform of $x[n]$ and $y[n]$, respectively. The value of the integral $\frac{1}{2\pi} \int_0^{2\pi} X(e^{j\omega}) Y(e^{-j\omega}) d\omega$ (rounded off to one decimal place) is _____



Q.42

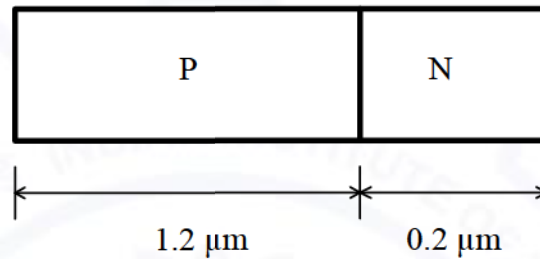
A silicon P-N junction is shown in the figure. The doping in the P region is $5 \times 10^{16} \text{ cm}^{-3}$ and doping in the N region is $10 \times 10^{16} \text{ cm}^{-3}$. The parameters given are

Built-in voltage (Φ_{bi}) = 0.8 V

Electron charge (q) = $1.6 \times 10^{-19} \text{ C}$

Vacuum permittivity (ϵ_0) = $8.85 \times 10^{-12} \text{ F/m}$

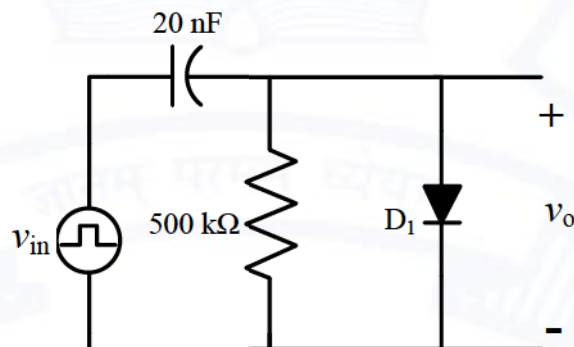
Relative permittivity of silicon (ϵ_{Si}) = 12



The magnitude of reverse bias voltage that would completely deplete one of the two regions (P or N) prior to the other (*rounded off to one decimal place*) is _____ V.

Q.43

An asymmetrical periodic pulse train v_{in} of 10 V amplitude with on-time $T_{ON} = 1 \text{ ms}$ and off-time $T_{OFF} = 1 \mu\text{s}$ is applied to the circuit shown in the figure. The diode D_1 is ideal.

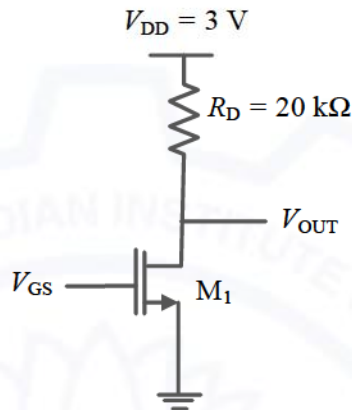


The difference between the maximum voltage and minimum voltage of the output waveform v_o (*in integer*) is _____ V.



Q.44

For the transistor M_1 in the circuit shown in the figure, $\mu_n C_{ox} = 100 \mu\text{A}/\text{V}^2$ and $(W/L) = 10$, where μ_n is the mobility of electron, C_{ox} is the oxide capacitance per unit area, W is the width and L is the length.

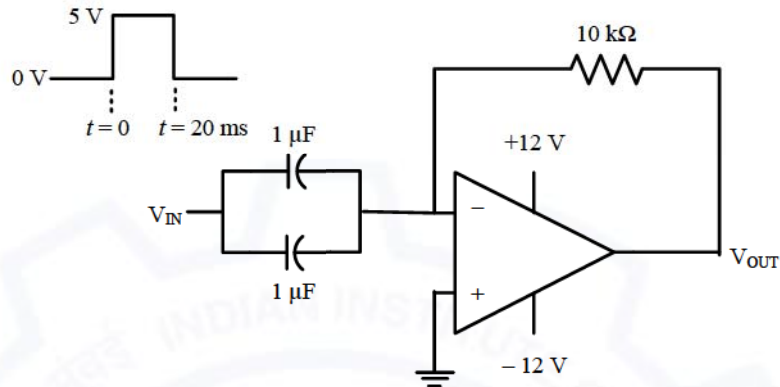


The channel length modulation coefficient is ignored. If the gate-to-source voltage V_{GS} is 1 V to keep the transistor at the edge of saturation, then the threshold voltage of the transistor (rounded off to one decimal place) is _____ V.



Q.45

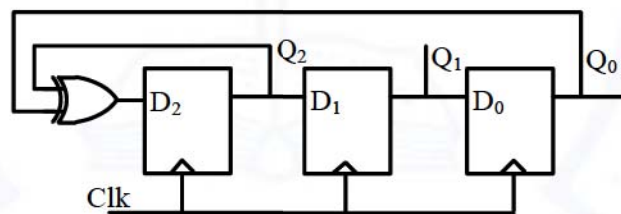
A circuit with an ideal OPAMP is shown in the figure. A pulse V_{IN} of 20 ms duration is applied to the input. The capacitors are initially uncharged.



The output voltage V_{OUT} of this circuit at $t = 0^+$ (in integer) is _____ V.

Q.46

The propagation delay of the exclusive-OR (XOR) gate in the circuit in the figure is 3 ns. The propagation delay of all the flip-flops is assumed to be zero. The clock (Clk) frequency provided to the circuit is 500 MHz.

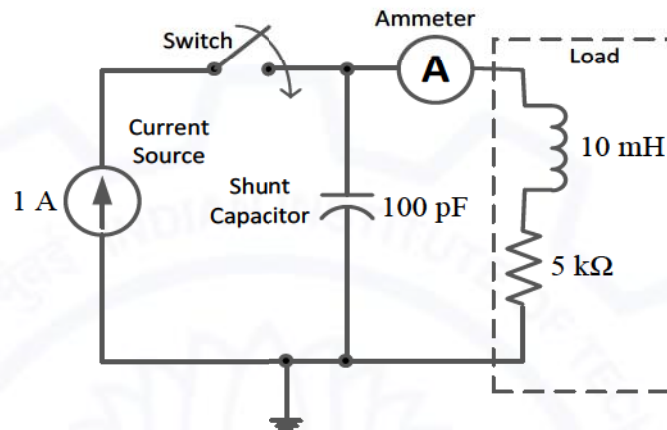


Starting from the initial value of the flip-flop outputs $Q_2Q_1Q_0 = 1\ 1\ 1$ with $D_2 = 1$, the minimum number of triggering clock edges after which the flip-flop outputs $Q_2Q_1Q_0$ becomes 1 0 0 (in integer) is _____



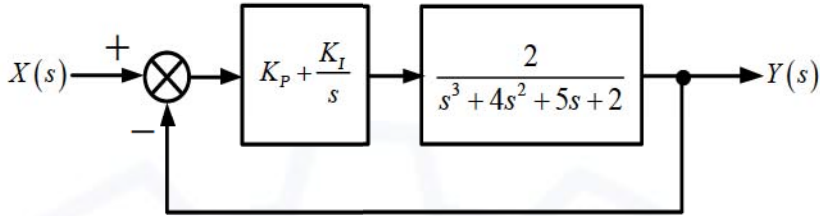
Q.47

The circuit in the figure contains a current source driving a load having an inductor and a resistor in series, with a shunt capacitor across the load. The ammeter is assumed to have zero resistance. The switch is closed at time $t = 0$.



Initially, when the switch is open, the capacitor is discharged and the ammeter reads zero ampere. After the switch is closed, the ammeter reading keeps fluctuating for some time till it settles to a final steady value. The maximum ammeter reading that one will observe after the switch is closed (rounded off to two decimal places) is _____ A.



Q.48	A unity feedback system that uses proportional-integral (PI) control is shown in the figure.  The stability of the overall system is controlled by tuning the PI control parameters K_P and K_I. The maximum value of K_I that can be chosen so as to keep the overall system stable or, in the worst case, marginally stable (rounded off to three decimal places) is _____
Q.49	A sinusoidal message signal having root mean square value of 4 V and frequency of 1 kHz is fed to a phase modulator with phase deviation constant 2 rad/volt. If the carrier signal is $c(t) = 2 \cos(2\pi 10^6 t)$, the maximum instantaneous frequency of the phase modulated signal (rounded off to one decimal place) is _____ Hz.
Q.50	Consider a superheterodyne receiver tuned to 600 kHz. If the local oscillator feeds a 1000 kHz signal to the mixer, the image frequency (in integer) is _____ kHz.
Q.51	In a high school having equal number of boy students and girl students, 75% of the students study Science and the remaining 25% students study Commerce. Commerce students are two times more likely to be a boy than are Science students. The amount of information gained in knowing that a randomly selected girl student studies Commerce (rounded off to three decimal places) is _____ bits.



Q.52	A message signal having peak-to-peak value of 2 V, root mean square value of 0.1 V and bandwidth of 5 kHz is sampled and fed to a pulse code modulation (PCM) system that uses a uniform quantizer. The PCM output is transmitted over a channel that can support a maximum transmission rate of 50 kbps. Assuming that the quantization error is uniformly distributed, the maximum signal to quantization noise ratio that can be obtained by the PCM system (<i>rounded off to two decimal places</i>) is _____
Q.53	Consider a polar non-return to zero (NRZ) waveform, using +2 V and −2 V for representing binary '1' and '0' respectively, is transmitted in the presence of additive zero-mean white Gaussian noise with variance 0.4 V^2 . If the <i>a priori</i> probability of transmission of a binary '1' is 0.4, the optimum threshold voltage for a maximum <i>a posteriori</i> (MAP) receiver (<i>rounded off to two decimal places</i>) is _____ V.
Q.54	A standard air-filled rectangular waveguide with dimensions $a = 8 \text{ cm}$, $b = 4 \text{ cm}$, operates at 3.4 GHz. For the dominant mode of wave propagation, the phase velocity of the signal is v_p . The value (<i>rounded off to two decimal places</i>) of v_p/c , where c denotes the velocity of light, is _____
Q.55	An antenna with a directive gain of 6 dB is radiating a total power of 16 kW. The amplitude of the electric field in free space at a distance of 8 km from the antenna in the direction of 6 dB gain (<i>rounded off to three decimal places</i>) is _____ V/m.

END OF THE QUESTION PAPER

Graduate Aptitude Test in Engineering (GATE 2021)

Subject/Paper: Electronics and Communication Engineering (EC)

Q. No.	Session	Question Type MCQ/MSQ/NAT	Section Name	Answer Key/Range	Marks	Negative Marks
1	4	MCQ	GA	C	1	1/3
2	4	MCQ	GA	B	1	1/3
3	4	MCQ	GA	C	1	1/3
4	4	MCQ	GA	C	1	1/3
5	4	MCQ	GA	B	1	1/3
6	4	MCQ	GA	C	2	2/3
7	4	MCQ	GA	D	2	2/3
8	4	MCQ	GA	A	2	2/3
9	4	MCQ	GA	C	2	2/3
10	4	MCQ	GA	A	2	2/3
1	4	MCQ	EC	A	1	1/3
2	4	MCQ	EC	B	1	1/3
3	4	MCQ	EC	B	1	1/3
4	4	MCQ	EC	B	1	1/3
5	4	MCQ	EC	C	1	1/3
6	4	MCQ	EC	D	1	1/3
7	4	MCQ	EC	A	1	1/3
8	4	MCQ	EC	D	1	1/3
9	4	MCQ	EC	MTA	1	1/3
10	4	MCQ	EC	A	1	1/3

GATE 2021 Answer Key for Electronics and Communication Engineering (EC)

Q. No.	Session	Question Type MCQ/MSQ/NAT	Section Name	Answer Key/Range	Marks	Negative Marks
11	4	MCQ	EC	B	1	1/3
12	4	MCQ	EC	C	1	1/3
13	4	MCQ	EC	C	1	1/3
14	4	MCQ	EC	D	1	1/3
15	4	MCQ	EC	A	1	1/3
16	4	NAT	EC	8.0 to 8.0	1	0
17	4	NAT	EC	0 to 0	1	0
18	4	NAT	EC	0.5 to 0.5	1	0
19	4	NAT	EC	1.0 to 1.0	1	0
20	4	NAT	EC	4.5 to 4.5	1	0
21	4	NAT	EC	2.0 to 2.0	1	0
22	4	NAT	EC	94.2 to 94.6	1	0
23	4	NAT	EC	9.24 to 9.28	1	0
24	4	NAT	EC	2.80 to 3.20	1	0
25	4	NAT	EC	9.30 to 9.44	1	0
26	4	MCQ	EC	MTA	2	2/3
27	4	MCQ	EC	B	2	2/3
28	4	MCQ	EC	C	2	2/3
29	4	MCQ	EC	C	2	2/3
30	4	MCQ	EC	A	2	2/3
31	4	MCQ	EC	C	2	2/3
32	4	MCQ	EC	A	2	2/3
33	4	MCQ	EC	D	2	2/3



GATE 2022 General Aptitude (GA)

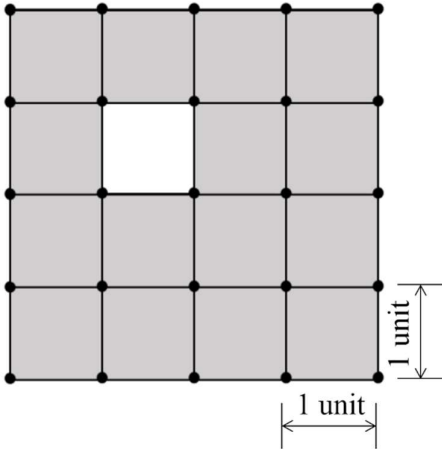
Q.1 – Q.5 Carry ONE mark each.

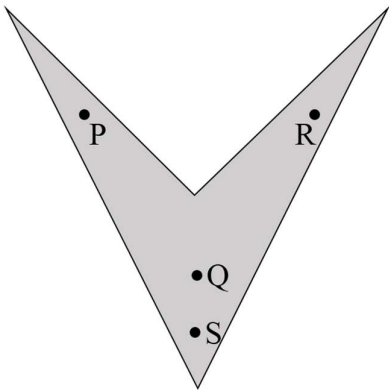
Q.1	Mr. X speaks _____ Japanese _____ Chinese.
(A)	neither / or
(B)	either / nor
(C)	neither / nor
(D)	also / but

Q.2	A sum of money is to be distributed among P, Q, R, and S in the proportion 5 : 2 : 4 : 3, respectively. If R gets ₹ 1000 more than S, what is the share of Q (in ₹)?
(A)	500
(B)	1000
(C)	1500
(D)	2000



Q.3	A trapezium has vertices marked as P, Q, R and S (in that order anticlockwise). The side PQ is parallel to side SR. Further, it is given that, $PQ = 11$ cm, $QR = 4$ cm, $RS = 6$ cm and $SP = 3$ cm. What is the shortest distance between PQ and SR (in cm)?
(A)	1.80
(B)	2.40
(C)	4.20
(D)	5.76

Q.4	The figure shows a grid formed by a collection of unit squares. The unshaded unit square in the grid represents a hole.  What is the maximum number of squares without a “hole in the interior” that can be formed within the 4×4 grid using the unit squares as building blocks?
(A)	15
(B)	20
(C)	21
(D)	26

Q.5	An art gallery engages a security guard to ensure that the items displayed are protected. The diagram below represents the plan of the gallery where the boundary walls are opaque. The location the security guard posted is identified such that all the inner space (shaded region in the plan) of the gallery is within the line of sight of the security guard. If the security guard does not move around the posted location and has a 360° view, which one of the following correctly represents the set of ALL possible locations among the locations P, Q, R and S, where the security guard can be posted to watch over the entire inner space of the gallery. 
(A)	P and Q
(B)	Q
(C)	Q and S
(D)	R and S

Q. 6 – Q. 10 Carry TWO marks each.

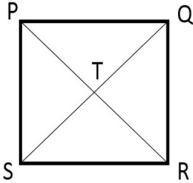
Q.6	Mosquitoes pose a threat to human health. Controlling mosquitoes using chemicals may have undesired consequences. In Florida, authorities have used genetically modified mosquitoes to control the overall mosquito population. It remains to be seen if this novel approach has unforeseen consequences. Which one of the following is the correct logical inference based on the information in the above passage?
(A)	Using chemicals to kill mosquitoes is better than using genetically modified mosquitoes because genetic engineering is dangerous
(B)	Using genetically modified mosquitoes is better than using chemicals to kill mosquitoes because they do not have any side effects
(C)	Both using genetically modified mosquitoes and chemicals have undesired consequences and can be dangerous
(D)	Using chemicals to kill mosquitoes may have undesired consequences but it is not clear if using genetically modified mosquitoes has any negative consequence



Q.7	Consider the following inequalities. (i) $2x - 1 > 7$ (ii) $2x - 9 < 1$ Which one of the following expressions below satisfies the above two inequalities?
(A)	$x \leq -4$
(B)	$-4 < x \leq 4$
(C)	$4 < x < 5$
(D)	$x \geq 5$

Q.8	Four points $P(0, 1)$, $Q(0, -3)$, $R(-2, -1)$, and $S(2, -1)$ represent the vertices of a quadrilateral. What is the area enclosed by the quadrilateral?
(A)	4
(B)	$4\sqrt{2}$
(C)	8
(D)	$8\sqrt{2}$

Q.9	In a class of five students P, Q, R, S and T, only one student is known to have copied in the exam. The disciplinary committee has investigated the situation and recorded the statements from the students as given below. Statement of P: R has copied in the exam. Statement of Q: S has copied in the exam. Statement of R: P did not copy in the exam. Statement of S: Only one of us is telling the truth. Statement of T: R is telling the truth. The investigating team had authentic information that S never lies. Based on the information given above, the person who has copied in the exam is
(A)	R
(B)	P
(C)	Q
(D)	T

Q.10	Consider the following square with the four corners and the center marked as P, Q, R, S and T respectively. Let X, Y and Z represent the following operations: X: rotation of the square by 180 degree with respect to the S-Q axis. Y: rotation of the square by 180 degree with respect to the P-R axis. Z: rotation of the square by 90 degree clockwise with respect to the axis perpendicular, going into the screen and passing through the point T. Consider the following three distinct sequences of operation (which are applied in the left to right order). (1) XYZZ (2) XY (3) ZZZZ Which one of the following statements is correct as per the information provided above?	
(A)	The sequence of operations (1) and (2) are equivalent	
(B)	The sequence of operations (1) and (3) are equivalent	
(C)	The sequence of operations (2) and (3) are equivalent	
(D)	The sequence of operations (1), (2) and (3) are equivalent	

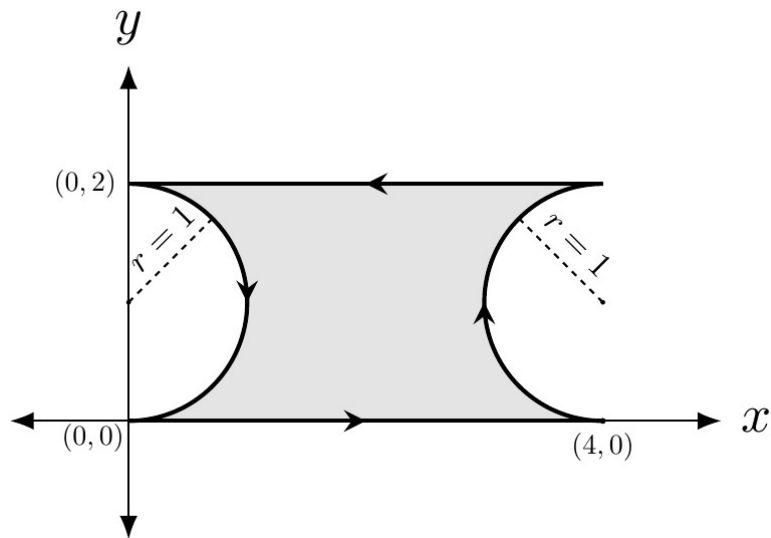
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Q.11 – Q.35 Carry ONE mark Each

Q.11 Consider the two-dimensional vector field $\vec{F}(x, y) = x \vec{i} + y \vec{j}$, where $\vec{i}$ and $\vec{j}$ denote the unit vectors along the x -axis and the y -axis, respectively. A contour C in the x - y plane, as shown in the figure, is composed of two horizontal lines connected at the two ends by two semicircular arcs of unit radius. The contour is traversed in the counter-clockwise sense. The value of the closed path integral

$$\oint_C \vec{F}(x, y) \cdot (dx \vec{i} + dy \vec{j})$$

is _____.



(A) 0

(B) 1

(C) $8 + 2\pi$

(D) -1

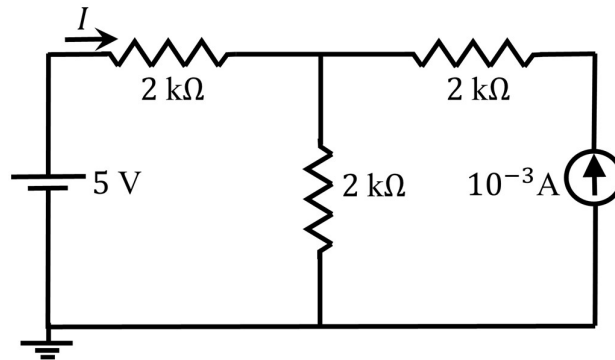


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Q.12	Consider a system of linear equations $Ax = b$, where $A = \begin{bmatrix} 1 & -\sqrt{2} & 3 \\ -1 & \sqrt{2} & -3 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$ This system of equations admits _____.
(A)	a unique solution for x
(B)	infinitely many solutions for x
(C)	no solutions for x
(D)	exactly two solutions for x

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Q.13 The current I in the circuit shown is _____.



(A) $1.25 \times 10^{-3}\text{ A}$

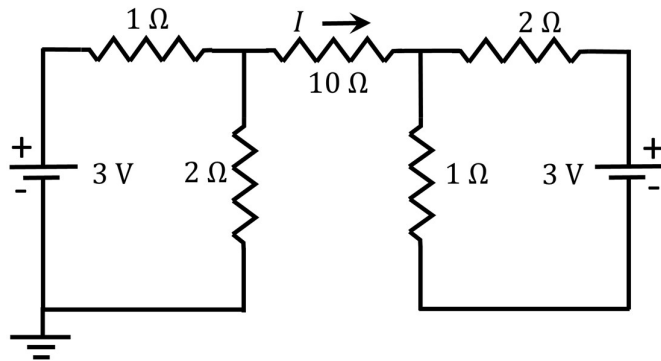
(B) $0.75 \times 10^{-3}\text{ A}$

(C) $-0.5 \times 10^{-3}\text{ A}$

(D) $1.16 \times 10^{-3}\text{ A}$

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Q.14 Consider the circuit shown in the figure. The current I flowing through the $10\ \Omega$ resistor is _____.



(A) 1 A

(B) 0 A

(C) 0.1 A

(D) -0.1 A

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Q.15	The Fourier transform $X(j\omega)$ of the signal $x(t) = \frac{t}{(1 + t^2)^2}$ is _____.
(A)	$\frac{\pi}{2j} \omega e^{- \omega }$
(B)	$\frac{\pi}{2} \omega e^{- \omega }$
(C)	$\frac{\pi}{2j} e^{- \omega }$
(D)	$\frac{\pi}{2} e^{- \omega }$

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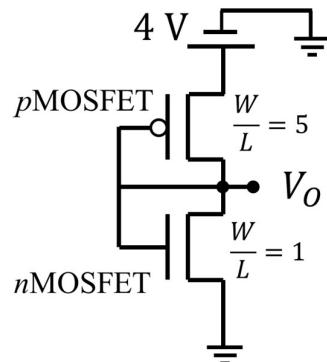
Q.16	Consider a long rectangular bar of direct bandgap p-type semiconductor. The equilibrium hole density is 10^{17} cm^{-3} and the intrinsic carrier concentration is 10^{10} cm^{-3}. Electron and hole diffusion lengths are $2 \mu\text{m}$ and $1 \mu\text{m}$, respectively. The left side of the bar ($x = 0$) is uniformly illuminated with a laser having photon energy greater than the bandgap of the semiconductor. Excess electron-hole pairs are generated ONLY at $x = 0$ because of the laser. The steady state electron density at $x = 0$ is 10^{14} cm^{-3} due to laser illumination. Under these conditions and ignoring electric field, the closest approximation (among the given options) of the steady state electron density at $x = 2 \mu\text{m}$, is _____.
(A)	$0.37 \times 10^{14} \text{ cm}^{-3}$
(B)	$0.63 \times 10^{13} \text{ cm}^{-3}$
(C)	$3.7 \times 10^{14} \text{ cm}^{-3}$
(D)	10^3 cm^{-3}

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Q.17	In a non-degenerate bulk semiconductor with electron density $n = 10^{16} \text{ cm}^{-3}$, the value of $E_C - E_{Fn} = 200 \text{ meV}$, where E_C and E_{Fn} denote the bottom of the conduction band energy and electron Fermi level energy, respectively. Assume thermal voltage as 26 meV and the intrinsic carrier concentration is 10^{10} cm^{-3} . For $n = 0.5 \times 10^{16} \text{ cm}^{-3}$, the closest approximation of the value of $(E_C - E_{Fn})$, among the given options, is _____.
(A)	226 meV
(B)	174 meV
(C)	218 meV
(D)	182 meV

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Q.18 Consider the CMOS circuit shown in the figure (substrates are connected to their respective sources). The gate width (W) to gate length (L) ratios ($\frac{W}{L}$) of the transistors are as shown. Both the transistors have the same gate oxide capacitance per unit area. For the p MOSFET, the threshold voltage is -1 V and the mobility of holes is $40 \frac{\text{cm}^2}{\text{V.s}}$. For the n MOSFET, the threshold voltage is 1 V and the mobility of electrons is $300 \frac{\text{cm}^2}{\text{V.s}}$. The steady state output voltage V_O is _____.



(A) equal to 0 V

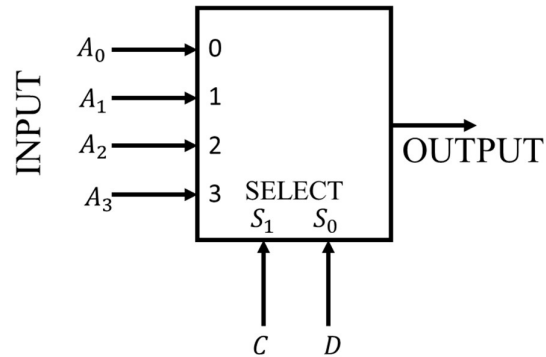
(B) more than 2 V

(C) less than 2 V

(D) equal to 2 V

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Q.19 Consider the 2-bit multiplexer (MUX) shown in the figure. For OUTPUT to be the XOR of C and D, the values for A_0 , A_1 , A_2 , and A_3 are _____.



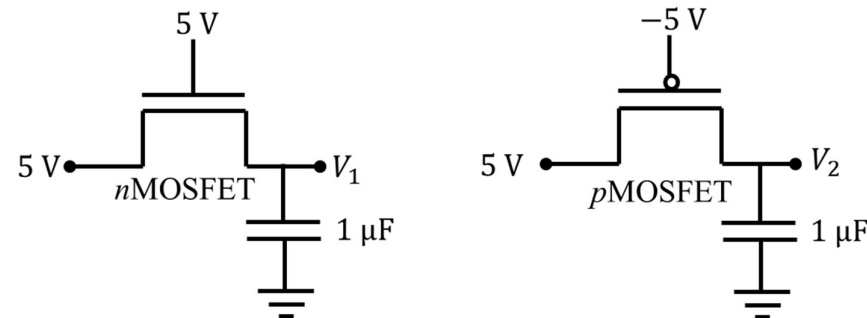
(A) $A_0 = 0, A_1 = 0, A_2 = 1, A_3 = 1$

(B) $A_0 = 1, A_1 = 0, A_2 = 1, A_3 = 0$

(C) $A_0 = 0, A_1 = 1, A_2 = 1, A_3 = 0$

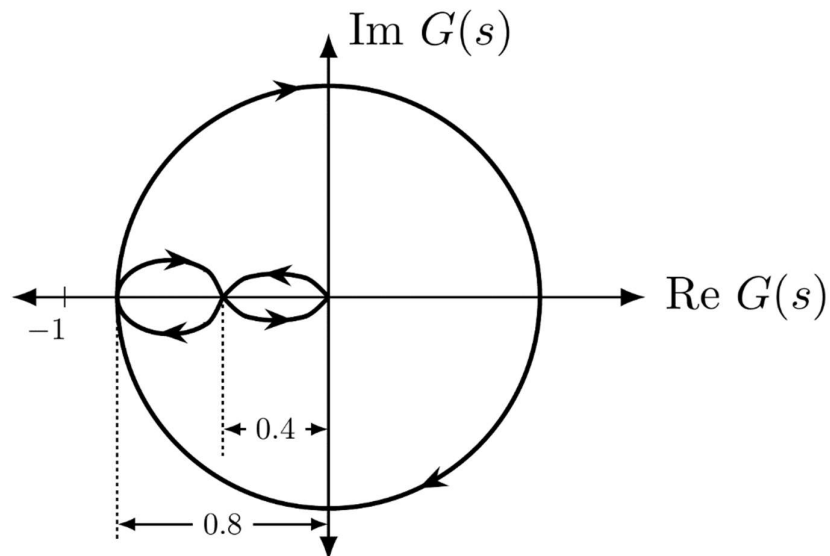
(D) $A_0 = 1, A_1 = 1, A_2 = 0, A_3 = 0$

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Q.20	The ideal long channel nMOSFET and pMOSFET devices shown in the circuits have threshold voltages of 1 V and -1 V, respectively. The MOSFET substrates are connected to their respective sources. Ignore leakage currents and assume that the capacitors are initially discharged. For the applied voltages as shown, the steady state voltages are _____.
	
(A)	$V_1 = 5$ V, $V_2 = 5$ V
(B)	$V_1 = 5$ V, $V_2 = 4$ V
(C)	$V_1 = 4$ V, $V_2 = 5$ V
(D)	$V_1 = 4$ V, $V_2 = -5$ V

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Q.21 Consider a closed-loop control system with unity negative feedback and $KG(s)$ in the forward path, where the gain $K = 2$. The complete Nyquist plot of the transfer function $G(s)$ is shown in the figure. Note that the Nyquist contour has been chosen to have the clockwise sense. Assume $G(s)$ has no poles on the closed right-half of the complex plane. The number of poles of the closed-loop transfer function in the closed right-half of the complex plane is _____.



(A) 0

(B) 1

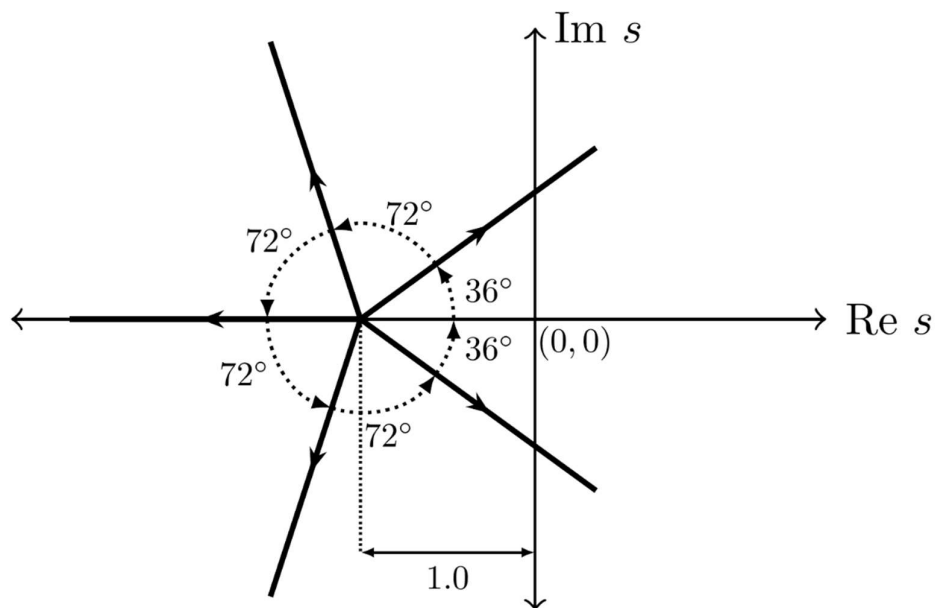
(C) 2

(D) 3

GATE 2022 Electronics and Communications Engineering (EC)

Q.22 The root-locus plot of a closed-loop system with unity negative feedback and transfer function $KG(s)$ in the forward path is shown in the figure. Note that K is varied from 0 to ∞ .

Select the transfer function $G(s)$ that results in the root-locus plot of the closed-loop system as shown in the figure.



(A) $G(s) = \frac{1}{(s+1)^5}$

(B) $G(s) = \frac{1}{s^5 + 1}$

(C) $G(s) = \frac{s-1}{(s+1)^6}$

(D) $G(s) = \frac{s+1}{s^6 + 1}$

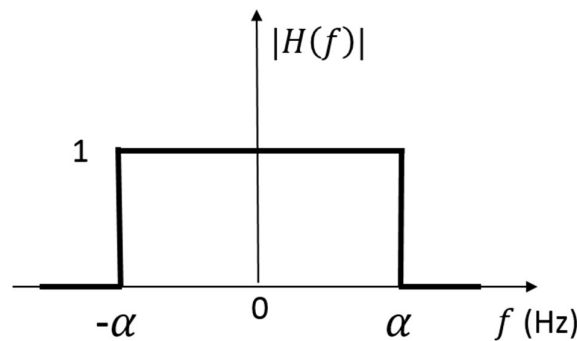
GATE 2022 Electronics and Communications Engineering (EC)

Q.23 The frequency response $H(f)$ of a linear time-invariant system has magnitude as shown in the figure.

Statement I: The system is necessarily a pure delay system for inputs which are bandlimited to $-\alpha \leq f \leq \alpha$.

Statement II: For any wide-sense stationary input process with power spectral density $S_X(f)$, the output power spectral density $S_Y(f)$ obeys $S_Y(f) = S_X(f)$ for $-\alpha \leq f \leq \alpha$.

Which one of the following combinations is true?



(A) Statement I is correct, Statement II is correct

(B) Statement I is correct, Statement II is incorrect

(C) Statement I is incorrect, Statement II is correct

(D) Statement I is incorrect, Statement II is incorrect



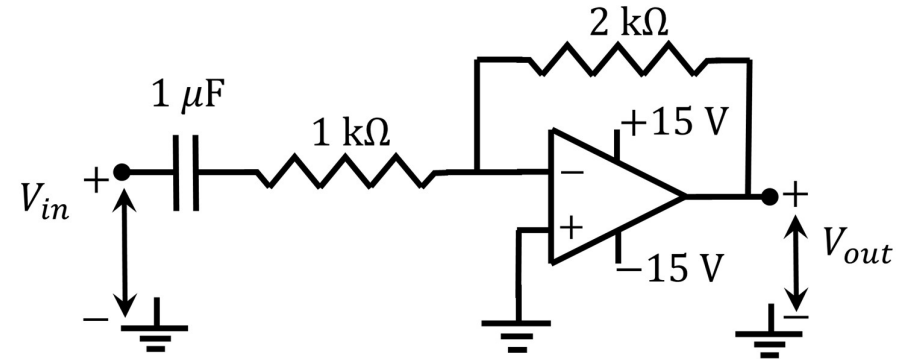
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Q.24	In a circuit, there is a series connection of an ideal resistor and an ideal capacitor. The conduction current (in Amperes) through the resistor is $2\sin(t + \pi/2)$. The displacement current (in Amperes) through the capacitor is _____.
(A)	$2\sin(t)$
(B)	$2\sin(t + \pi)$
(C)	$2\sin(t + \pi/2)$
(D)	0

GATE 2022 Electronics and Communications Engineering (EC)

Q.25	Consider the following partial differential equation (PDE) $a \frac{\partial^2 f(x, y)}{\partial x^2} + b \frac{\partial^2 f(x, y)}{\partial y^2} = f(x, y),$ where a and b are distinct positive real numbers. Select the combination(s) of values of the real parameters ξ and η such that $f(x, y) = e^{(\xi x + \eta y)}$ is a solution of the given PDE.
(A)	$\xi = \frac{1}{\sqrt{2a}}, \eta = \frac{1}{\sqrt{2b}}$
(B)	$\xi = \frac{1}{\sqrt{a}}, \eta = 0$
(C)	$\xi = 0, \eta = 0$
(D)	$\xi = \frac{1}{\sqrt{a}}, \eta = \frac{1}{\sqrt{b}}$

GATE 2022 Electronics and Communications Engineering (EC)

Q.26	An ideal OPAMP circuit with a sinusoidal input is shown in the figure. The 3 dB frequency is the frequency at which the magnitude of the voltage gain decreases by 3 dB from the maximum value. Which of the options is/are correct?
	
(A)	The circuit is a low pass filter.
(B)	The circuit is a high pass filter.
(C)	The 3 dB frequency is 1000 rad/s.
(D)	The 3 dB frequency is $\frac{1000}{3}$ rad/s.

GATE 2022 Electronics and Communications Engineering (EC)

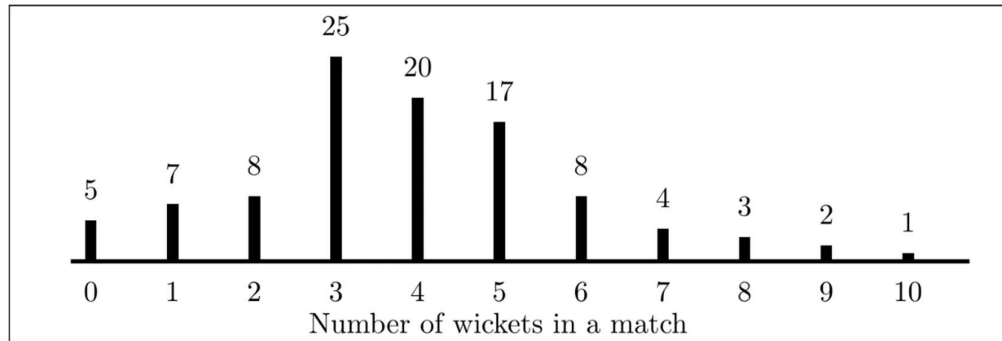
Q.27	Select the Boolean function(s) equivalent to $x + yz$, where x, y , and z are Boolean variables, and $+$ denotes logical OR operation.
(A)	$x + z + xy$
(B)	$(x + y)(x + z)$
(C)	$x + xy + yz$
(D)	$x + xz + xy$
Q.28	Select the correct statement(s) regarding CMOS implementation of NOT gates.
(A)	Noise Margin High (NM_H) is always equal to the Noise Margin Low (NM_L), irrespective of the sizing of transistors.
(B)	Dynamic power consumption during switching is zero.
(C)	For a logical high input under steady state, the n MOSFET is in the linear regime of operation.
(D)	Mobility of electrons never influences the switching speed of the NOT gate.

GATE 2022 Electronics and Communications Engineering (EC)

Q.29	Let $H(X)$ denote the entropy of a discrete random variable X taking K possible distinct real values. Which of the following statements is/are necessarily true?
(A)	$H(X) \leq \log_2 K$ bits
(B)	$H(X) \leq H(2X)$
(C)	$H(X) \leq H(X^2)$
(D)	$H(X) \leq H(2^X)$
Q.30	Consider the following wave equation, $\frac{\partial^2 f(x, t)}{\partial t^2} = 10000 \frac{\partial^2 f(x, t)}{\partial x^2}$ Which of the given options is/are solution(s) to the given wave equation?
(A)	$f(x, t) = e^{-(x-100t)^2} + e^{-(x+100t)^2}$
(B)	$f(x, t) = e^{-(x-100t)} + 0.5e^{-(x+1000t)}$
(C)	$f(x, t) = e^{-(x-100t)} + \sin(x + 100t)$
(D)	$f(x, t) = e^{j100\pi(-100x+t)} + e^{j100\pi(100x+t)}$

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Q.31 The bar graph shows the frequency of the number of wickets taken in a match by a bowler in her career. For example, in 17 of her matches, the bowler has taken 5 wickets each. The median number of wickets taken by the bowler in a match is _____ (rounded off to one decimal place).

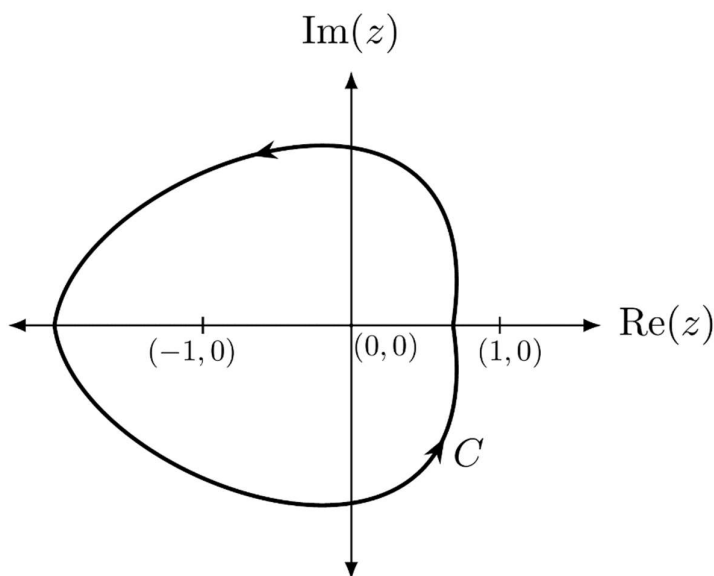


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Q.32 A simple closed path C in the complex plane is shown in the figure. If

$$\oint_C \frac{2^z}{z^2 - 1} dz = -i\pi A,$$

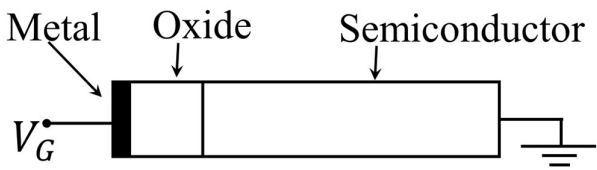
where $i = \sqrt{-1}$, then the value of A is _____ (rounded off to two decimal places).



Q.33 Let $x_1(t) = e^{-t}u(t)$ and $x_2(t) = u(t) - u(t - 2)$, where $u(\cdot)$ denotes the unit step function.

If $y(t)$ denotes the convolution of $x_1(t)$ and $x_2(t)$, then $\lim_{t \rightarrow \infty} y(t) = \underline{\hspace{2cm}}$ (rounded off to one decimal place).

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Q.34	An ideal MOS capacitor (p-type semiconductor) is shown in the figure. The MOS capacitor is under strong inversion with $V_G = 2$ V. The corresponding inversion charge density (Q_{IN}) is $2.2 \mu\text{C}/\text{cm}^2$. Assume oxide capacitance per unit area as $C_{OX} = 1.7 \mu\text{F}/\text{cm}^2$. For $V_G = 4$ V, the value of Q_{IN} is _____ $\mu\text{C}/\text{cm}^2$ (rounded off to one decimal place).
	
Q.35	A symbol stream contains alternate QPSK and 16-QAM symbols. If symbols from this stream are transmitted at the rate of 1 mega-symbols per second, the raw (uncoded) data rate is _____ mega-bits per second (rounded off to one decimal place).



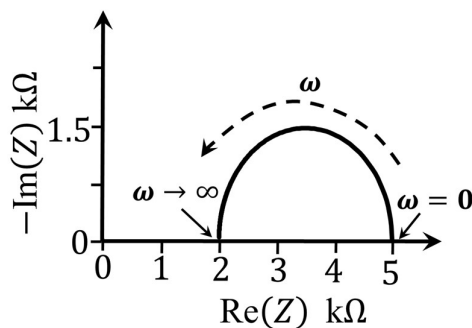
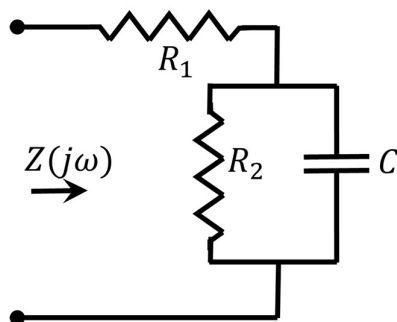
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Q.36 – Q.65 Carry TWO marks Each

Q.36	The function $f(x) = 8 \log_e x - x^2 + 3$ attains its minimum over the interval $[1, e]$ at $x = \underline{\hspace{2cm}}$. (Here $\log_e x$ is the natural logarithm of x.)
(A)	2
(B)	1
(C)	e
(D)	$\frac{1+e}{2}$
Q.37	Let α, β be two non-zero real numbers and v_1, v_2 be two non-zero real vectors of size 3×1. Suppose that v_1 and v_2 satisfy $v_1^T v_2 = 0$, $v_1^T v_1 = 1$, and $v_2^T v_2 = 1$. Let A be the 3×3 matrix given by: $A = \alpha v_1 v_1^T + \beta v_2 v_2^T$ The eigenvalues of A are $\underline{\hspace{2cm}}$.
(A)	0, α , β
(B)	0, $\alpha + \beta$, $\alpha - \beta$
(C)	0, $\frac{\alpha + \beta}{2}$, $\sqrt{\alpha\beta}$
(D)	0, 0, $\sqrt{\alpha^2 + \beta^2}$

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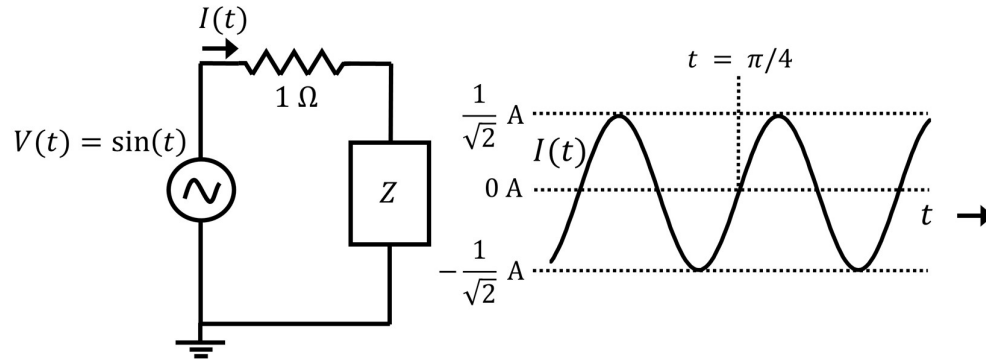
Q.38 For the circuit shown, the locus of the impedance $Z(j\omega)$ is plotted as ω increases from zero to infinity. The values of R_1 and R_2 are:



- (A) $R_1 = 2 \text{ k}\Omega$, $R_2 = 3 \text{ k}\Omega$
- (B) $R_1 = 5 \text{ k}\Omega$, $R_2 = 2 \text{ k}\Omega$
- (C) $R_1 = 5 \text{ k}\Omega$, $R_2 = 2.5 \text{ k}\Omega$
- (D) $R_1 = 2 \text{ k}\Omega$, $R_2 = 5 \text{ k}\Omega$

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Q.39 Consider the circuit shown in the figure with input $V(t)$ in volts. The sinusoidal steady state current $I(t)$ flowing through the circuit is shown graphically (where t is in seconds). The circuit element Z can be _____.



(A) a capacitor of 1 F

(B) an inductor of 1 H

(C) a capacitor of $\sqrt{3}$ F

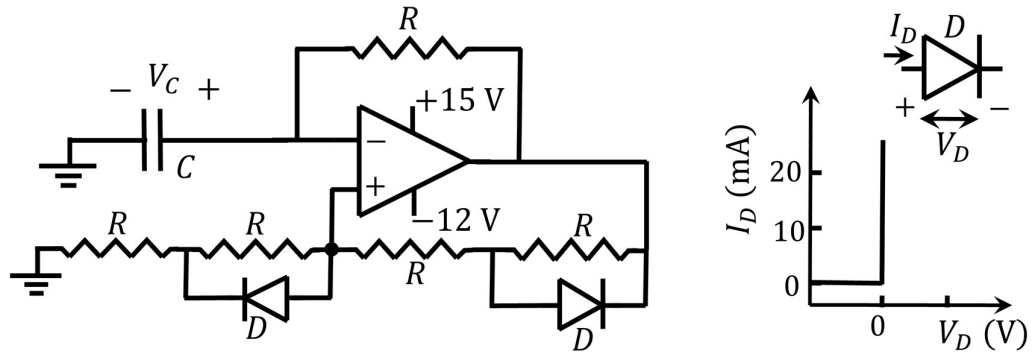
(D) an inductor of $\sqrt{3}$ H

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Q.40	Consider an ideal long channel n MOSFET (enhancement-mode) with gate length $10\text{ }\mu\text{m}$ and width $100\text{ }\mu\text{m}$. The product of electron mobility (μ_n) and oxide capacitance per unit area (C_{OX}) is $\mu_n C_{OX} = 1\text{ mA/V}^2$. The threshold voltage of the transistor is 1 V . For a gate-to-source voltage $V_{GS} = [2 - \sin(2t)]\text{ V}$ and drain-to-source voltage $V_{DS} = 1\text{ V}$ (substrate connected to the source), the maximum value of the drain-to-source current is _____.
(A)	40 mA
(B)	20 mA
(C)	15 mA
(D)	5 mA

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Q.41 For the following circuit with an ideal OPAMP, the difference between the maximum and the minimum values of the capacitor voltage (V_c) is _____.



(A) 15 V

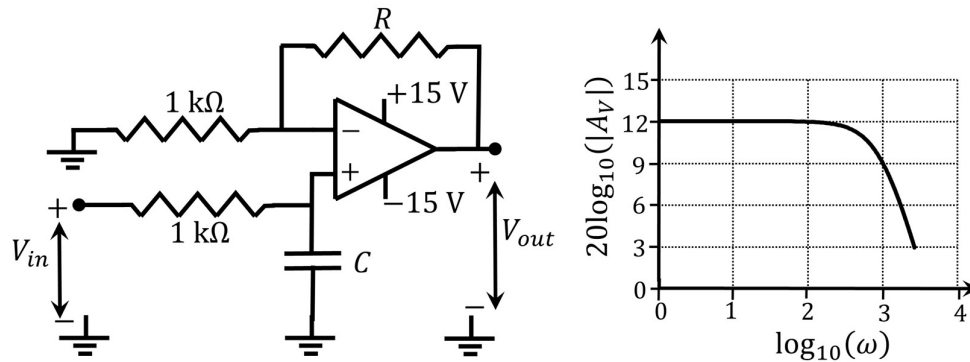
(B) 27 V

(C) 13 V

(D) 14 V

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Q.42 A circuit with an ideal OPAMP is shown. The Bode plot for the magnitude (in dB) of the gain transfer function ($A_V(j\omega) = V_{out}(j\omega)/V_{in}(j\omega)$) of the circuit is also provided (here, ω is the angular frequency in rad/s). The values of R and C are _____.



(A) $R = 3\text{ k}\Omega$, $C = 1\text{ }\mu\text{F}$

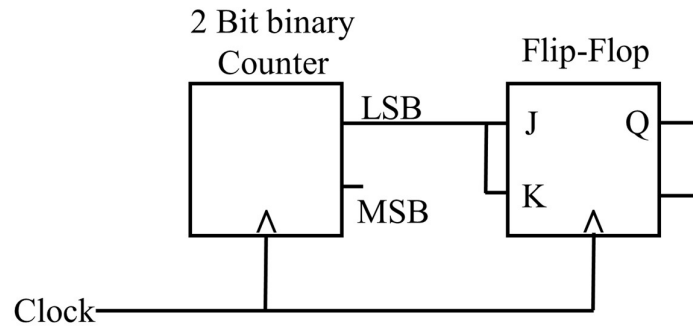
(B) $R = 1\text{ k}\Omega$, $C = 3\text{ }\mu\text{F}$

(C) $R = 4\text{ k}\Omega$, $C = 1\text{ }\mu\text{F}$

(D) $R = 3\text{ k}\Omega$, $C = 2\text{ }\mu\text{F}$

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Q.43 For the circuit shown, the clock frequency is f_0 and the duty cycle is 25%. For the signal at the Q output of the Flip-Flop, _____.



(A) frequency is $f_0/4$ and duty cycle is 50%

(B) frequency is $f_0/4$ and duty cycle is 25%

(C) frequency is $f_0/2$ and duty cycle is 50%

(D) frequency is f_0 and duty cycle is 25%



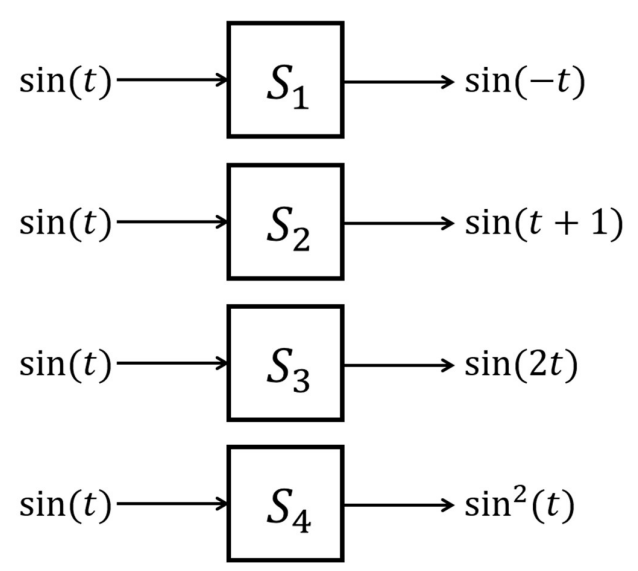
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Q.44	Consider an even polynomial $p(s)$ given by $p(s) = s^4 + 5s^2 + 4 + K ,$ where K is an unknown real parameter. The complete range of K for which $p(s)$ has all its roots on the imaginary axis is _____.
(A)	$-4 \leq K \leq \frac{9}{4}$
(B)	$-3 \leq K \leq \frac{9}{2}$
(C)	$-6 \leq K \leq \frac{5}{4}$
(D)	$-5 \leq K \leq 0$

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Q.45	Consider the following series: $\sum_{n=1}^{\infty} \frac{n^d}{c^n}$ For which of the following combinations of c, d values does this series converge?
(A)	$c = 1, d = -1$
(B)	$c = 2, d = 1$
(C)	$c = 0.5, d = -10$
(D)	$c = 1, d = -2$

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Q.46	The outputs of four systems (S_1, S_2, S_3, and S_4) corresponding to the input signal $\sin(t)$, for all time t, are shown in the figure. Based on the given information, which of the four systems is/are definitely NOT LTI (linear and time-invariant)?
	 The diagram shows four systems, S_1, S_2, S_3, and S_4, each represented by a square block. Each block has an input arrow from the left and an output arrow to the right. The input for all systems is $\sin(t)$. The outputs are as follows: S_1: Input $\sin(t)$ → Output $\sin(-t)$ S_2: Input $\sin(t)$ → Output $\sin(t + 1)$ S_3: Input $\sin(t)$ → Output $\sin(2t)$ S_4: Input $\sin(t)$ → Output $\sin^2(t)$
(A)	S_1
(B)	S_2
(C)	S_3
(D)	S_4

**GATE 2022 Electronics and Communications Engineering (EC)**

Q.47	Select the CORRECT statement(s) regarding semiconductor devices.
(A)	Electrons and holes are of equal density in an intrinsic semiconductor at equilibrium.
(B)	Collector region is generally more heavily doped than Base region in a BJT.
(C)	Total current is spatially constant in a two terminal electronic device in dark under steady state condition.
(D)	Mobility of electrons always increases with temperature in Silicon beyond 300 K.

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Q.48	A state transition diagram with states A, B, and C, and transition probabilities $p_1, p_2, \dots, p_7$ is shown in the figure (e.g., p_1 denotes the probability of transition from state A to B). For this state diagram, select the statement(s) which is/are universally true.
	<pre> graph LR A((A)) -- p7 --> A A -- p1 --> B((B)) B -- p3 --> A B -- p2 --> B A -- p4 --> C((C)) C -- p6 --> A C -- p5 --> C </pre>
(A)	$p_2 + p_3 = p_5 + p_6$
(B)	$p_1 + p_3 = p_4 + p_6$
(C)	$p_1 + p_4 + p_7 = 1$
/(D)	$p_2 + p_5 + p_7 = 1$



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Q.49	Consider a Boolean gate (D) where the output Y is related to the inputs A and B as, $Y = A + \bar{B}$, where $+$ denotes logical OR operation. The Boolean inputs '0' and '1' are also available separately. Using instances of only D gates and inputs '0' and '1', _____ (select the correct option(s)).
(A)	NAND logic can be implemented
(B)	OR logic cannot be implemented
(C)	NOR logic can be implemented
(D)	AND logic cannot be implemented
Q.50	Two linear time-invariant systems with transfer functions $G_1(s) = \frac{10}{s^2 + s + 1} \quad \text{and} \quad G_2(s) = \frac{10}{s^2 + s\sqrt{10} + 10}$ have unit step responses $y_1(t)$ and $y_2(t)$, respectively. Which of the following statements is/are true?
(A)	$y_1(t)$ and $y_2(t)$ have the same percentage peak overshoot.
(B)	$y_1(t)$ and $y_2(t)$ have the same steady-state value.
(C)	$y_1(t)$ and $y_2(t)$ have the same damped frequency of oscillation.
(D)	$y_1(t)$ and $y_2(t)$ have the same 2% settling time.

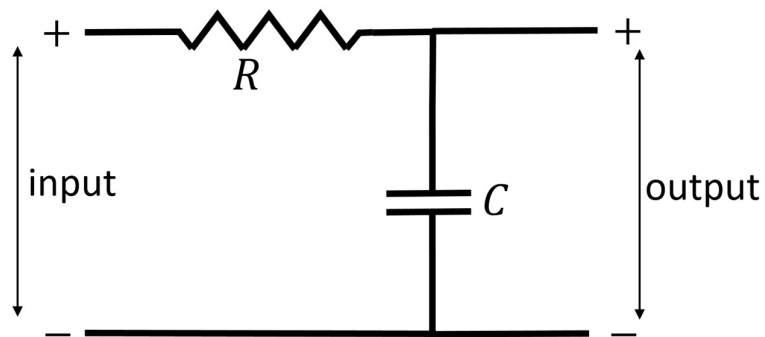
GATE 2022 Electronics and Communications Engineering (EC)

Q.51 Consider an FM broadcast that employs the pre-emphasis filter with frequency response

$$H_{pe}(\omega) = 1 + \frac{j\omega}{\omega_0},$$

where $\omega_0 = 10^4$ rad/sec.

For the network shown in the figure to act as a corresponding de-emphasis filter, the appropriate pair(s) of (R, C) values is/are _____.



(A) $R = 1\text{ k}\Omega, C = 0.1\text{ }\mu\text{F}$

(B) $R = 2\text{ k}\Omega, C = 1\text{ }\mu\text{F}$

(C) $R = 1\text{ k}\Omega, C = 2\text{ }\mu\text{F}$

(D) $R = 2\text{ k}\Omega, C = 0.5\text{ }\mu\text{F}$

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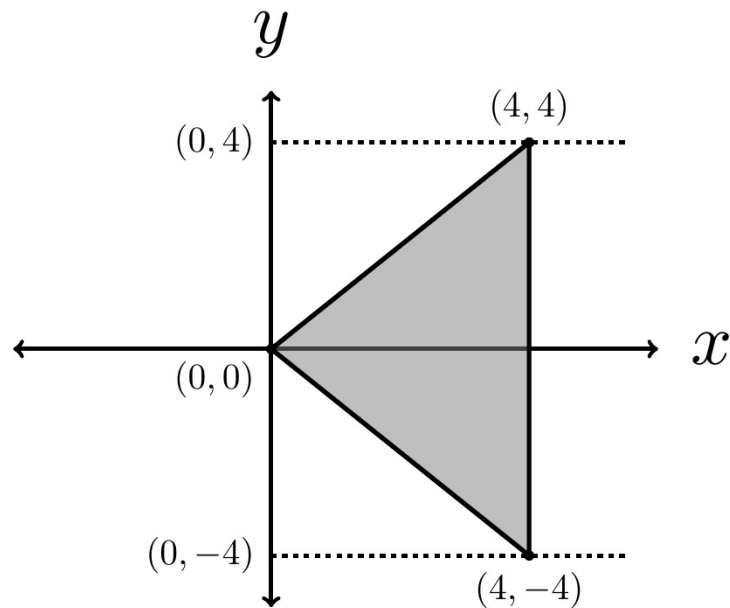
Q.52	A waveguide consists of two infinite parallel plates (perfect conductors) at a separation of 10^{-4} cm, with air as the dielectric. Assume the speed of light in air to be 3×10^8 m/s. The frequency/frequencies of TM waves which can propagate in this waveguide is/are _____.
(A)	6×10^{15} Hz
(B)	0.5×10^{12} Hz
(C)	8×10^{14} Hz
(D)	1×10^{13} Hz

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Q.53 The value of the integral

$$\iint_D 3(x^2 + y^2) dx dy,$$

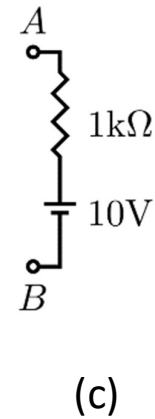
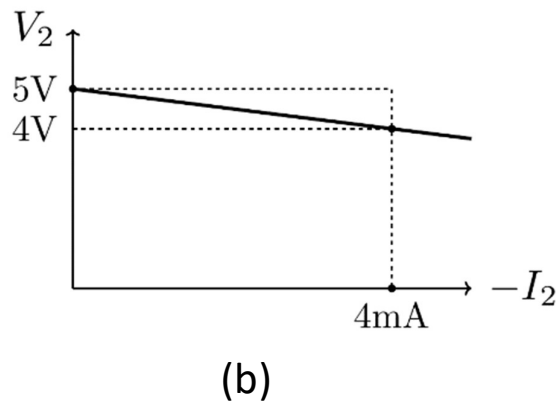
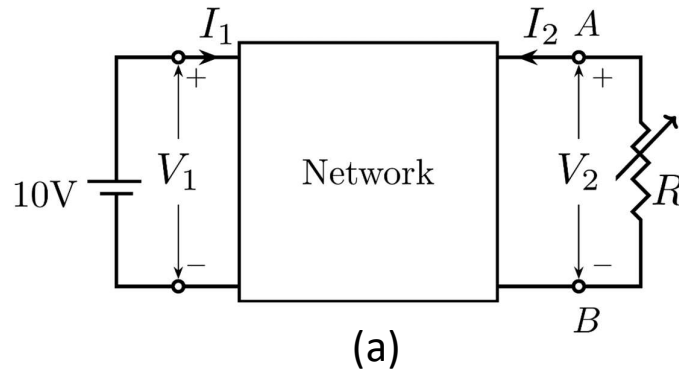
where D is the shaded triangular region shown in the diagram, is _____ (rounded off to the nearest integer).



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Q.54 A linear 2-port network is shown in Fig. (a). An ideal DC voltage source of **10 V** is connected across Port 1. A variable resistance R is connected across Port 2. As R is varied, the measured voltage and current at Port 2 is shown in Fig. (b) as a V_2 versus $-I_2$ plot. Note that for $V_2 = 5 \text{ V}$, $I_2 = 0 \text{ mA}$, and for $V_2 = 4 \text{ V}$, $I_2 = -4 \text{ mA}$.

When the variable resistance R at Port 2 is replaced by the load shown in Fig. (c), the current I_2 is _____ mA (rounded off to one decimal place).



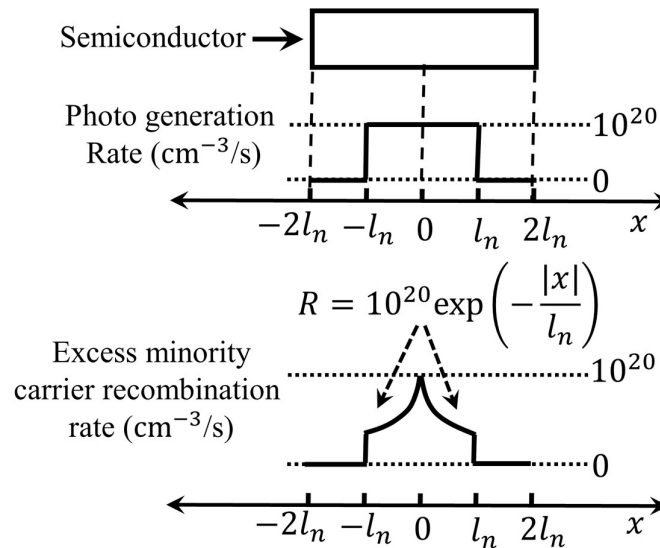
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- Q.55 For a vector $\bar{x} = [x[0], x[1], \dots, x[7]]$, the 8-point discrete Fourier transform (DFT) is denoted by $\bar{X} = DFT(\bar{x}) = [X[0], X[1], \dots, X[7]]$, where

$$X[k] = \sum_{n=0}^7 x[n] \exp\left(-j \frac{2\pi}{8} nk\right).$$

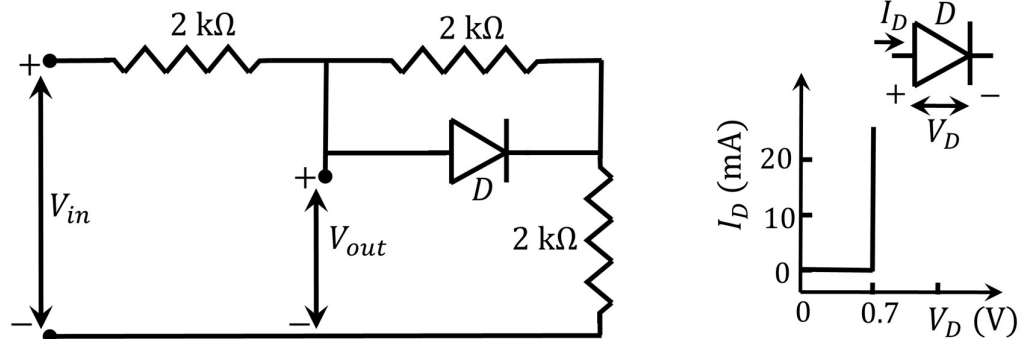
Here, $j = \sqrt{-1}$. If $\bar{x} = [1, 0, 0, 0, 2, 0, 0, 0]$ and $\bar{y} = DFT(DFT(\bar{x}))$, then the value of $y[0]$ is _____ (rounded off to one decimal place).

- Q.56 A p -type semiconductor with zero electric field is under illumination (low level injection) in steady state condition. Excess minority carrier density is zero at $x = \pm 2l_n$, where $l_n = 10^{-4}$ cm is the diffusion length of electrons. Assume electronic charge, $q = -1.6 \times 10^{-19}$ C. The profiles of photo-generation rate of carriers and the recombination rate of excess minority carriers (R) are shown. Under these conditions, the magnitude of the current density due to the photo-generated electrons at $x = +2l_n$ is _____ mA/cm² (rounded off to two decimal places).

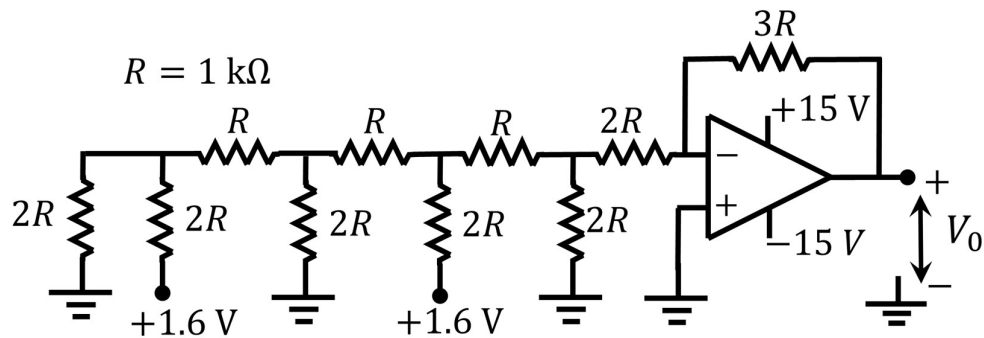


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Q.57 A circuit and the characteristics of the diode (D) in it are shown. The ratio of the minimum to the maximum small signal voltage gain $\frac{\partial V_{out}}{\partial V_{in}}$ is _____ (rounded off to two decimal places).

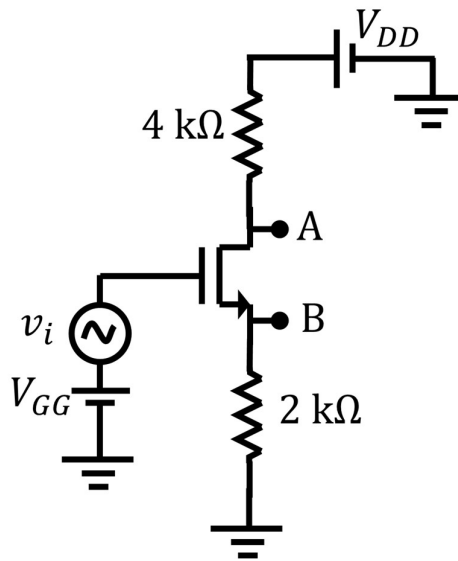


Q.58 Consider the circuit shown with an ideal OPAMP. The output voltage V_0 is _____ V (rounded off to two decimal places).

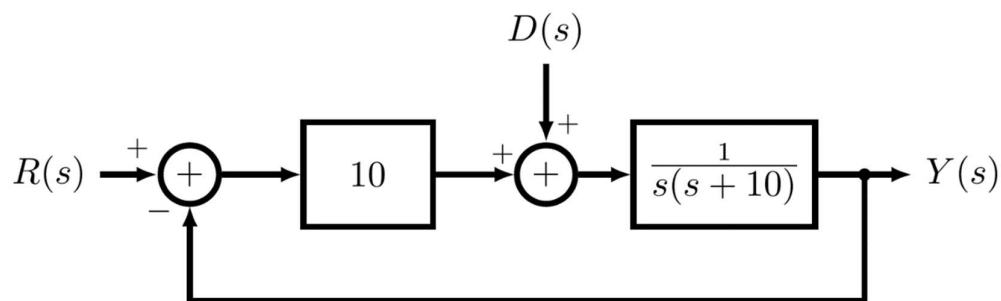


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- Q.59 Consider the circuit shown with an ideal long channel n MOSFET (enhancement-mode, substrate is connected to the source). The transistor is appropriately biased in the saturation region with V_{GG} and V_{DD} such that it acts as a linear amplifier. v_i is the small-signal ac input voltage. v_A and v_B represent the small-signal voltages at the nodes A and B, respectively. The value of $\frac{v_A}{v_B}$ is _____ (rounded off to one decimal place).



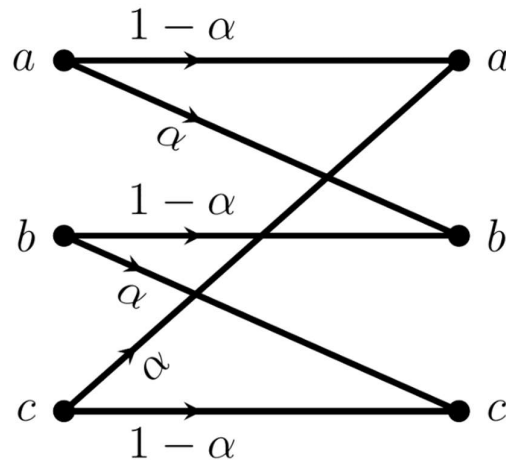
- Q.60 The block diagram of a closed-loop control system is shown in the figure. $R(s)$, $Y(s)$, and $D(s)$ are the Laplace transforms of the time-domain signals $r(t)$, $y(t)$, and $d(t)$, respectively. Let the error signal be defined as $e(t) = r(t) - y(t)$. Assuming the reference input $r(t) = 0$ for all t , the steady-state error $e(\infty)$, due to a unit step disturbance $d(t)$, is _____ (rounded off to two decimal places).



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Q.61 The transition diagram of a discrete memoryless channel with three input symbols and three output symbols is shown in the figure. The transition probabilities are as marked.

The parameter α lies in the interval $[0.25, 1]$. The value of α for which the capacity of this channel is maximized, is _____ (rounded off to two decimal places).



Q.62 Consider communication over a memoryless binary symmetric channel using a $(7, 4)$ Hamming code. Each transmitted bit is received correctly with probability $(1 - \epsilon)$, and flipped with probability ϵ . For each codeword transmission, the receiver performs minimum Hamming distance decoding, and correctly decodes the message bits if and only if the channel introduces at most one bit error.

For $\epsilon = 0.1$, the probability that a transmitted codeword is decoded correctly is _____ (rounded off to two decimal places).

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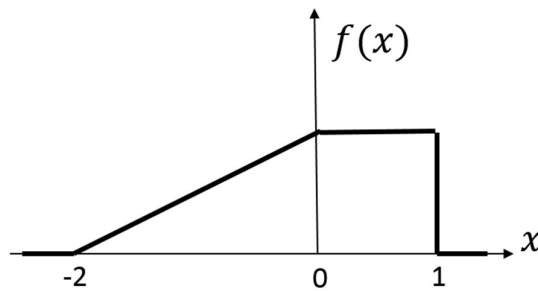
- Q.63 Consider a channel over which either symbol x_A or symbol x_B is transmitted. Let the output of the channel Y be the input to a maximum likelihood (ML) detector at the receiver. The conditional probability density functions for Y given x_A and x_B are:

$$f_{Y|x_A}(y) = e^{-(y+1)}u(y+1),$$

$$f_{Y|x_B}(y) = e^{(y-1)}(1 - u(y-1)),$$

where $u(\cdot)$ is the standard unit step function. The probability of symbol error for this system is _____ (rounded off to two decimal places).

- Q.64 Consider a real valued source whose samples are independent and identically distributed random variables with the probability density function, $f(x)$, as shown in the figure.



Consider a 1 bit quantizer that maps positive samples to value α and others to value β . If α^* and β^* are the respective choices for α and β that minimize the mean square quantization error, then $(\alpha^* - \beta^*) =$ _____ (rounded off to two decimal places).

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Q.65	In an electrostatic field, the electric displacement density vector, $\vec{D}$, is given by $\vec{D}(x, y, z) = (x^3 \vec{i} + y^3 \vec{j} + xy^2 \vec{k}) \text{ C/m}^2,$ where $\vec{i}$, $\vec{j}$, $\vec{k}$ are the unit vectors along x-axis, y-axis, and z-axis, respectively. Consider a cubical region R centered at the origin with each side of length 1 m, and vertices at $(\pm 0.5 \text{ m}, \pm 0.5 \text{ m}, \pm 0.5 \text{ m})$. The electric charge enclosed within R is _____ C (rounded off to two decimal places).

Q. No.	Session	Question Type	Subject Name	Key/Range	Mark
1	3	MCQ	GA	C	1
2	3	MCQ	GA	D	1
3	3	MCQ	GA	B	1
4	3	MCQ	GA	B	1
5	3	MCQ	GA	C	1
6	3	MCQ	GA	D	2
7	3	MCQ	GA	C	2
8	3	MCQ	GA	C	2
9	3	MCQ	GA	B	2
10	3	MCQ	GA	B	2
11	3	MCQ	EC	A	1
12	3	MCQ	EC	C	1
13	3	MCQ	EC	B	1
14	3	MCQ	EC	B	1
15	3	MCQ	EC	A	1
16	3	MCQ	EC	A	1
17	3	MCQ	EC	C	1
18	3	MCQ	EC	C	1
19	3	MCQ	EC	C	1
20	3	MCQ	EC	C	1
21	3	MCQ	EC	C	1
22	3	MCQ	EC	A	1
23	3	MCQ	EC	C	1
24	3	MCQ	EC	C	1
25	3	MSQ	EC	A,B	1
26	3	MSQ	EC	B,C	1
27	3	MSQ	EC	B,C	1
28	3	MSQ	EC	C	1
29	3	MSQ	EC	A,B,D	1
30	3	MSQ	EC	A,C	1
31	3	NAT	EC	4.0 to 4.0	1
32	3	NAT	EC	0.50 to 0.50	1
33	3	NAT	EC	0.0 to 0.0	1
34	3	NAT	EC	5.5 to 5.7	1
35	3	NAT	EC	2.99 to 3.01	1
36	3	MCQ	EC	B	2
37	3	MCQ	EC	A	2
38	3	MCQ	EC	A	2
39	3	MCQ	EC	B	2
40	3	MCQ	EC	C	2
41	3	MCQ	EC	C	2
42	3	MCQ	EC	A	2
43	3	MCQ	EC	A	2
44	3	MCQ	EC	A	2



45	3	MSQ	EC	B,D	2
46	3	MSQ	EC	C,D	2
47	3	MSQ	EC	A,C	2
48	3	MSQ	EC	A,C	2
49	3	MSQ	EC	A,C	2
50	3	MSQ	EC	A	2
51	3	MSQ	EC	A	2
52	3	MSQ	EC	MTA	2
53	3	NAT	EC	512 to 512	2
54	3	NAT	EC	3.9 to 4.1	2
55	3	NAT	EC	7.9 to 8.1	2
56	3	NAT	EC	0.57 to 0.61	2
57	3	NAT	EC	0.70 to 0.80	2
58	3	NAT	EC	-0.55 to -0.45	2
59	3	NAT	EC	-2.1 to -1.9	2
60	3	NAT	EC	-0.11 to -0.09	2
61	3	NAT	EC	1.00 to 1.00	2
62	3	NAT	EC	0.84 to 0.86	2
63	3	NAT	EC	0.22 to 0.25	2
64	3	NAT	EC	1.15 to 1.18	2
65	3	NAT	EC	0.48 to 0.52	2

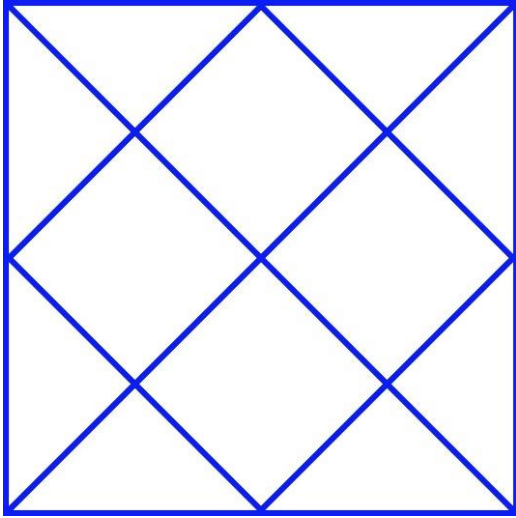
General Aptitude (GA)**Q.1 – Q.5 Carry ONE mark Each**

Q.1	“I cannot support this proposal. My _____ will not permit it.”
(A)	conscious
(B)	consensus
(C)	conscience
(D)	consent

Q.2	Courts : _____ : : Parliament : Legislature (By word meaning)
(A)	Judiciary
(B)	Executive
(C)	Governmental
(D)	Legal

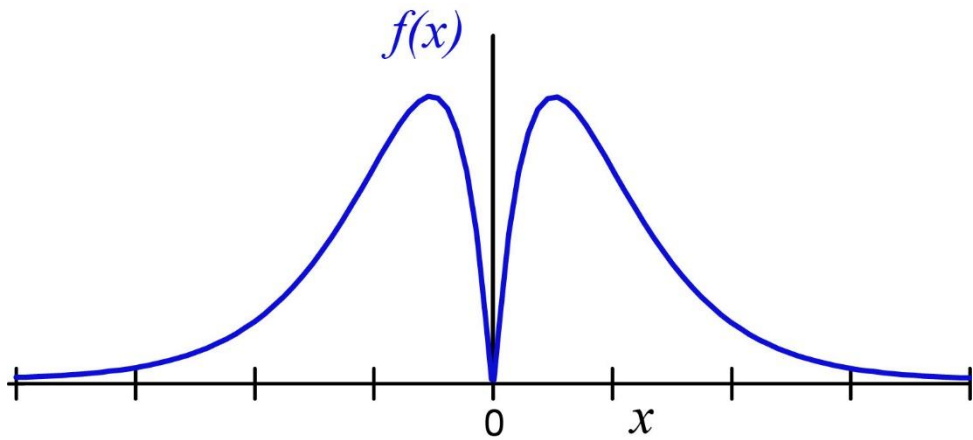
Q.3	What is the smallest number with distinct digits whose digits add up to 45?
(A)	123555789
(B)	123457869
(C)	123456789
(D)	99999

Q.4	In a class of 100 students, (i) there are 30 students who neither like romantic movies nor comedy movies, (ii) the number of students who like romantic movies is twice the number of students who like comedy movies, and (iii) the number of students who like both romantic movies and comedy movies is 20. How many students in the class like romantic movies?
(A)	40
(B)	20
(C)	60
(D)	30

Q.5	How many rectangles are present in the given figure?
	
(A)	8
(B)	9
(C)	10
(D)	12

Q.6 – Q.10 Carry TWO marks Each

Q.6	Forestland is a planet inhabited by different kinds of creatures. Among other creatures, it is populated by animals all of whom are ferocious. There are also creatures that have claws, and some that do not. All creatures that have claws are ferocious. Based only on the information provided above, which one of the following options can be logically inferred with <i>certainty</i>?
(A)	All creatures with claws are animals.
(B)	Some creatures with claws are non-ferocious.
(C)	Some non-ferocious creatures have claws.
(D)	Some ferocious creatures are creatures with claws.

Q.7	Which one of the following options represents the given graph?
	
(A)	$f(x) = x^2 2^{- x }$
(B)	$f(x) = x 2^{- x }$
(C)	$f(x) = x 2^{-x}$
(D)	$f(x) = x 2^{-x}$

Q.8	Which one of the following options can be inferred from the given passage alone? When I was a kid, I was partial to stories about other worlds and interplanetary travel. I used to imagine that I could just gaze off into space and be whisked to another planet. [Excerpt from <i>The Truth about Stories</i> by T. King]
(A)	It is a child's description of what he or she likes.
(B)	It is an adult's memory of what he or she liked as a child.
(C)	The child in the passage read stories about interplanetary travel only in parts.
(D)	It teaches us that stories are good for children.

Q.9	Out of 1000 individuals in a town, 100 unidentified individuals are covid positive. Due to lack of adequate covid-testing kits, the health authorities of the town devised a strategy to identify these covid-positive individuals. The strategy is to: (i) Collect saliva samples from all 1000 individuals and randomly group them into sets of 5. (ii) Mix the samples within each set and test the mixed sample for covid. (iii) If the test done in (ii) gives a negative result, then declare all the 5 individuals to be covid negative. (iv) If the test done in (ii) gives a positive result, then all the 5 individuals are separately tested for covid. Given this strategy, no more than _____ testing kits will be required to identify all the 100 covid positive individuals irrespective of how they are grouped.
(A)	700
(B)	600
(C)	800
(D)	1000

Q.10	A $100\text{ cm} \times 32\text{ cm}$ rectangular sheet is folded 5 times. Each time the sheet is folded, the long edge aligns with its opposite side. Eventually, the folded sheet is a rectangle of dimensions $100\text{ cm} \times 1\text{ cm}$. The total number of creases visible when the sheet is unfolded is _____.
(A)	32
(B)	5
(C)	31
(D)	63

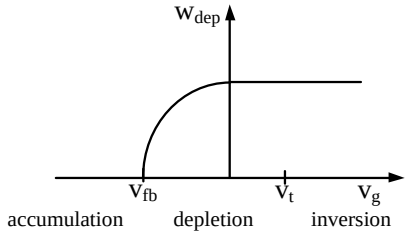
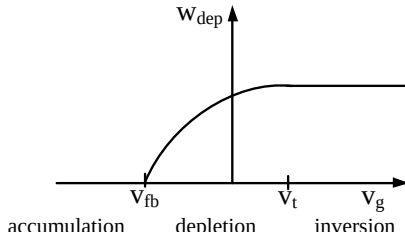
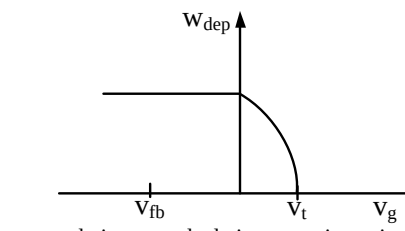
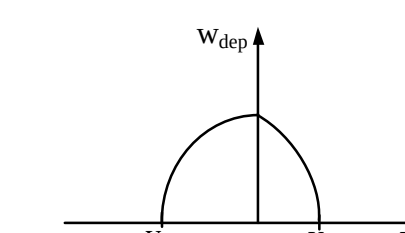
Q.11 – Q.35 Carry ONE mark Each

Q.11	Let $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$ be two vectors. The value of the coefficient α in the expression $\mathbf{v}_1 = \alpha \mathbf{v}_2 + \mathbf{e}$, which minimizes the length of the error vector $\mathbf{e}$, is
(A)	$\frac{7}{2}$
(B)	$-\frac{2}{7}$
(C)	$\frac{2}{7}$
(D)	$-\frac{7}{2}$
Q.12	The rate of increase, of a scalar field $f(x, y, z) = xyz$, in the direction $\mathbf{v} = (2, 1, 2)$ at a point $(0, 2, 1)$ is
(A)	$\frac{2}{3}$
(B)	$\frac{4}{3}$
(C)	2
(D)	4

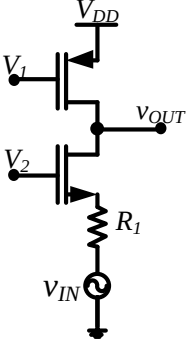
Q.13	Let $w^4 = 16j$. Which of the following cannot be a value of w ?
(A)	$2e^{\frac{j2\pi}{8}}$
(B)	$2e^{\frac{j\pi}{8}}$
(C)	$2e^{\frac{j5\pi}{8}}$
(D)	$2e^{\frac{j9\pi}{8}}$
Q.14	The value of the contour integral, $\oint_C \left(\frac{z+2}{z^2+2z+2} \right) dz$, where the contour C is $\left\{ z: \left z + 1 - \frac{3}{2}j \right = 1 \right\}$, taken in the counter clockwise direction, is
(A)	$-\pi(1+j)$
(B)	$\pi(1+j)$
(C)	$\pi(1-j)$
(D)	$-\pi(1-j)$

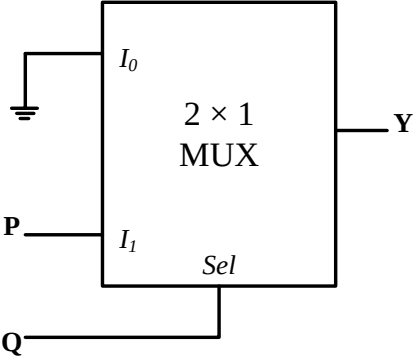
Q.15	Let the sets of eigenvalues and eigenvectors of a matrix B be $\{\lambda_k \mid 1 \leq k \leq n\}$ and $\{\mathbf{v}_k \mid 1 \leq k \leq n\}$, respectively. For any invertible matrix P , the sets of eigenvalues and eigenvectors of the matrix A , where $B = P^{-1}AP$, respectively, are
(A)	$\{\lambda_k \det(A) \mid 1 \leq k \leq n\}$ and $\{P\mathbf{v}_k \mid 1 \leq k \leq n\}$
(B)	$\{\lambda_k \mid 1 \leq k \leq n\}$ and $\{\mathbf{v}_k \mid 1 \leq k \leq n\}$
(C)	$\{\lambda_k \mid 1 \leq k \leq n\}$ and $\{P\mathbf{v}_k \mid 1 \leq k \leq n\}$
(D)	$\{\lambda_k \mid 1 \leq k \leq n\}$ and $\{P^{-1}\mathbf{v}_k \mid 1 \leq k \leq n\}$
Q.16	In a semiconductor, if the Fermi energy level lies in the conduction band, then the semiconductor is known as
(A)	degenerate n-type.
(B)	degenerate p-type.
(C)	non-degenerate n-type.
(D)	non-degenerate p-type.

Q.17	For an intrinsic semiconductor at temperature $T = 0\text{ K}$, which of the following statement is true?
(A)	All energy states in the valence band are filled with electrons and all energy states in the conduction band are empty of electrons.
(B)	All energy states in the valence band are empty of electrons and all energy states in the conduction band are filled with electrons.
(C)	All energy states in the valence and conduction band are filled with holes.
(D)	All energy states in the valence and conduction band are filled with electrons.
Q.18	A series RLC circuit has a quality factor Q of 1000 at a center frequency of 10^6 rad/s . The possible values of R, L and C are
(A)	$R = 1\ \Omega, L = 1\ \mu\text{H}$ and $C = 1\ \mu\text{F}$
(B)	$R = 0.1\ \Omega, L = 1\ \mu\text{H}$ and $C = 1\ \mu\text{F}$
(C)	$R = 0.01\ \Omega, L = 1\ \mu\text{H}$ and $C = 1\ \mu\text{F}$
(D)	$R = 0.001\ \Omega, L = 1\ \mu\text{H}$ and $C = 1\ \mu\text{F}$

Q.19	For a MOS capacitor, V_{fb} and V_t are the flat-band voltage and the threshold voltage, respectively. The variation of the depletion width (W_{dep}) for varying gate voltage (V_g) is best represented by
(A)	 Graph (A) shows the depletion width W_{dep} on the y-axis versus gate voltage V_g on the x-axis. The x-axis is divided into three regions: accumulation (left of V_{fb}), depletion (between V_{fb} and V_t), and inversion (right of V_t). The curve starts at V_{fb}, rises in the depletion region, and levels off in the inversion region.
(B)	 Graph (B) shows the depletion width W_{dep} on the y-axis versus gate voltage V_g on the x-axis. The x-axis is divided into three regions: accumulation (left of V_{fb}), depletion (between V_{fb} and V_t), and inversion (right of V_t). The curve starts at V_{fb}, rises in the depletion region, and levels off in the inversion region at a higher value than in graph (A).
(C)	 Graph (C) shows the depletion width W_{dep} on the y-axis versus gate voltage V_g on the x-axis. The x-axis is divided into three regions: accumulation (left of V_{fb}), depletion (between V_{fb} and V_t), and inversion (right of V_t). The curve is constant in the accumulation and depletion regions and then drops to zero at V_t.
(D)	 Graph (D) shows the depletion width W_{dep} on the y-axis versus gate voltage V_g on the x-axis. The x-axis is divided into three regions: accumulation (left of V_{fb}), depletion (between V_{fb} and V_t), and inversion (right of V_t). The curve is a semi-circle starting at V_{fb} and ending at V_t.

Q.20	Consider a narrow band signal, propagating in a lossless dielectric medium ($\epsilon_r = 4, \mu_r = 1$), with phase velocity v_p and group velocity v_g . Which of the following statement is true? (c is the velocity of light in vacuum.)
(A)	$v_p > c, v_g > c$
(B)	$v_p < c, v_g > c$
(C)	$v_p > c, v_g < c$
(D)	$v_p < c, v_g < c$

Q.21	In the circuit shown below, V_1 and V_2 are bias voltages. Based on input and output impedances, the circuit behaves as a
	
(A)	voltage controlled voltage source.
(B)	voltage controlled current source.
(C)	current controlled voltage source.
(D)	current controlled current source.
Q.22	A cascade of common-source amplifiers in a unity gain feedback configuration oscillates when
(A)	the closed loop gain is less than 1 and the phase shift is less than 180° .
(B)	the closed loop gain is greater than 1 and the phase shift is less than 180° .
(C)	the closed loop gain is less than 1 and the phase shift is greater than 180° .
(D)	the closed loop gain is greater than 1 and the phase shift is greater than 180° .

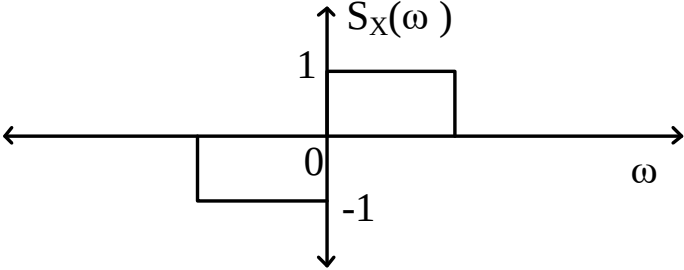
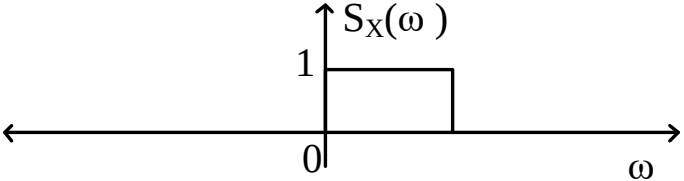
Q.23	In the circuit shown below, P and Q are the inputs. The logical function realized by the circuit shown below is
	 The diagram shows a 2 × 1 MUX block. The input I_0 is connected to ground. The input I_1 is connected to input P. The select input Sel is connected to input Q. The output of the MUX is Y.
(A)	$Y = PQ$
(B)	$Y = P + Q$
(C)	$Y = \overline{PQ}$
(D)	$Y = \overline{P + Q}$

Q.24	The synchronous sequential circuit shown below works at a clock frequency of 1 GHz. The throughput, in Mbits/s, and the latency, in ns, respectively, are
(A)	1000, 3
(B)	333.33, 1
(C)	2000, 3
(D)	333.33, 3

Q.25	The open loop transfer function of a unity negative feedback system is $G(s) = \frac{k}{s(1+sT_1)(1+sT_2)}$, where k , T_1 and T_2 are positive constants. The phase cross-over frequency, in rad/s, is
(A)	$\frac{1}{\sqrt{T_1 T_2}}$
(B)	$\frac{1}{T_1 T_2}$
(C)	$\frac{1}{T_1 \sqrt{T_2}}$
(D)	$\frac{1}{T_2 \sqrt{T_1}}$
Q.26	Consider a system with input $x(t)$ and output $y(t) = x(e^t)$. The system is
(A)	Causal and time invariant.
(B)	Non-causal and time varying.
(C)	Causal and time varying.
(D)	Non-causal and time invariant.

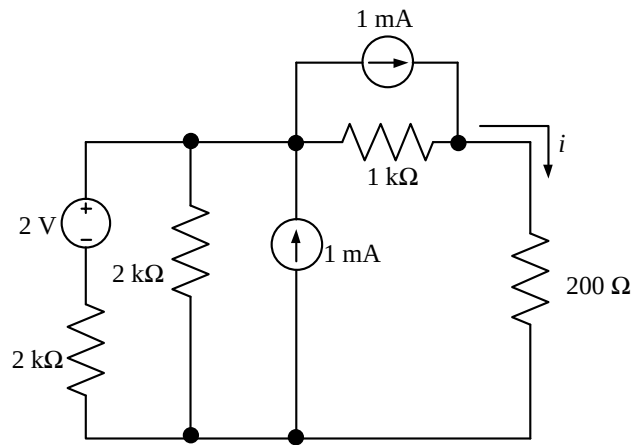
Q.27	Let $m(t)$ be a strictly band-limited signal with bandwidth B and energy E . Assuming $\omega_0 = 10B$, the energy in the signal $m(t) \cos \omega_0 t$ is
(A)	$\frac{E}{4}$
(B)	$\frac{E}{2}$
(C)	E
(D)	$2E$
Q.28	The Fourier transform $X(\omega)$ of $x(t) = e^{-t^2}$ is
	Note: $\int_{-\infty}^{\infty} e^{-y^2} dy = \sqrt{\pi}$
(A)	$\sqrt{\pi} e^{\frac{\omega^2}{2}}$
(B)	$\frac{e^{-\frac{\omega^2}{4}}}{2\sqrt{\pi}}$
(C)	$\sqrt{\pi} e^{-\frac{\omega^2}{4}}$
(D)	$\sqrt{\pi} e^{-\frac{\omega^2}{2}}$

Q.29	In the table shown below, match the signal type with its spectral characteristics. <table border="1" data-bbox="320 293 1385 801"> <thead> <tr> <th data-bbox="320 293 853 394">Signal type</th><th data-bbox="853 293 1385 394">Spectral characteristics</th></tr> </thead> <tbody> <tr> <td data-bbox="320 394 853 495">(i) Continuous, aperiodic</td><td data-bbox="853 394 1385 495">(a) Continuous, aperiodic</td></tr> <tr> <td data-bbox="320 495 853 595">(ii) Continuous, periodic</td><td data-bbox="853 495 1385 595">(b) Continuous, periodic</td></tr> <tr> <td data-bbox="320 595 853 696">(iii) Discrete, aperiodic</td><td data-bbox="853 595 1385 696">(c) Discrete, aperiodic</td></tr> <tr> <td data-bbox="320 696 853 801">(iv) Discrete, periodic</td><td data-bbox="853 696 1385 801">(d) Discrete, periodic</td></tr> </tbody> </table>	Signal type	Spectral characteristics	(i) Continuous, aperiodic	(a) Continuous, aperiodic	(ii) Continuous, periodic	(b) Continuous, periodic	(iii) Discrete, aperiodic	(c) Discrete, aperiodic	(iv) Discrete, periodic	(d) Discrete, periodic
Signal type	Spectral characteristics										
(i) Continuous, aperiodic	(a) Continuous, aperiodic										
(ii) Continuous, periodic	(b) Continuous, periodic										
(iii) Discrete, aperiodic	(c) Discrete, aperiodic										
(iv) Discrete, periodic	(d) Discrete, periodic										
(A)	(i) $\rightarrow$ (a), (ii) $\rightarrow$ (b), (iii) $\rightarrow$ (c), (iv) $\rightarrow$ (d)										
(B)	(i) $\rightarrow$ (a), (ii) $\rightarrow$ (c), (iii) $\rightarrow$ (b), (iv) $\rightarrow$ (d)										
(C)	(i) $\rightarrow$ (d), (ii) $\rightarrow$ (b), (iii) $\rightarrow$ (c), (iv) $\rightarrow$ (a)										
(D)	(i) $\rightarrow$ (a), (ii) $\rightarrow$ (c), (iii) $\rightarrow$ (d), (iv) $\rightarrow$ (b)										

Q.30	For a real signal, which of the following is/are valid power spectral density/densities?
(A)	$S_X(\omega) = \frac{2}{9 + \omega^2}$
(B)	$S_X(\omega) = e^{-\omega^2} \cos^2 \omega$
(C)	
(D)	
Q.31	The signal-to-noise ratio (SNR) of an ADC with a full-scale sinusoidal input is given to be 61.96 dB. The resolution of the ADC is _____ bits (<i>rounded off to the nearest integer</i>).

Q.32

In the circuit shown below, the current i flowing through $200\ \Omega$ resistor is _____ mA (rounded off to two decimal places).

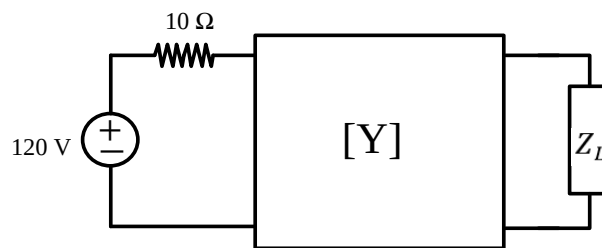


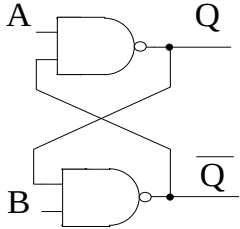
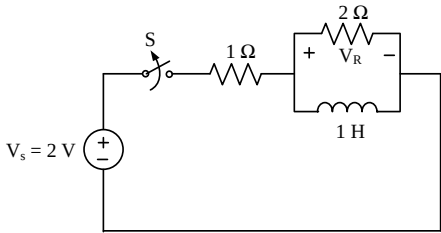
Q.33

For the two port network shown below, the $[Y]$ -parameters is given as

$$[Y] = \frac{1}{100} \begin{bmatrix} 2 & -1 \\ -1 & 4/3 \end{bmatrix} S$$

The value of load impedance Z_L , in Ω , for maximum power transfer will be _____ (rounded off to the nearest integer).



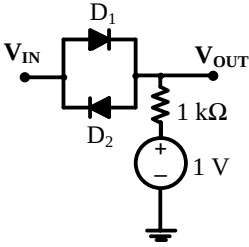
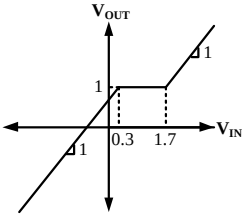
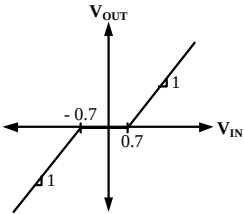
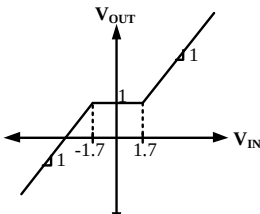
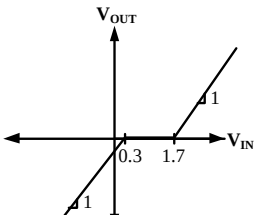
Q.34	For the circuit shown below, the propagation delay of each NAND gate is 1 ns. The critical path delay, in ns, is _____ (rounded off to the nearest integer).
	
Q.35	In the circuit shown below, switch S was closed for a long time. If the switch is opened at $t = 0$, the maximum magnitude of the voltage V_R , in volts, is _____ (rounded off to the nearest integer).
	

Q.36 – Q.65 Carry TWO marks Each

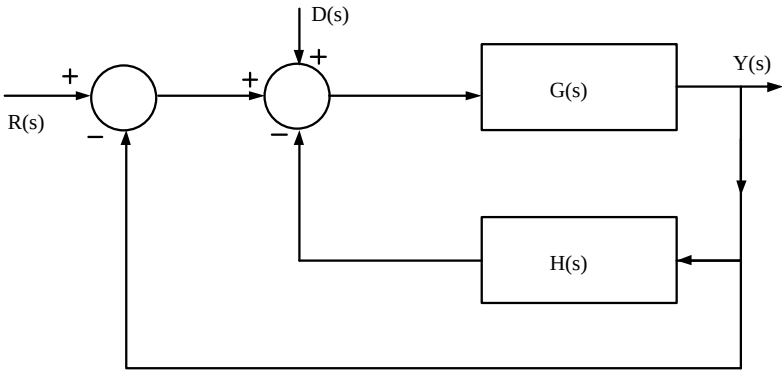
Q.36	A random variable X, distributed normally as $N(0,1)$, undergoes the transformation $Y = h(X)$, given in the figure. The form of the probability density function of Y is (In the options given below, a, b, c are non-zero constants and $g(y)$ is piece-wise continuous function)
(A)	$a\delta(y - 1) + b\delta(y + 1) + g(y)$
(B)	$a\delta(y + 1) + b\delta(y) + c\delta(y - 1) + g(y)$
(C)	$a\delta(y + 2) + b\delta(y) + c\delta(y - 2) + g(y)$
(D)	$a\delta(y + 2) + b\delta(y - 2) + g(y)$

Q.37	The value of the line integral $\int_P^Q (z^2 dx + 3y^2 dy + 2xz dz)$ along the straight line joining the points $P (1,1,2)$ and $Q (2,3,1)$ is
(A)	20
(B)	24
(C)	29
(D)	-5
Q.38	Let $\mathbf{x}$ be an $n \times 1$ real column vector with length $l = \sqrt{\mathbf{x}^T \mathbf{x}}$. The trace of the matrix $P = \mathbf{x} \mathbf{x}^T$ is
(A)	l^2
(B)	$\frac{l^2}{4}$
(C)	l
(D)	$\frac{l^2}{2}$

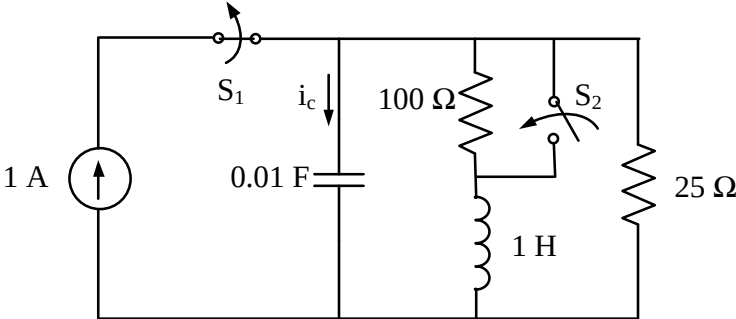
Q.39	The $\frac{V_{OUT}}{V_{IN}}$ of the circuit shown below is
(A)	$-\frac{R_4}{R_3}$
(B)	$\frac{R_4}{R_3}$
(C)	$1 + \frac{R_4}{R_3}$
(D)	$1 - \frac{R_4}{R_3}$

Q.40	In the circuit shown below, D_1 and D_2 are silicon diodes with cut-in voltage of 0.7 V. V_{IN} and V_{OUT} are input and output voltages in volts. The transfer characteristic is
	
(A)	
(B)	
(C)	
(D)	

Q.41	A closed loop system is shown in the figure where $k > 0$ and $\alpha > 0$. The steady state error due to a ramp input ($R(s) = \alpha/s^2$) is given by
(A)	$\frac{2\alpha}{k}$
(B)	$\frac{\alpha}{k}$
(C)	$\frac{\alpha}{2k}$
(D)	$\frac{\alpha}{4k}$

Q.42	In the following block diagram, $R(s)$ and $D(s)$ are two inputs. The output $Y(s)$ is expressed as $Y(s) = G_1(s)R(s) + G_2(s)D(s)$. $G_1(s)$ and $G_2(s)$ are given by
	
(A)	$G_1(s) = \frac{G(s)}{1 + G(s) + G(s)H(s)} \text{ and } G_2(s) = \frac{G(s)}{1 + G(s) + G(s)H(s)}$
(B)	$G_1(s) = \frac{G(s)}{1 + G(s) + H(s)} \text{ and } G_2(s) = \frac{G(s)}{1 + G(s) + H(s)}$
(C)	$G_1(s) = \frac{G(s)}{1 + G(s) + H(s)} \text{ and } G_2(s) = \frac{G(s)}{1 + G(s) + G(s)H(s)}$
(D)	$G_1(s) = \frac{G(s)}{1 + G(s) + G(s)H(s)} \text{ and } G_2(s) = \frac{G(s)}{1 + G(s) + H(s)}$

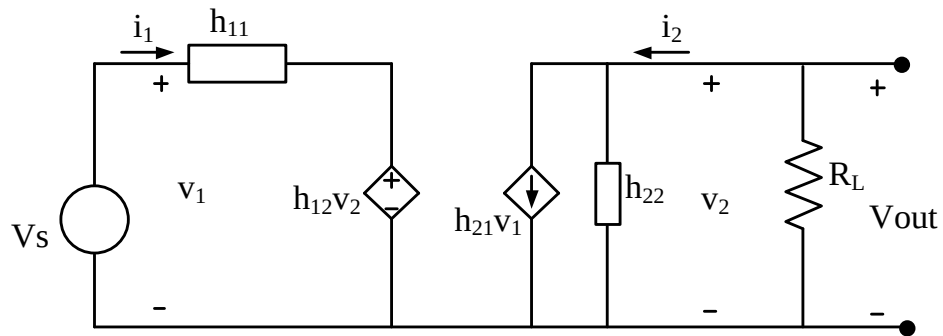
Q.43	The state equation of a second order system is $\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t), \quad \mathbf{x}(0) \text{ is the initial condition.}$ Suppose λ_1 and λ_2 are two distinct eigenvalues of $\mathbf{A}$ and $\mathbf{v}_1$ and $\mathbf{v}_2$ are the corresponding eigenvectors. For constants α_1 and α_2, the solution, $\mathbf{x}(t)$, of the state equation is
(A)	$\sum_{i=1}^2 \alpha_i e^{\lambda_i t} \mathbf{v}_i$
(B)	$\sum_{i=1}^2 \alpha_i e^{2\lambda_i t} \mathbf{v}_i$
(C)	$\sum_{i=1}^2 \alpha_i e^{3\lambda_i t} \mathbf{v}_i$
(D)	$\sum_{i=1}^2 \alpha_i e^{4\lambda_i t} \mathbf{v}_i$

Q.44	The switch S_1 was closed and S_2 was open for a long time. At $t = 0$, switch S_1 is opened and S_2 is closed, simultaneously. The value of $i_c(0^+)$, in amperes, is
	
(A)	1
(B)	-1
(C)	0.2
(D)	0.8

Q.45	Let a frequency modulated (FM) signal $x(t) = A \cos(\omega_c t + k_f \int_{-\infty}^t m(\lambda) d\lambda),$ where $m(t)$ is a message signal of bandwidth W. It is passed through a non-linear system with output $y(t) = 2x(t) + 5(x(t))^2$. Let B_T denote the FM bandwidth. The minimum value of ω_c required to recover $x(t)$ from $y(t)$ is
(A)	$B_T + W$
(B)	$\frac{3}{2} B_T$
(C)	$2B_T + W$
(D)	$\frac{5}{2} B_T$

Q.46

The h-parameters of a two port network are shown below. The condition for the maximum small signal voltage gain $\frac{V_{out}}{V_s}$ is



(A)

$h_{11} = 0, h_{12} = 0, h_{21} = \text{very high and } h_{22} = 0$

(B)

$h_{11} = \text{very high}, h_{12} = 0, h_{21} = \text{very high and } h_{22} = 0$

(C)

$h_{11} = 0, h_{12} = \text{very high}, h_{21} = \text{very high and } h_{22} = 0$

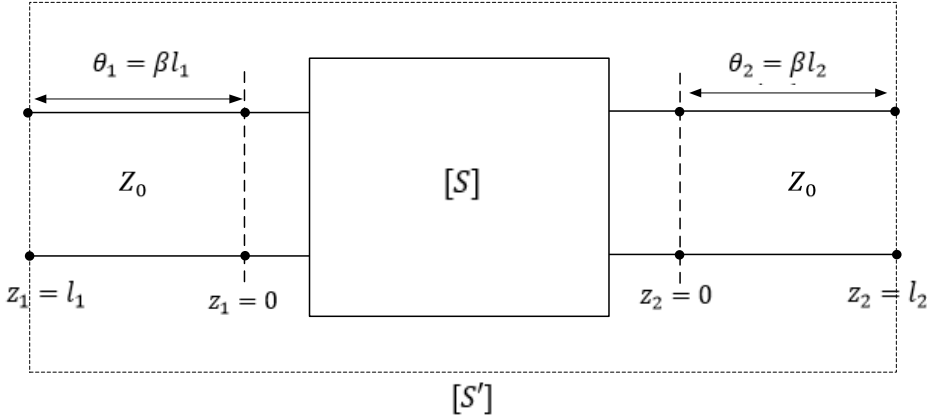
(D)

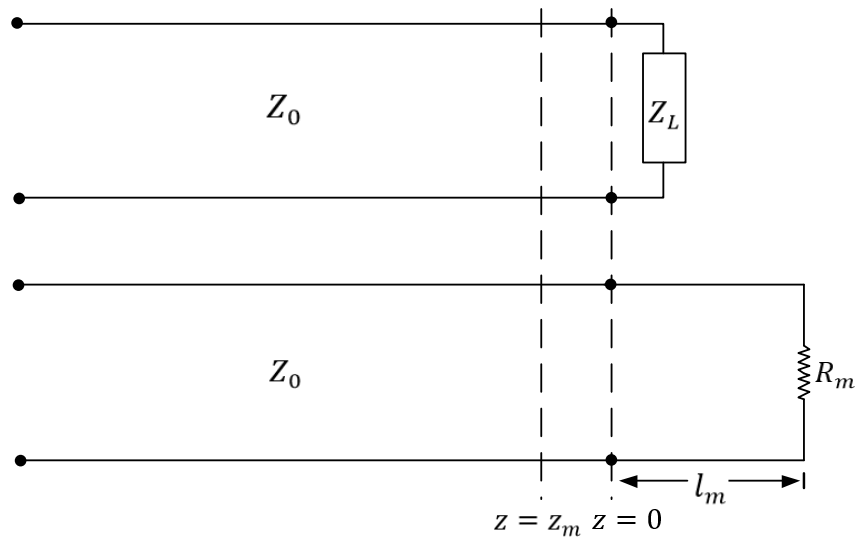
$h_{11} = 0, h_{12} = 0, h_{21} = \text{very high and } h_{22} = \text{very high}$

Q.47	Consider a discrete-time periodic signal with period $N = 5$. Let the discrete-time Fourier series (DTFS) representation be $x[n] = \sum_{k=0}^4 a_k e^{jk2\pi n/5}$, where $a_0 = 1, a_1 = 3j, a_2 = 2j, a_3 = -2j$ and $a_4 = -3j$. The value of the sum $\sum_{n=0}^4 x[n] \sin \frac{4\pi n}{5}$ is
(A)	-10
(B)	10
(C)	-2
(D)	2
Q.48	Let an input $x[n]$ having discrete time Fourier transform $X(e^{j\Omega}) = 1 - e^{-j\Omega} + 2e^{-3j\Omega}$ be passed through an LTI system. The frequency response of the LTI system is $H(e^{j\Omega}) = 1 - \frac{1}{2}e^{-j2\Omega}$. The output $y[n]$ of the system is
(A)	$\delta[n] + \delta[n-1] - \frac{1}{2}\delta[n-2] - \frac{5}{2}\delta[n-3] + \delta[n-5]$
(B)	$\delta[n] - \delta[n-1] - \frac{1}{2}\delta[n-2] - \frac{5}{2}\delta[n-3] + \delta[n-5]$
(C)	$\delta[n] - \delta[n-1] - \frac{1}{2}\delta[n-2] + \frac{5}{2}\delta[n-3] - \delta[n-5]$
(D)	$\delta[n] + \delta[n-1] + \frac{1}{2}\delta[n-2] + \frac{5}{2}\delta[n-3] + \delta[n-5]$

Q.49	Let $x(t) = 10 \cos(10.5Wt)$ be passed through an LTI system having impulse response $h(t) = \pi \left(\frac{\sin Wt}{\pi t} \right)^2 \cos 10Wt$. The output of the system is
(A)	$\left(\frac{15W}{4} \right) \cos(10.5Wt)$
(B)	$\left(\frac{15W}{2} \right) \cos(10.5Wt)$
(C)	$\left(\frac{15W}{8} \right) \cos(10.5Wt)$
(D)	$(15W) \cos(10.5Wt)$

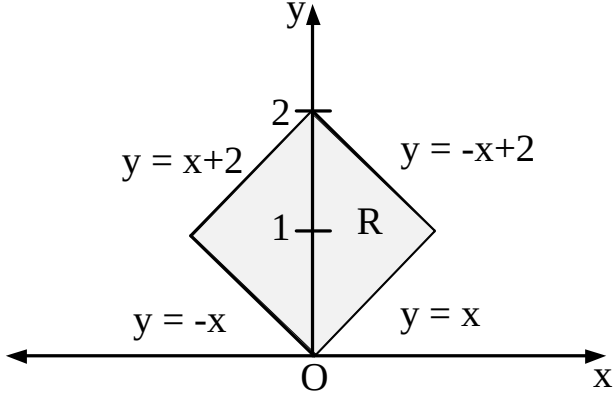
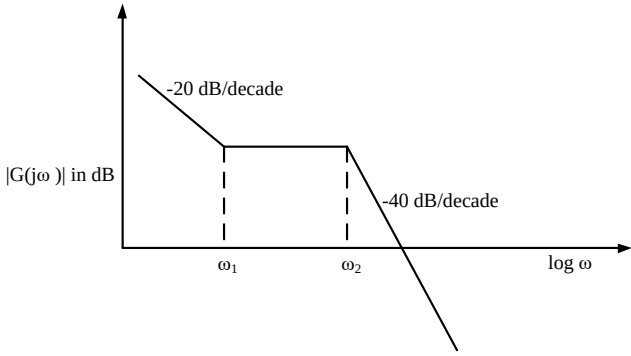
Q.50	Let $x_1(t)$ and $x_2(t)$ be two band-limited signals having bandwidth $B = 4\pi \times 10^3$ rad/s each. In the figure below, the Nyquist sampling frequency, in rad/s, required to sample $y(t)$, is
	The diagram shows a system where two signals, $x_1(t)$ and $x_2(t)$, are processed. $x_1(t)$ is multiplied by $\cos(4\pi \times 10^3 t)$, and $x_2(t)$ is multiplied by $\cos(12\pi \times 10^3 t)$. The results are then summed to produce $y(t)$.
(A)	$20\pi \times 10^3$
(B)	$40\pi \times 10^3$
(C)	$8\pi \times 10^3$
(D)	$32\pi \times 10^3$

Q.51	The S-parameters of a two port network is given as $[S] = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}$ with reference to Z_0. Two lossless transmission line sections of electrical lengths $\theta_1 = \beta l_1$ and $\theta_2 = \beta l_2$ are added to the input and output ports for measurement purposes, respectively. The S-parameters $[S']$ of the resultant two port network is
	
(A)	$\begin{bmatrix} S_{11}e^{-j2\theta_1} & S_{12}e^{-j(\theta_1+\theta_2)} \\ S_{21}e^{-j(\theta_1+\theta_2)} & S_{22}e^{-j2\theta_2} \end{bmatrix}$
(B)	$\begin{bmatrix} S_{11}e^{j2\theta_1} & S_{12}e^{-j(\theta_1+\theta_2)} \\ S_{21}e^{-j(\theta_1+\theta_2)} & S_{22}e^{j2\theta_2} \end{bmatrix}$
(C)	$\begin{bmatrix} S_{11}e^{j2\theta_1} & S_{12}e^{j(\theta_1+\theta_2)} \\ S_{21}e^{j(\theta_1+\theta_2)} & S_{22}e^{j2\theta_2} \end{bmatrix}$
(D)	$\begin{bmatrix} S_{11}e^{-j2\theta_1} & S_{12}e^{j(\theta_1+\theta_2)} \\ S_{21}e^{j(\theta_1+\theta_2)} & S_{22}e^{-j2\theta_2} \end{bmatrix}$

Q.52	The standing wave ratio on a $50\ \Omega$ lossless transmission line terminated in an unknown load impedance is found to be 2.0. The distance between successive voltage minima is 30 cm and the first minimum is located at 10 cm from the load. Z_L can be replaced by an equivalent length l_m and terminating resistance R_m of the same line. The value of R_m and l_m, respectively, are
	
(A)	$R_m = 100\ \Omega$, $l_m = 20\text{ cm}$
(B)	$R_m = 25\ \Omega$, $l_m = 20\text{ cm}$
(C)	$R_m = 100\ \Omega$, $l_m = 5\text{ cm}$
(D)	$R_m = 25\ \Omega$, $l_m = 5\text{ cm}$

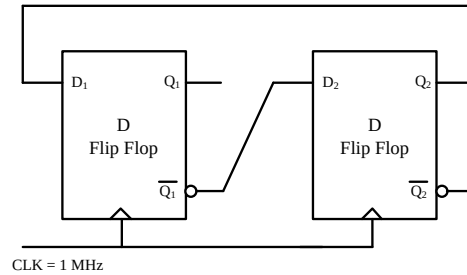
Q.53	The electric field of a plane electromagnetic wave is $\mathbf{E} = \mathbf{a}_x C_{1x} \cos(\omega t - \beta z) + \mathbf{a}_y C_{1y} \cos(\omega t - \beta z + \theta) \quad \text{V/m.}$ Which of the following combination(s) will give rise to a left handed elliptically polarized (LHEP) wave?
(A)	$C_{1x} = 1, C_{1y} = 1, \theta = \pi/4$
(B)	$C_{1x} = 2, C_{1y} = 1, \theta = \pi/2$
(C)	$C_{1x} = 1, C_{1y} = 2, \theta = 3\pi/2$
(D)	$C_{1x} = 2, C_{1y} = 1, \theta = 3\pi/4$

Q.54	The following circuit(s) representing a lumped element equivalent of an infinitesimal section of a transmission line is/are
(A)	
(B)	
(C)	
(D)	

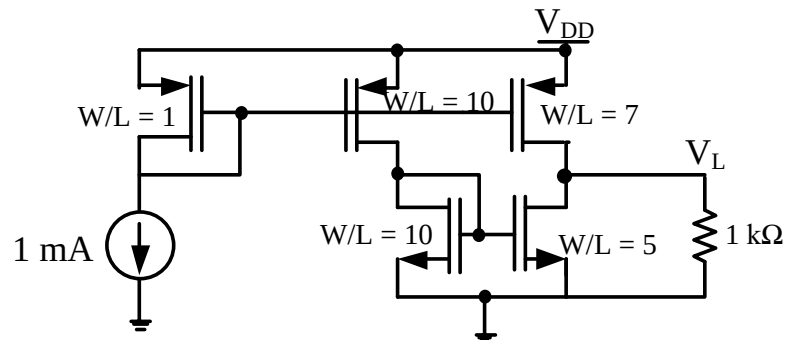
Q.55	The value of the integral $\iint_R xy \, dx \, dy$ over the region R, given in the figure, is _____ (rounded off to the nearest integer).
	
Q.56	In an extrinsic semiconductor, the hole concentration is given to be $1.5n_i$ where n_i is the intrinsic carrier concentration of $1 \times 10^{10} \text{ cm}^{-3}$. The ratio of electron to hole mobility for equal hole and electron drift current is given as _____ (rounded off to two decimal places).
Q.57	The asymptotic magnitude Bode plot of a minimum phase system is shown in the figure. The transfer function of the system is $(s) = \frac{k(s+z)^a}{s^b(s+p)^c}$, where k, z, p, a, b and c are positive constants. The value of $(a + b + c)$ is _____ (rounded off to the nearest integer).
	

Q.58	Let $x_1(t) = u(t + 1.5) - u(t - 1.5)$ and $x_2(t)$ is shown in the figure below. For $y(t) = x_1(t) * x_2(t)$, the $\int_{-\infty}^{\infty} y(t)dt$ is _____ (rounded off to the nearest integer).
Q.59	Let $X(t)$ be a white Gaussian noise with power spectral density $\frac{1}{2}$ W/Hz. If $X(t)$ is input to an LTI system with impulse response $e^{-t}u(t)$. The average power of the system output is _____ W (rounded off to two decimal places).
Q.60	A transparent dielectric coating is applied to glass ($\epsilon_r = 4, \mu_r = 1$) to eliminate the reflection of red light ($\lambda_0 = 0.75 \mu\text{m}$). The minimum thickness of the dielectric coating, in μm , that can be used is _____ (rounded off to two decimal places).
Q.61	In a semiconductor device, the Fermi-energy level is 0.35 eV above the valence band energy. The effective density of states in the valence band at $T = 300$ K is $1 \times 10^{19} \text{cm}^{-3}$. The thermal equilibrium hole concentration in silicon at 400 K is _____ $\times 10^{13} \text{cm}^{-3}$ (rounded off to two decimal places). Given kT at 300 K is 0.026 eV.
Q.62	A sample and hold circuit is implemented using a resistive switch and a capacitor with a time constant of $1 \mu\text{s}$. The time for the sampling switch to stay closed to charge a capacitor adequately to a full scale voltage of 1 V with 12-bit accuracy is _____ μs (rounded off to two decimal places).

Q.63 In a given sequential circuit, initial states are $Q_1 = 1$ and $Q_2 = 0$. For a clock frequency of 1 MHz, the frequency of signal Q_2 in kHz, is _____ (*rounded off to the nearest integer*).



Q.64 In the circuit below, the voltage V_L is _____ V (rounded off to two decimal places).



Q.65	The frequency of occurrence of 8 symbols (a-h) is shown in the table below. A symbol is chosen and it is determined by asking a series of “yes/no” questions which are assumed to be truthfully answered. The average number of questions when asked in the most efficient sequence, to determine the chosen symbol, is _____ (<i>rounded off to two decimal places</i>).								
	Symbols	a	b	c	d	e	f	g	h
	Frequency of occurrence	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{128}$	$\frac{1}{128}$

END OF QUESTION PAPER

General Aptitude (GA)

Q.1 – Q.5 Carry ONE mark Each

Q.1 If '→' denotes increasing order of intensity, then the meaning of the words
[charm → enamor → bewitch] is analogous to [bored → _____ → weary].
Which one of the given options is appropriate to fill the blank?

- (A) jaded
- (B) baffled
- (C) dead
- (D) worsted

Q.2 P, Q, R, S, and T have launched a new startup. Two of them are siblings. The office of the startup has just three rooms. All of them agree that the siblings should not share the same room.

If S and Q are single children, and the room allocations shown below are acceptable to all,

P R	T S	Q
-----	-----	---

P Q	R T	S
-----	-----	---

then, which one of the given options is the siblings?

- (A) P and T
- (B) P and S
- (C) T and Q
- (D) T and R

Q.3 Five years ago, the ratio of Aman's age to his father's age was 1:4, and five years from now, the ratio will be 2:5. What was his father's age when Aman was born?

- (A) 28 years
- (B) 30 years
- (C) 35 years
- (D) 32 years

Q.4 For a real number $x > 1$,

$$\frac{1}{\log_2 x} + \frac{1}{\log_3 x} + \frac{1}{\log_4 x} = 1$$

The value of x is

- (A) 4
- (B) 12
- (C) 24
- (D) 36

Q.5 The greatest prime factor of $(3^{199} - 3^{196})$ is

(A) 13

(B) 17

(C) 3

(D) 11

GATE 2024

Q.6 – Q.10 Carry TWO marks Each

Q.6 Sequence the following sentences (P, Q, R, S) in a coherent passage:

P: Shifu's student exclaimed, "Why do you run since the bull is an illusion?"

Q: Shifu said, "Surely my running away from the bull is also an illusion."

R: Shifu once proclaimed that all life is illusion.

S: One day, when a bull gave him chase, Shifu began running for his life.

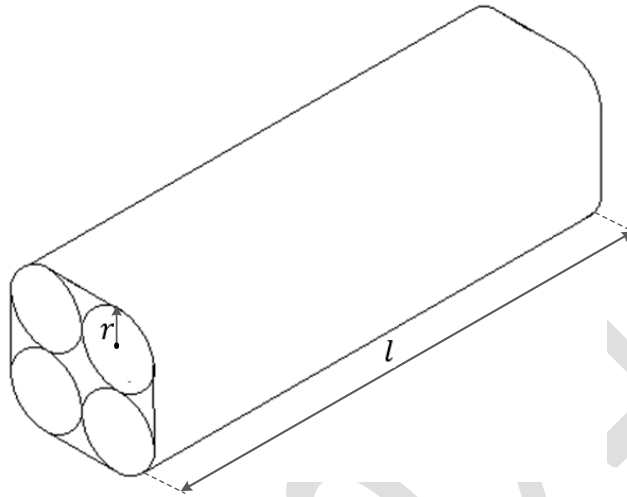
(A) SPRQ

(B) SRPQ

(C) RSPQ

(D) RPQS

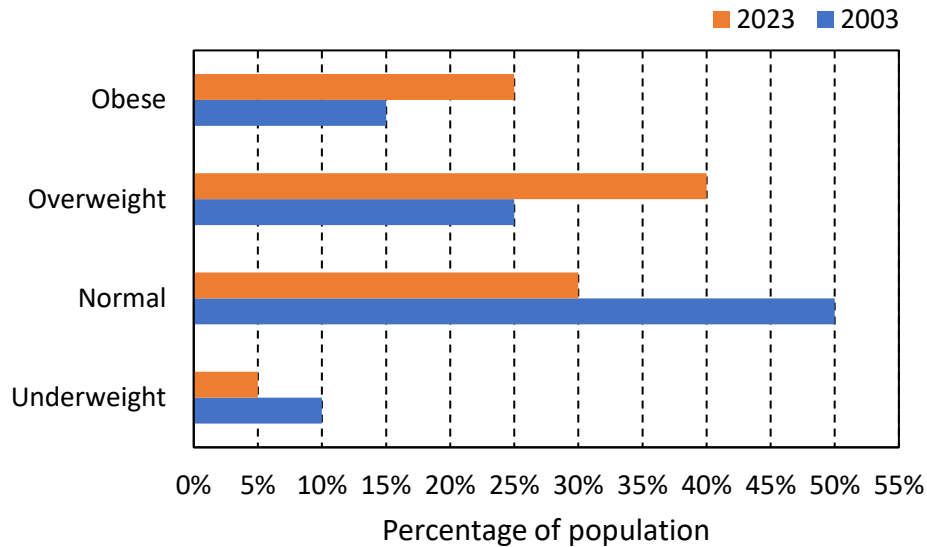
- Q.7 Four identical cylindrical chalk-sticks, each of radius $r = 0.5$ cm and length $l = 10$ cm, are bound tightly together using a duct tape as shown in the following figure.



The width of the duct tape is equal to the length of the chalk-stick. The area (in cm^2) of the duct tape required to wrap the bundle of chalk-sticks once, is

- (A) $20(4 + \pi)$
- (B) $20(8 + \pi)$
- (C) $10(8 + \pi)$
- (D) $10(4 + \pi)$

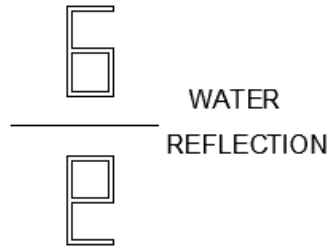
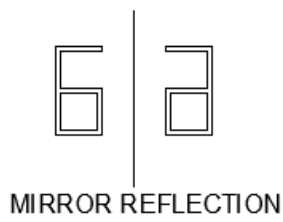
Q.8 The bar chart shows the data for the percentage of population falling into different categories based on Body Mass Index (BMI) in 2003 and 2023.



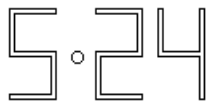
Based on the data provided, which one of the following options is INCORRECT?

- (A) The ratio of the percentage of population falling into overweight category to the percentage of population falling into normal category has increased in 20 years.
- (B) The ratio of the percentage of population falling into underweight category to the percentage of population falling into normal category has decreased in 20 years.
- (C) The ratio of the percentage of population falling into obese category to the percentage of population falling into normal category has decreased in 20 years.
- (D) The percentage of population falling into normal category has decreased in 20 years.

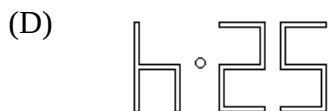
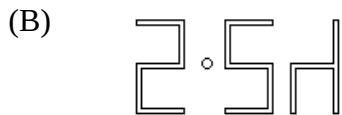
Q.9 Examples of mirror and water reflections are shown in the figures below:



An object appears as the following image after first reflecting in a mirror and then reflecting on water.



The original object is



Q.10 Two identical sheets A and B, of dimensions $24 \text{ cm} \times 16 \text{ cm}$, can be folded into half using two distinct operations, FO1 or FO2.

In FO1, the axis of folding remains parallel to the initial long edge, and in FO2, the axis of folding remains parallel to the initial short edge.

If sheet A is folded twice using FO1, and sheet B is folded twice using FO2, the ratio of the perimeters of the final shapes of A and B is

- (A) 14:11
- (B) 11:14
- (C) 18:11
- (D) 11:18

Q.11 – Q.35 Carry ONE mark Each

Q.11 The general form of the complementary function of a differential equation is given by $y(t) = (At + B)e^{-2t}$, where A and B are real constants determined by the initial condition. The corresponding differential equation is _____.

(A) $\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 4y = f(t)$

(B) $\frac{d^2y}{dt^2} + 4y = f(t)$

(C) $\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = f(t)$

(D) $\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = f(t)$

Q.12 In the context of Bode magnitude plots, 40 dB/decade is the same as _____.

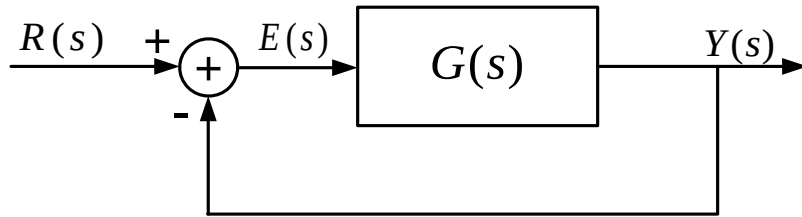
(A) 12 dB/octave

(B) 6 dB/octave

(C) 20 dB/octave

(D) 10 dB/octave

Q.13 In the feedback control system shown in the figure below $G(s) = \frac{6}{s(s+1)(s+2)}$.



$R(s)$, $Y(s)$, and $E(s)$ are the Laplace transforms of $r(t)$, $y(t)$, and $e(t)$, respectively. If the input $r(t)$ is a unit step function, then _____.

- (A) $\lim_{t \rightarrow \infty} e(t) = 0$
- (B) $\lim_{t \rightarrow \infty} e(t) = \frac{1}{3}$
- (C) $\lim_{t \rightarrow \infty} e(t) = \frac{1}{4}$
- (D) $\lim_{t \rightarrow \infty} e(t)$ does not exist, $e(t)$ is oscillatory

- Q.14 A digital communication system transmits through a noiseless bandlimited channel $[-W \ W]$. The received signal $z(t)$ at the output of the receiving filter is given by $z(t) = \sum_n b[n]x(t - nT)$ where $b[n]$ are the symbols and $x(t)$ is the overall system response to a single symbol. The received signal is sampled at $t = mT$. The Fourier transform of $x(t)$ is $X(f)$. The Nyquist condition that $X(f)$ must satisfy for zero intersymbol interference at the receiver is _____.

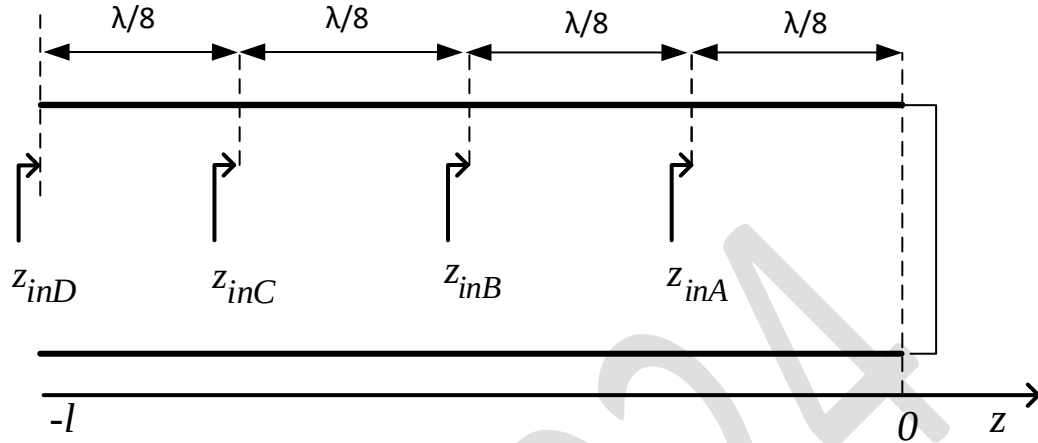
(A) $\sum_{m=-\infty}^{\infty} X\left(f + \frac{m}{T}\right) = T$

(B) $\sum_{m=-\infty}^{\infty} X\left(f + \frac{m}{T}\right) = \frac{1}{T}$

(C) $\sum_{m=-\infty}^{\infty} X(f + mT) = T$

(D) $\sum_{m=-\infty}^{\infty} X(f + mT) = \frac{1}{T}$

- Q.15 Consider a lossless transmission line terminated with a short circuit as shown in the figure below. As one moves towards the generator from the load, the normalized impedances z_{inA} , z_{inB} , z_{inC} , and z_{inD} (indicated in the figure) are _____.

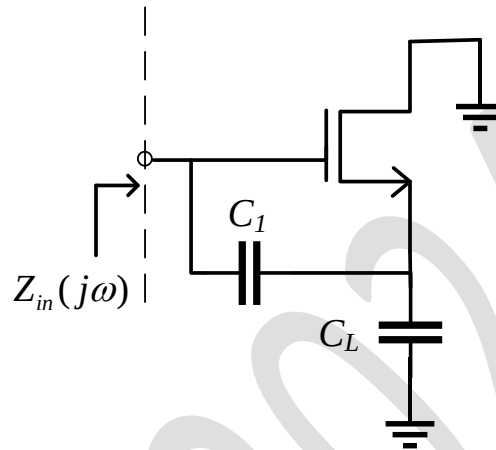


- (A) $z_{inA} = +1j \Omega$, $z_{inB} = \infty$, $z_{inC} = -1j \Omega$, $z_{inD} = 0$
- (B) $z_{inA} = \infty$, $z_{inB} = +0.4j \Omega$, $z_{inC} = 0$, $z_{inD} = +0.4j \Omega$
- (C) $z_{inA} = -1j \Omega$, $z_{inB} = 0$, $z_{inC} = +1j \Omega$, $z_{inD} = \infty$
- (D) $z_{inA} = +0.4j \Omega$, $z_{inB} = \infty$, $z_{inC} = -0.4j \Omega$, $z_{inD} = 0$

Q.16 Let $\hat{i}$ and $\hat{j}$ be the unit vectors along x and y axes, respectively and let A be a positive constant. Which one of the following statements is true for the vector fields $\vec{F}_1 = A(\hat{i}y + \hat{j}x)$ and $\vec{F}_2 = A(\hat{i}y - \hat{j}x)$?

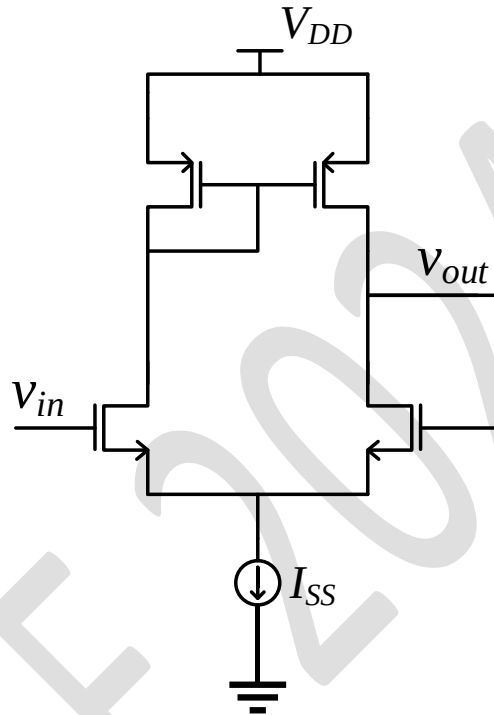
- (A) Both $\vec{F}_1$ and $\vec{F}_2$ are electrostatic fields.
- (B) Only $\vec{F}_1$ is an electrostatic field.
- (C) Only $\vec{F}_2$ is an electrostatic field.
- (D) Neither $\vec{F}_1$ nor $\vec{F}_2$ is an electrostatic field.

- Q.17 In the circuit below, assume that the long channel NMOS transistor is biased in saturation. The small signal trans-conductance of the transistor is g_m . Neglect body effect, channel length modulation and intrinsic device capacitances. The small signal input impedance $Z_{in}(j\omega)$ is _____.



- (A) $\frac{-g_m}{C_1 C_L \omega^2} + \frac{1}{j\omega C_1} + \frac{1}{j\omega C_L}$
- (B) $\frac{g_m}{C_1 C_L \omega^2} + \frac{1}{j\omega C_1} + \frac{1}{j\omega C_L}$
- (C) $\frac{1}{j\omega C_1} + \frac{1}{j\omega C_L}$
- (D) $\frac{-g_m}{C_1 C_L \omega^2} + \frac{1}{j\omega C_1 + j\omega C_L}$

- Q.18 For the closed loop amplifier circuit shown below, the magnitude of open loop low frequency small signal voltage gain is 40. All the transistors are biased in saturation. The current source I_{SS} is ideal. Neglect body effect, channel length modulation and intrinsic device capacitances. The closed loop low frequency small signal voltage gain $\frac{v_{out}}{v_{in}}$ (rounded off to three decimal places) is ____.



- (A) 0.976
(B) 1.000
(C) 1.025
(D) 0.488

Q.19 For the Boolean function

$$F(A, B, C, D) = \sum m(0, 2, 5, 7, 8, 10, 12, 13, 14, 15),$$

the essential prime implicants are _____.

- (A) $BD, \bar{B}\bar{D}$
- (B) BD, AB
- (C) $AB, \bar{B}\bar{D}$
- (D) $BD, \bar{B}\bar{D}, AB$

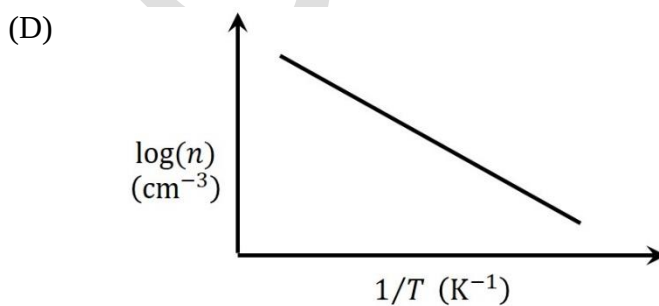
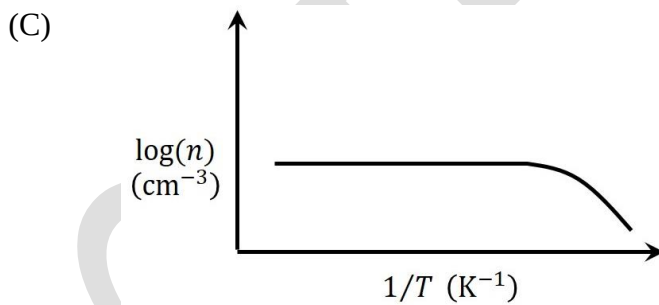
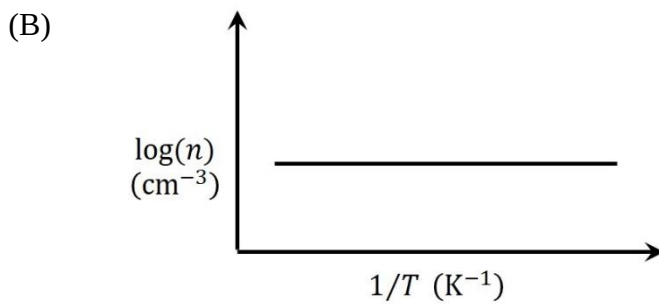
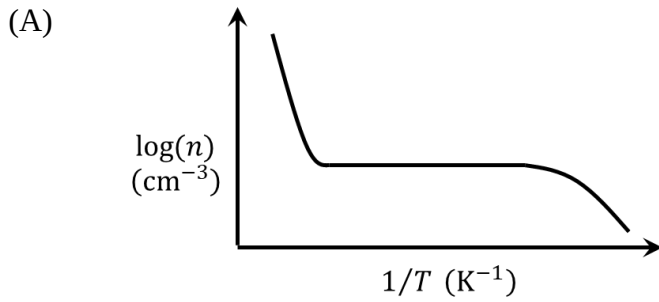
Q.20 A white Gaussian noise $w(t)$ with zero mean and power spectral density $\frac{N_0}{2}$, when applied to a first-order RC low pass filter produces an output $n(t)$. At a particular time $t = t_k$, the variance of the random variable $n(t_k)$ is _____.

- (A) $\frac{N_0}{4RC}$
- (B) $\frac{N_0}{2RC}$
- (C) $\frac{N_0}{RC}$
- (D) $\frac{2N_0}{RC}$

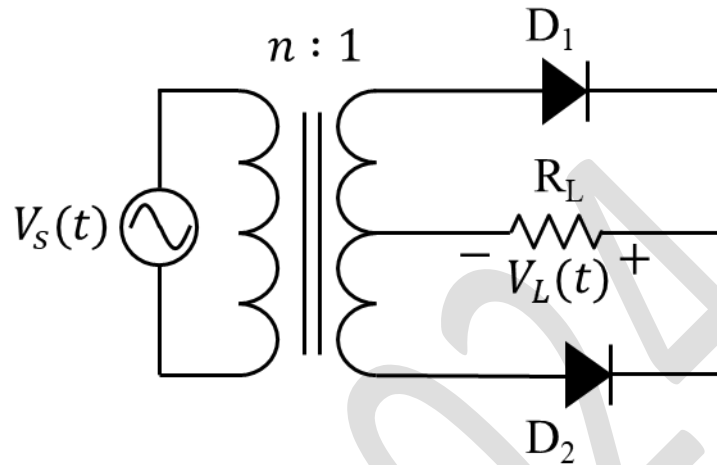
Q.21 A causal and stable LTI system with impulse response $h(t)$ produces an output $y(t)$ for an input signal $x(t)$. A signal $x(0.5t)$ is applied to another causal and stable LTI system with impulse response $h(0.5t)$. The resulting output is _____.

- (A) $2y(0.5t)$
- (B) $4y(0.5t)$
- (C) $0.25y(2t)$
- (D) $0.25y(0.25t)$

Q.22 For non-degenerately doped n-type silicon, which one of the following plots represents the temperature (T) dependence of free electron concentration (n)?



- Q.23 In the circuit shown, the $n:1$ step-down transformer and the diodes are ideal. The diodes have no voltage drop in forward biased condition. If the input voltage (in Volts) is $V_s(t) = 10\sin\omega t$ and the average value of load voltage $V_L(t)$ (in Volts) is $2.5/\pi$, the value of n is _____.



- (A) 4
(B) 8
(C) 12
(D) 16

Q.24 For a causal discrete-time LTI system with transfer function

$$H(z) = \frac{2z^2 + 3}{\left(z + \frac{1}{3}\right)\left(z - \frac{1}{3}\right)}$$

which of the following statements is/are true?

- (A) The system is stable.
- (B) The system is a minimum phase system.
- (C) The initial value of the impulse response is 2.
- (D) The final value of the impulse response is 0.

- Q.25 Let $\rho(x, y, z, t)$ and $u(x, y, z, t)$ represent density and velocity, respectively, at a point (x, y, z) and time t . Assume $\frac{\partial \rho}{\partial t}$ is continuous. Let V be an arbitrary volume in space enclosed by the closed surface S and $\hat{n}$ be the outward unit normal of S . Which of the following equations is/are equivalent to $\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho u) = 0$?

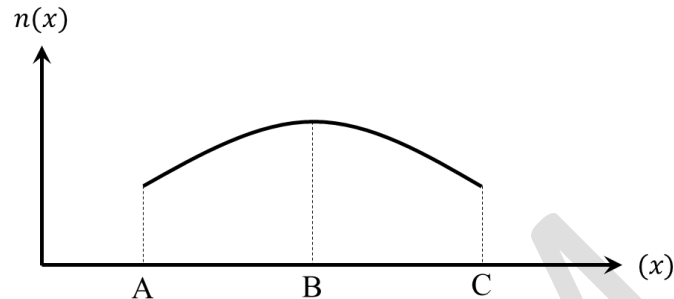
(A) $\int_V \frac{\partial \rho}{\partial t} dv = - \oint_S \rho u \cdot \hat{n} ds$

(B) $\int_V \frac{\partial \rho}{\partial t} dv = \oint_S \rho u \cdot \hat{n} ds$

(C) $\int_V \frac{\partial \rho}{\partial t} dv = - \int_V \nabla \cdot (\rho u) dv$

(D) $\int_V \frac{\partial \rho}{\partial t} dv = \int_V \nabla \cdot (\rho u) dv$

- Q.26 The free electron concentration profile $n(x)$ in a doped semiconductor at equilibrium is shown in the figure, where the points A, B, and C mark three different positions. Which of the following statements is/are true?



- (A) For x between B and C, the electron diffusion current is directed from C to B.
- (B) For x between B and A, the electron drift current is directed from B to A.
- (C) For x between B and C, the electric field is directed from B to C.
- (D) For x between B and A, the electric field is directed from A to B.

- Q.27 A machine has a 32-bit architecture with 1-word long instructions. It has 24 registers and supports an instruction set of size 40. Each instruction has five distinct fields, namely opcode, two source register identifiers, one destination register identifier, and an immediate value. Assuming that the immediate operand is an unsigned integer, its maximum value is _____.

Q.28 An amplitude modulator has output (in Volts)

$$s(t) = A \cos(400\pi t) + B \cos(360\pi t) + B \cos(440\pi t).$$

The carrier power normalized to 1Ω resistance is 50 Watts. The ratio of the total sideband power to the total power is $1/9$. The value of B (in Volts, rounded off to two decimal places) is _____.

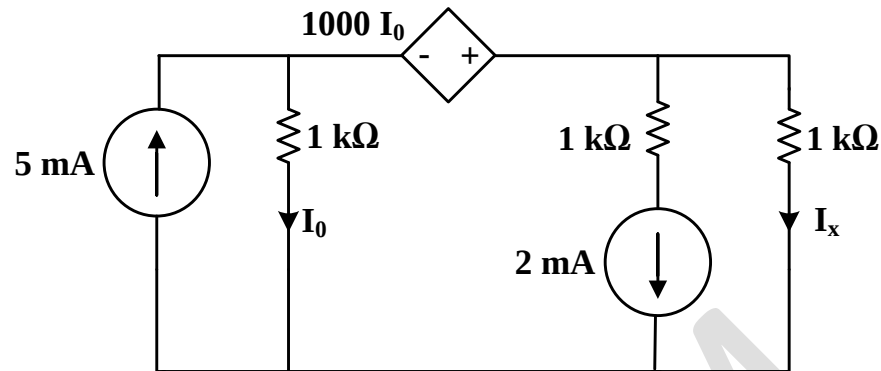
Q.29 In a number system of base r , the equation $x^2 - 12x + 37 = 0$ has $x = 8$ as one of its solutions. The value of r is _____.

Q.30 Let $\mathbb{R}$ and $\mathbb{R}^3$ denote the set of real numbers and the three dimensional vector space over it, respectively. The value of α for which the set of vectors

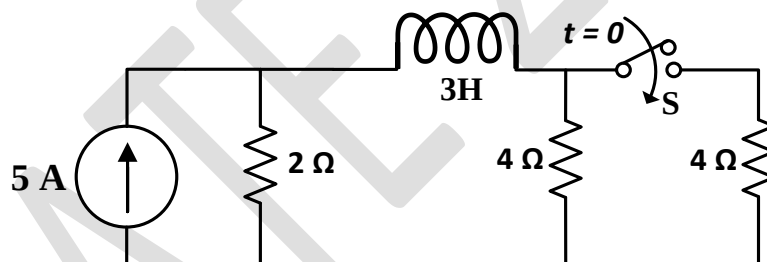
$$\{[2 \ -3 \ \alpha], [3 \ -1 \ 3], [1 \ -5 \ 7]\}$$

does not form a basis of $\mathbb{R}^3$ is _____.

Q.31 In the given circuit, the current I_x (in mA) is _____.



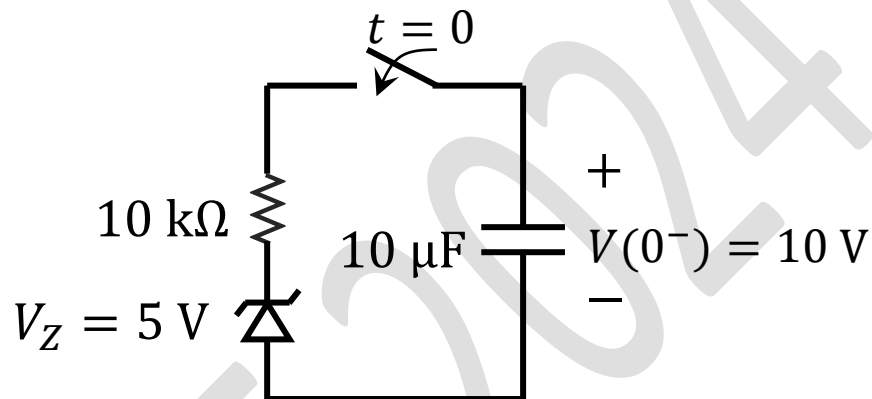
Q.32 In the circuit given below, the switch S was kept open for a sufficiently long time and is closed at time $t = 0$. The time constant (in seconds) of the circuit for $t > 0$ is _____.



Q.33 Suppose X and Y are independent and identically distributed random variables that are distributed uniformly in the interval $[0,1]$. The probability that $X \geq Y$ is _____.

Q.34 A source transmits symbols from an alphabet of size 16. The value of maximum achievable entropy (in bits) is _____.

Q.35 As shown in the circuit, the initial voltage across the capacitor is 10 V, with the switch being open. The switch is then closed at $t = 0$. The total energy dissipated in the ideal Zener diode ($V_Z = 5\text{ V}$) after the switch is closed (in mJ, rounded off to three decimal places) is _____.



Q.36 – Q.65 Carry TWO marks Each

Q.36 Consider the Earth to be a perfect sphere of radius R . Then the surface area of the region, enclosed by the 60°N latitude circle, that contains the north pole in its interior is _____.

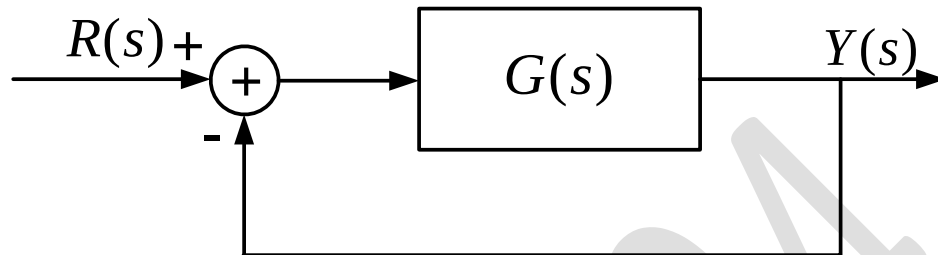
(A) $(2 - \sqrt{3})\pi R^2$

(B) $\frac{(\sqrt{2} - 1)\pi R^2}{2}$

(C) $\frac{2\pi R^2}{3}$

(D) $\frac{(2 + \sqrt{3})\pi R^2}{8\sqrt{2}}$

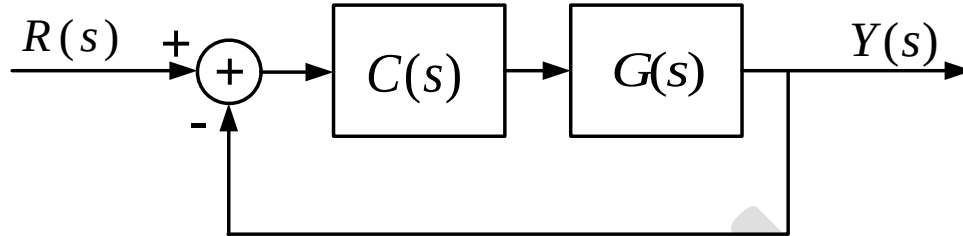
- Q.37 Consider a unity negative feedback control system with forward path gain $G(s) = \frac{K}{(s+1)(s+2)(s+3)}$ as shown.



The impulse response of the closed-loop system decays faster than e^{-t} if _____.

- (A) $1 \leq K \leq 5$
- (B) $7 \leq K \leq 21$
- (C) $-4 \leq K \leq -1$
- (D) $-24 \leq K \leq -6$

- Q.38 A satellite attitude control system, as shown below, has a plant with transfer function $G(s) = \frac{1}{s^2}$ cascaded with a compensator $C(s) = \frac{K(s+\alpha)}{s+4}$, where K and α are positive real constants.



In order for the closed-loop system to have poles at $-1 \pm j\sqrt{3}$, the value of α must be _____.

- (A) 0
- (B) 1
- (C) 2
- (D) 3

Q.39 A uniform plane wave with electric field $\vec{E}(x) = A_y \hat{a}_y e^{-j\frac{2\pi x}{3}}$ V/m is travelling in the air (relative permittivity, $\epsilon_r = 1$ and relative permeability, $\mu_r = 1$) in the $+x$ direction (A_y is a positive constant, $\hat{a}_y$ is the unit vector along the y axis). It is incident normally on an ideal electric conductor (conductivity, $\sigma = \infty$) at $x = 0$. The position of the first null of the total magnetic field in the air (measured from $x = 0$, in metres) is _____.

(A) $-\frac{3}{4}$

(B) $-\frac{3}{2}$

(C) -6

(D) -3

Q.40 A 4-bit priority encoder has inputs D_3, D_2, D_1 , and D_0 in descending order of priority. The two-bit output AB is generated as 00, 01, 10, and 11 corresponding to inputs D_3, D_2, D_1 , and D_0 , respectively. The Boolean expression of the output bit B is _____.

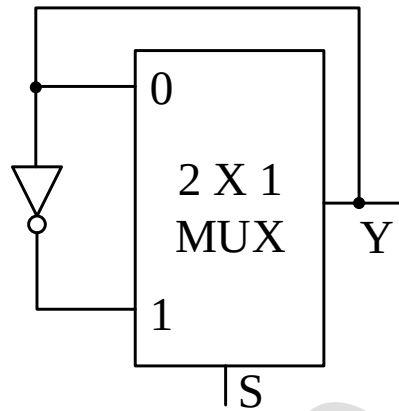
(A) $\overline{D_3} \overline{D_2}$

(B) $\overline{D_3} D_2 + \overline{D_3} \overline{D_1}$

(C) $D_3 \overline{D_2} + \overline{D_3} D_1$

(D) $\overline{D_3} \overline{D_1}$

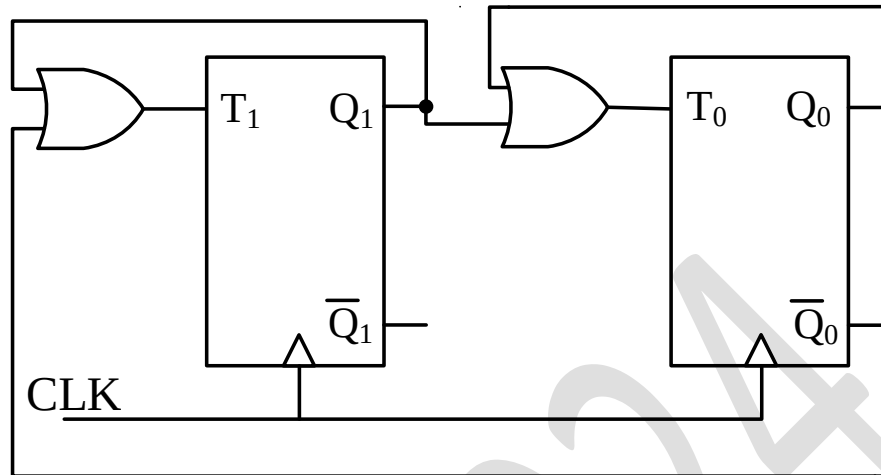
- Q.41 The propagation delay of the 2×1 MUX shown in the circuit is 10 ns. Consider the propagation delay of the inverter as 0 ns.



If S is set to 1 then the output Y is _____.

- (A) a square wave of frequency 100 MHz
- (B) a square wave of frequency 50 MHz
- (C) constant at 0
- (D) constant at 1

Q.42 The sequence of states (Q_1Q_0) of the given synchronous sequential circuit is _____.



- (A) $00 \rightarrow 10 \rightarrow 11 \rightarrow 00$
- (B) $11 \rightarrow 00 \rightarrow 10 \rightarrow 01 \rightarrow 00$
- (C) $01 \rightarrow 10 \rightarrow 11 \rightarrow 00 \rightarrow 01$
- (D) $00 \rightarrow 01 \rightarrow 10 \rightarrow 00$

Q.43 Let z be a complex variable. If $f(z) = \frac{\sin(\pi z)}{z^2(z-2)}$ and C is the circle in the complex plane with $|z| = 3$ then $\oint_C f(z) dz$ is _____.

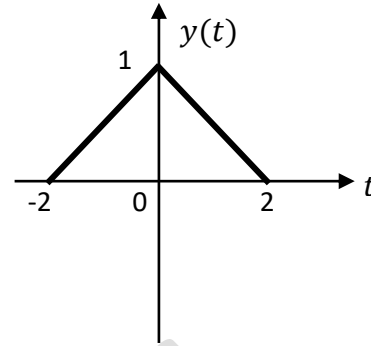
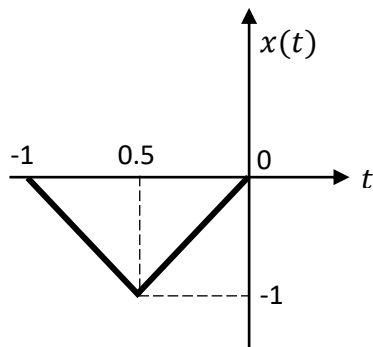
(A) $\pi^2 j$

(B) $j\pi\left(\frac{1}{2} - \pi\right)$

(C) $j\pi\left(\frac{1}{2} + \pi\right)$

(D) $-\pi^2 j$

Q.44 Consider two continuous time signals $x(t)$ and $y(t)$ as shown below



If $X(f)$ denotes the Fourier transform of $x(t)$, then the Fourier transform of $y(t)$ is _____.

- (A) $-4X(4f)e^{-j\pi f}$
- (B) $-4X(4f)e^{-j4\pi f}$
- (C) $-\frac{1}{4}X(f/4)e^{-j\pi f}$
- (D) $-\frac{1}{4}X(f/4)e^{-j4\pi f}$

- Q.45 A source transmits a symbol s , taken from $\{-4, 0, 4\}$ with equal probability, over an additive white Gaussian noise channel. The received noisy symbol r is given by $r = s + w$, where the noise w is zero mean with variance 4 and is independent of s .

Using $Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-\frac{t^2}{2}} dt$, the optimum symbol error probability is _____.

- (A) $\frac{2}{3}Q(2)$
- (B) $\frac{4}{3}Q(1)$
- (C) $\frac{2}{3}Q(1)$
- (D) $\frac{4}{3}Q(2)$

- Q.46 A full scale sinusoidal signal is applied to a 10-bit ADC. The fundamental signal component in the ADC output has a normalized power of 1 W, and the total noise and distortion normalized power is $10 \mu\text{W}$. The effective number of bits (*rounded off to the nearest integer*) of the ADC is _____.

- (A) 7
- (B) 8
- (C) 9
- (D) 10

Q.47 The information bit sequence $\{1\ 1\ 1\ 0\ 1\ 0\ 1\ 0\ 1\}$ is to be transmitted by encoding with Cyclic Redundancy Check 4 (CRC-4) code, for which the generator polynomial is $C(x) = x^4 + x + 1$. The encoded sequence of bits is _____.

- (A) $\{1\ 1\ 1\ 0\ 1\ 0\ 1\ 0\ 1\ 1\ 1\ 0\ 0\}$
- (B) $\{1\ 1\ 1\ 0\ 1\ 0\ 1\ 0\ 1\ 1\ 1\ 0\ 1\}$
- (C) $\{1\ 1\ 1\ 0\ 1\ 0\ 1\ 0\ 1\ 1\ 1\ 1\ 0\}$
- (D) $\{1\ 1\ 1\ 0\ 1\ 0\ 1\ 0\ 1\ 0\ 1\ 0\ 0\}$

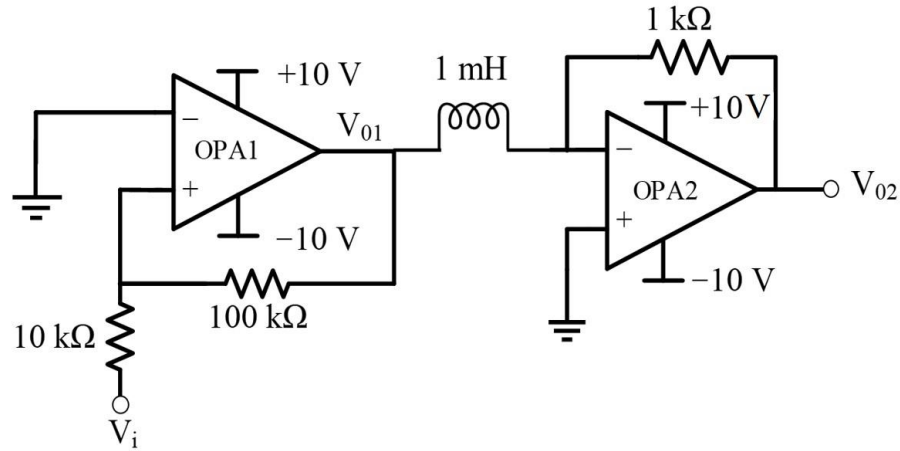
Q.48 A continuous time signal $x(t) = 2 \cos(8\pi t + \pi/3)$ is sampled at a rate of 15 Hz. The sampled signal $x_s(t)$ when passed through an LTI system with impulse response

$$h(t) = \left(\frac{\sin 2\pi t}{\pi t}\right) \cos(38\pi t - \pi/2)$$

produces an output $x_o(t)$. The expression for $x_o(t)$ is _____.

- (A) $15 \sin(38\pi t + \pi/3)$
- (B) $15 \sin(38\pi t - \pi/3)$
- (C) $15 \cos(38\pi t - \pi/6)$
- (D) $15 \cos(38\pi t + \pi/6)$

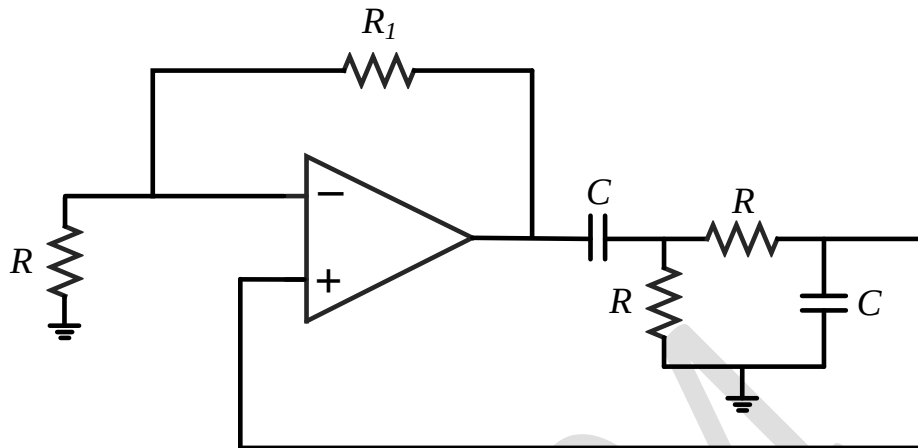
Q.49 The opamps in the circuit shown are ideal, but have saturation voltages of ± 10 V.



Assume that the initial inductor current is 0 A. The input voltage (V_i) is a triangular signal with peak voltages of ± 2 V and time period of $8 \mu\text{s}$. Which one of the following statements is true?

- (A) V_{01} is delayed by $2 \mu\text{s}$ relative to V_i , and V_{02} is a triangular waveform.
- (B) V_{01} is not delayed relative to V_i , and V_{02} is a trapezoidal waveform.
- (C) V_{01} is not delayed relative to V_i , and V_{02} is a triangular waveform.
- (D) V_{01} is delayed by $1 \mu\text{s}$ relative to V_i , and V_{02} is a trapezoidal waveform.

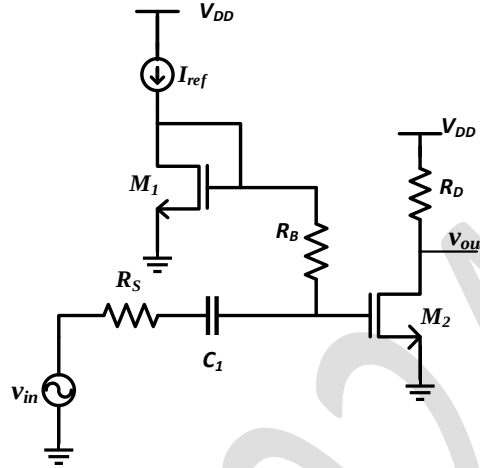
Q.50 In the circuit below, the opamp is ideal.



If the circuit is to show sustained oscillations, the respective values of R_1 and the corresponding frequency of oscillation are ____.

- (A) $29R$ and $1/(2\pi\sqrt{6}RC)$
- (B) $2R$ and $1/(2\pi RC)$
- (C) $29R$ and $1/(2\pi RC)$
- (D) $2R$ and $1/(2\pi\sqrt{6}RC)$

- Q.51 In the circuit shown below, the transistors M_1 and M_2 are biased in saturation. Their small signal transconductances are g_{m1} and g_{m2} respectively. Neglect body effect, channel length modulation and intrinsic device capacitances.



Assuming that capacitor C_1 is a short circuit for AC analysis, the exact magnitude of small signal voltage gain $\left| \frac{v_{out}}{v_{in}} \right|$ is _____.

- (A) $g_{m2} R_D$
- (B) $\frac{g_{m2} R_D \left(R_B + \frac{1}{g_{m1}} \right)}{R_B + \frac{1}{g_{m1}} + R_S}$
- (C) $\frac{g_{m2} R_D \left(R_B + \frac{1}{g_{m1}} + R_S \right)}{R_B + \frac{1}{g_{m1}}}$
- (D) $\frac{g_{m2} R_D \left(\frac{1}{g_{m1}} \right)}{\frac{1}{g_{m1}} + R_S}$

- Q.52 Which of the following statements is/are true for a BJT with respect to its DC current gain β ?
- (A) Under high-level injection condition in forward active mode, β will decrease with increase in the magnitude of collector current.
 - (B) Under low-level injection condition in forward active mode, where the current at the emitter-base junction is dominated by recombination-generation process, β will decrease with increase in the magnitude of collector current.
 - (C) β will be lower when the BJT is in saturation region compared to when it is in active region.
 - (D) A higher value of β will lead to a lower value of the collector-to-emitter breakdown voltage.

Q.53 Consider a system S represented in state space as

$$\frac{dx}{dt} = \begin{bmatrix} 0 & -2 \\ 1 & -3 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} r, \quad y = \begin{bmatrix} 2 & -5 \end{bmatrix} x.$$

Which of the state space representations given below has/have the same transfer function as that of S ?

(A) $\frac{dx}{dt} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} r, \quad y = \begin{bmatrix} 1 & 2 \end{bmatrix} x$

(B) $\frac{dx}{dt} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} r, \quad y = \begin{bmatrix} 0 & 2 \end{bmatrix} x$

(C) $\frac{dx}{dt} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} x + \begin{bmatrix} -1 \\ 3 \end{bmatrix} r, \quad y = \begin{bmatrix} 1 & 1 \end{bmatrix} x$

(D) $\frac{dx}{dt} = \begin{bmatrix} -1 & 0 \\ 0 & -2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} r, \quad y = \begin{bmatrix} 1 & 2 \end{bmatrix} x$

Q.54 Let F_1, F_2 , and F_3 be functions of (x, y, z) . Suppose that for every given pair of points A and B in space, the line integral $\int_C (F_1 dx + F_2 dy + F_3 dz)$ evaluates to the same value along any path C that starts at A and ends at B . Then which of the following is/are true?

- (A) For every closed path Γ , we have $\oint_{\Gamma} (F_1 dx + F_2 dy + F_3 dz) = 0$.
- (B) There exists a differentiable scalar function $f(x, y, z)$ such that $F_1 = \frac{\partial f}{\partial x}, F_2 = \frac{\partial f}{\partial y}, F_3 = \frac{\partial f}{\partial z}$.
- (C) $\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = 0$.
- (D) $\frac{\partial F_3}{\partial y} = \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x}, \frac{\partial F_2}{\partial x} = \frac{\partial F_1}{\partial y}$.

Q.55 Consider the matrix $\begin{bmatrix} 1 & k \\ 2 & 1 \end{bmatrix}$, where k is a positive real number. Which of the following vectors is/are eigenvector(s) of this matrix?

(A) $\begin{bmatrix} 1 \\ -\sqrt{2/k} \end{bmatrix}$

(B) $\begin{bmatrix} 1 \\ \sqrt{2/k} \end{bmatrix}$

(C) $\begin{bmatrix} \sqrt{2k} \\ 1 \end{bmatrix}$

(D) $\begin{bmatrix} \sqrt{2k} \\ -1 \end{bmatrix}$

Q.56 The radian frequency value(s) for which the discrete time sinusoidal signal $x[n] = A \cos(\Omega n + \pi/3)$ has a period of 40 is/are _____.

(A) 0.15π

(B) 0.225π

(C) 0.3π

(D) 0.45π

Q.57 Let $X(t) = A\cos(2\pi f_0 t + \theta)$ be a random process, where amplitude A and phase θ are independent of each other, and are uniformly distributed in the intervals $[-2, 2]$ and $[0, 2\pi]$, respectively. $X(t)$ is fed to an 8-bit uniform mid-rise type quantizer. Given that the autocorrelation of $X(t)$ is $R_X(\tau) = \frac{2}{3}\cos(2\pi f_0 \tau)$, the signal to quantization noise ratio (in dB, rounded off to two decimal places) at the output of the quantizer is _____.

Q.58 A lossless transmission line with characteristic impedance $Z_0 = 50 \Omega$ is terminated with an unknown load. The magnitude of the reflection co-efficient is $|\Gamma| = 0.6$. As one moves towards the generator from the load, the maximum value of the input impedance magnitude looking towards the load (in Ω) is _____.

Q.59 The relationship between any N-length sequence $x[n]$ and its corresponding N-point discrete Fourier transform $X[k]$ is defined as

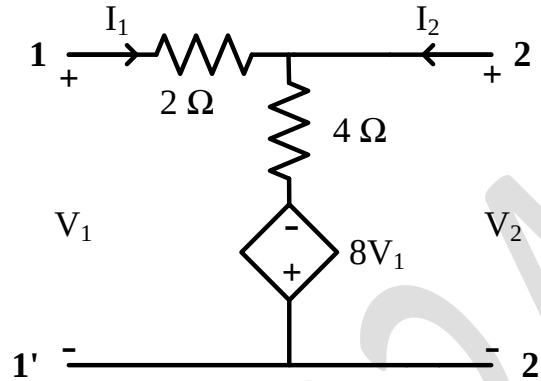
$$X[k] = \mathcal{F}\{x[n]\}.$$

Another sequence $y[n]$ is formed as below

$$y[n] = \mathcal{F}\left\{\mathcal{F}\left\{\mathcal{F}\left\{\mathcal{F}\{x[n]\}\right\}\right\}\right\}.$$

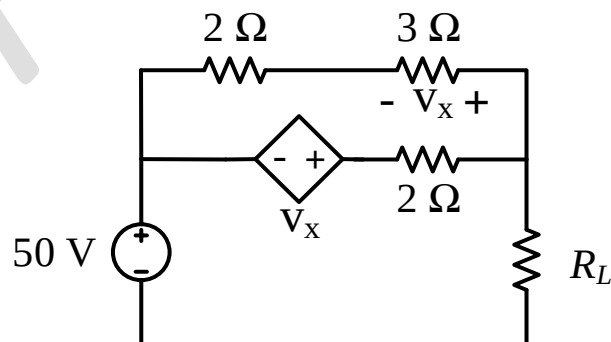
For the sequence $x[n] = \{1, 2, 1, 3\}$, the value of $Y[0]$ is _____.

- Q.60 For the two port network shown below, the value of the Y_{21} parameter (in Siemens) is _____.



- Q.61 Consider a MOS capacitor made with p-type silicon. It has an oxide thickness of 100 nm, a fixed positive oxide charge of 10^{-8} C/cm² at the oxide-silicon interface, and a metal work function of 4.6 eV. Assume that the relative permittivity of the oxide is 4 and the absolute permittivity of free space is 8.85×10^{-14} F/cm. If the flatband voltage is 0 V, the work function of the p-type silicon (in eV, rounded off to two decimal places) is _____.

- Q.62 In the network shown below, maximum power is to be transferred to the load R_L .



The value of R_L (in Ω) is _____.

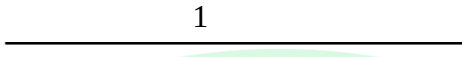
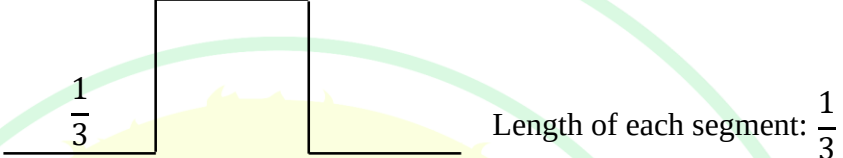
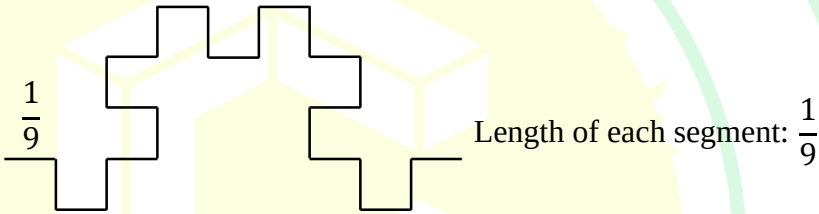
- Q.63 A non-degenerate n-type semiconductor has 5 % neutral dopant atoms. Its Fermi level is located at 0.25 eV below the conduction band (E_C) and the donor energy level (E_D) has a degeneracy of 2. Assuming the thermal voltage to be 20 mV, the difference between E_C and E_D (in eV, rounded off to two decimal places) is _____.
- Q.64 An NMOS transistor operating in the linear region has I_{DS} of 5 μA at V_{DS} of 0.1 V. Keeping V_{GS} constant, the V_{DS} is increased to 1.5 V.
- Given that $\mu_n C_{ox} \frac{W}{L} = 50 \mu\text{A/V}^2$, the transconductance at the new operating point (in $\mu\text{A/V}$, rounded off to two decimal places) is _____.
- Q.65 The photocurrent of a PN junction diode solar cell is 1 mA. The voltage corresponding to its maximum power point is 0.3 V. If the thermal voltage is 30 mV, the reverse saturation current of the diode (in nA, rounded off to two decimal places) is _____.

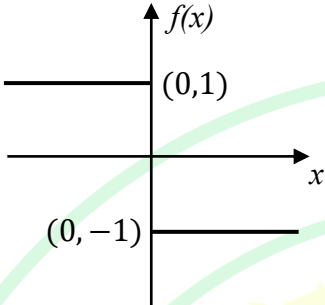
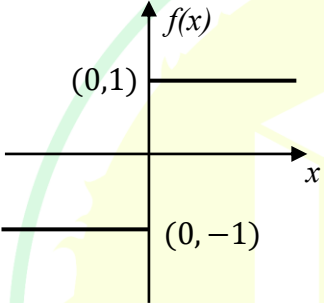
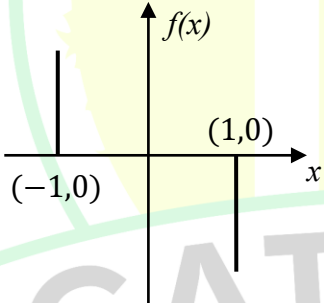
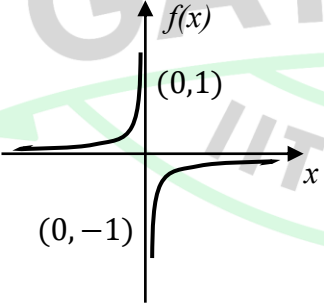
General Aptitude

Q.1 – Q.5 Carry ONE mark Each

Q.1	Here are two analogous groups, Group-I and Group-II, that list words in their decreasing order of intensity. Identify the missing word in Group-II. Group-I: Abuse → Insult → Ridicule Group-II: _____ → Praise → Appreciate
(A)	Extol
(B)	Prize
(C)	Appropriate
(D)	Espouse
Q.2	Had I learnt acting as a child, I _____ a famous film star. Select the most appropriate option to complete the above sentence.
(A)	will be
(B)	can be
(C)	am going to be
(D)	could have been

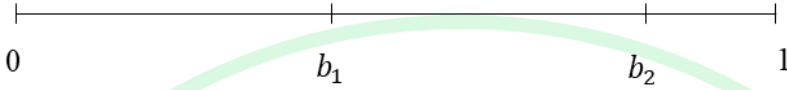
Q.3	The 12 musical notes are given as $C, C^\#, D, D^\#, E, F, F^\#, G, G^\#, A, A^\#,$ and B . Frequency of each note is $\sqrt[12]{2}$ times the frequency of the previous note. If the frequency of the note C is 130.8 Hz, then the ratio of frequencies of notes $F^\#$ and C is:
(A)	$\sqrt[6]{2}$
(B)	$\sqrt{2}$
(C)	$\sqrt[4]{2}$
(D)	2

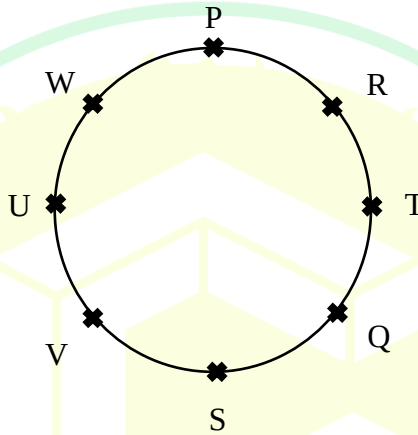
Q.4	The following figures show three curves generated using an iterative algorithm. The total length of the curve generated after 'Iteration n' is: Note: The figures shown are representative.
	Iteration 0:  Iteration 1:  Length of each segment: $\frac{1}{3}$ Iteration 2:  Length of each segment: $\frac{1}{9}$
(A)	$\left(\frac{5}{3}\right)^{\frac{n}{2}}$
(B)	$\left(\frac{5}{3}\right)^n$
(C)	$\left(\frac{5}{3}\right)^{2n}$
(D)	$\left(\frac{5}{3}\right)^{n(2n-1)}$

Q.5	Which one of the following plots represents $f(x) = -\frac{ x }{x}$, where x is a non-zero real number? Note: The figures shown are representative.
(A)	
(B)	
(C)	
(D)	

Q.6 – Q.10 Carry TWO marks Each

Q.6	Identify the option that has the most appropriate sequence such that a coherent paragraph is formed: P. Over time, such adaptations lead to significant evolutionary changes with the potential to shape the development of new species. Q. In natural world, organisms constantly adapt to their environments in response to challenges and opportunities. R. This process of adaptation is driven by the principle of natural selection, where favorable traits increase an organism's chances of survival and reproduction. S. As environments change, organisms that can adapt their behavior, structure and physiology to such changes are more likely to survive.
(A)	$P \rightarrow Q \rightarrow R \rightarrow S$
(B)	$Q \rightarrow S \rightarrow R \rightarrow P$
(C)	$R \rightarrow S \rightarrow Q \rightarrow P$
(D)	$S \rightarrow P \rightarrow R \rightarrow Q$

Q.7	A stick of length one meter is broken at two locations at distances of b_1 and b_2 from the origin (0), as shown in the figure. Note that $0 < b_1 < b_2 < 1$. Which one of the following is NOT a necessary condition for forming a triangle using the three pieces? Note: All lengths are in meter. The figure shown is representative.
	
(A)	$b_1 < 0.5$
(B)	$b_2 > 0.5$
(C)	$b_2 < b_1 + 0.5$
(D)	$b_1 + b_2 < 1$

Q.8	Eight students (P, Q, R, S, T, U, V, and W) are playing musical chairs. The figure indicates their order of position at the start of the game. They play the game by moving forward in a circle in the clockwise direction. After the 1st round, 4th student behind P leaves the game. After 2nd round, 5th student behind Q leaves the game. After 3rd round, 3rd student behind V leaves the game. After 4th round, 4th student behind U leaves the game. Who all are left in the game after the 4th round? Note: The figure shown is representative.
	
(A)	P; T; Q; S
(B)	V; P; T; Q
(C)	W; R; Q; V
(D)	Q; T; V; W

Q.9	The table lists the top 5 nations according to the number of gold medals won in a tournament; also included are the number of silver and the bronze medals won by them. Based only on the data provided in the table, which one of the following statements is INCORRECT?				
	Nation	Gold	Silver	Bronze	
	USA	40	44	41	
	Canada	39	27	24	
	Japan	20	12	13	
	Australia	17	19	16	
	France	16	26	22	
(A)	France will occupy the third place if the list were made on the basis of the total number of medals won.				
(B)	The order of the top two nations will not change even if the list is made on the basis of the total number of medals won.				
(C)	USA and Canada together have less than 50% of the medals awarded to the nations in the above table.				
(D)	Canada has won twice as many total medals as Japan.				

Q.10	An organization allows its employees to work independently on consultancy projects but charges an overhead on the consulting fee. The overhead is 20% of the consulting fee, if the fee is up to ₹ 5,00,000. For higher fees, the overhead is ₹ 1,00,000 plus 10% of the amount by which the fee exceeds ₹ 5,00,000. The government charges a Goods and Services Tax of 18% on the total amount (the consulting fee plus the overhead). An employee of the organization charges this entire amount, i.e., the consulting fee, overhead, and tax, to the client. If the client cannot pay more than ₹ 10,00,000, what is the maximum consulting fee that the employee can charge?
(A)	₹ 7,01,438
(B)	₹ 7,24,961
(C)	₹ 7,51,232
(D)	₹ 7,75,784

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Q.11 – Q.35 Carry ONE mark Each

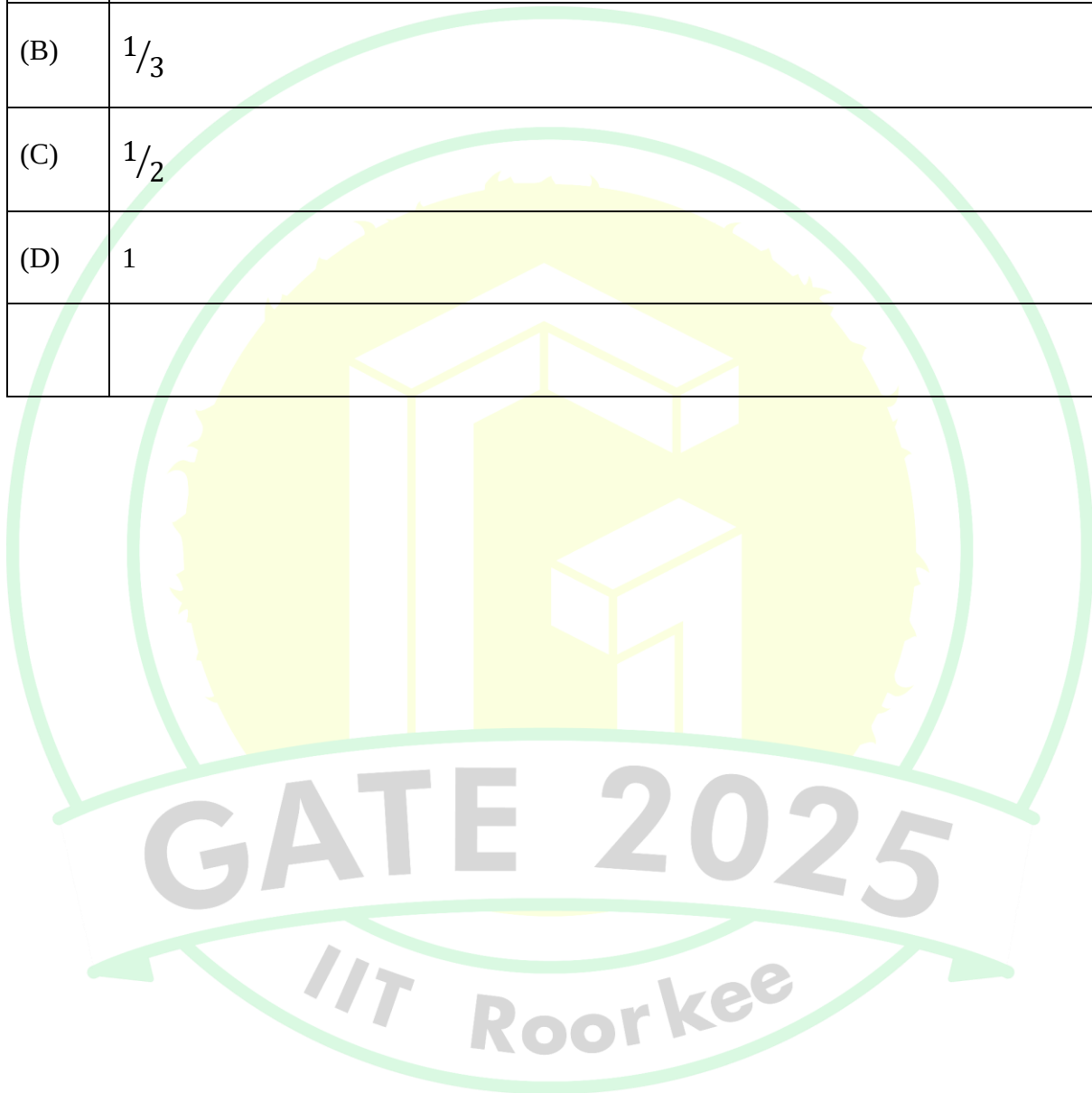
Q.11	Consider the matrix A below: $A = \begin{bmatrix} 2 & 3 & 4 & 5 \\ 0 & 6 & 7 & 8 \\ 0 & 0 & \alpha & \beta \\ 0 & 0 & 0 & \gamma \end{bmatrix}$ For which of the following combinations of α, β, and γ, is the rank of A at least three? (i) $\alpha = 0$ and $\beta = \gamma \neq 0$. (ii) $\alpha = \beta = \gamma = 0$. (iii) $\beta = \gamma = 0$ and $\alpha \neq 0$. (iv) $\alpha = \beta = \gamma \neq 0$.
(A)	Only (i), (iii), and (iv)
(B)	Only (iv)
(C)	Only (ii)
(D)	Only (i) and (iii)

Q.12	Consider the following series: (i) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ (ii) $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ (iii) $\sum_{n=1}^{\infty} \frac{1}{n!}$ Choose the correct option.
(A)	Only (ii) converges
(B)	Only (ii) and (iii) converge
(C)	Only (iii) converges
(D)	All three converge

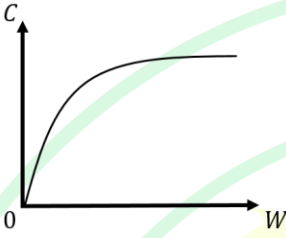
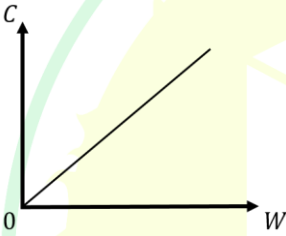
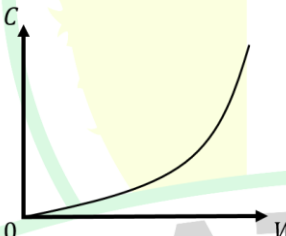
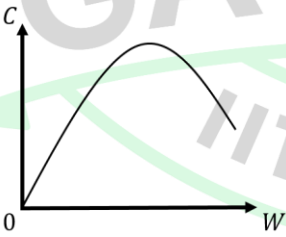
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Q.13	A pot contains two red balls and two blue balls. Two balls are drawn from this pot randomly without replacement. What is the probability that the two balls drawn have different colours?
(A)	$\frac{2}{3}$
(B)	$\frac{1}{3}$
(C)	$\frac{1}{2}$
(D)	1



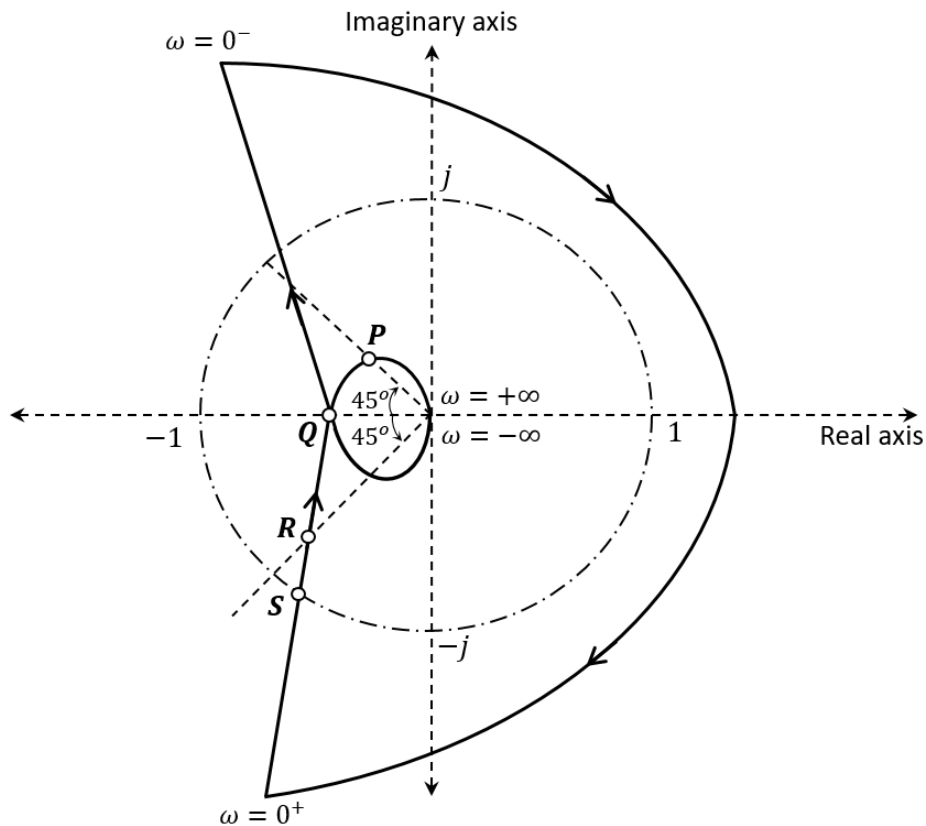
Q.14	Consider a frequency-modulated (FM) signal $f(t) = A_c \cos(2\pi f_c t + 3 \sin(2\pi f_1 t) + 4 \sin(6\pi f_1 t)),$ where A_c and f_c are, respectively, the amplitude and frequency (in Hz) of the carrier waveform. The frequency f_1 is in Hz, and assume that $f_c > 100f_1$. The peak frequency deviation of the FM signal in Hz is _____.
(A)	$15f_1$
(B)	$12f_1$
(C)	$4f_1$
(D)	$2f_1$

Q.15	Consider an additive white Gaussian noise (AWGN) channel with bandwidth W and noise power spectral density $\frac{N_0}{2}$. Let P_{av} denote the average transmit power constraint. Which one of the following plots illustrates the dependence of the channel capacity C on the bandwidth W (keeping P_{av} and N_0 fixed)?
(A)	
(B)	
(C)	
(D)	

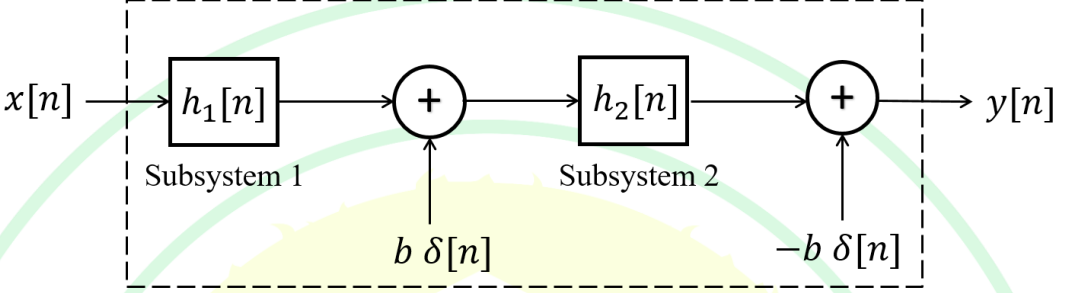
Q.16

The Nyquist plot of a system is given in the figure below. Let ω_P , ω_Q , ω_R , and ω_S be the positive frequencies at the points P , Q , R , and S , respectively.

Which one of the following statements is TRUE?



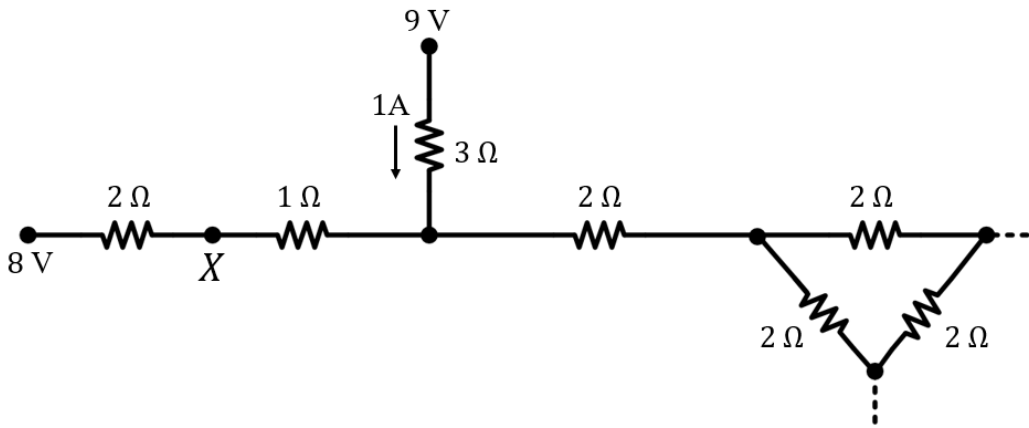
- (A) ω_S is the gain crossover frequency and ω_P is the phase crossover frequency
- (B) ω_Q is the gain crossover frequency and ω_R is the phase crossover frequency
- (C) ω_Q is the gain crossover frequency and ω_S is the phase crossover frequency
- (D) ω_S is the gain crossover frequency and ω_Q is the phase crossover frequency

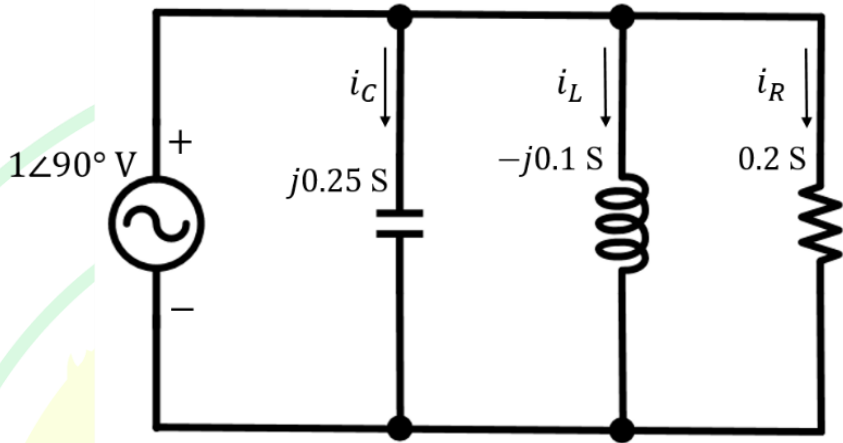
Q.17	Consider the discrete-time system below with input $x[n]$ and output $y[n]$. In the figure, $h_1[n]$ and $h_2[n]$ denote the impulse responses of LTI Subsystems 1 and 2, respectively. Also, $\delta[n]$ is the unit impulse, and $b > 0$. Assuming $h_2[n] \neq \delta[n]$, the overall system (denoted by the dashed box) is _____.
	
(A)	linear and time invariant
(B)	linear and time variant
(C)	nonlinear and time invariant
(D)	nonlinear and time variant

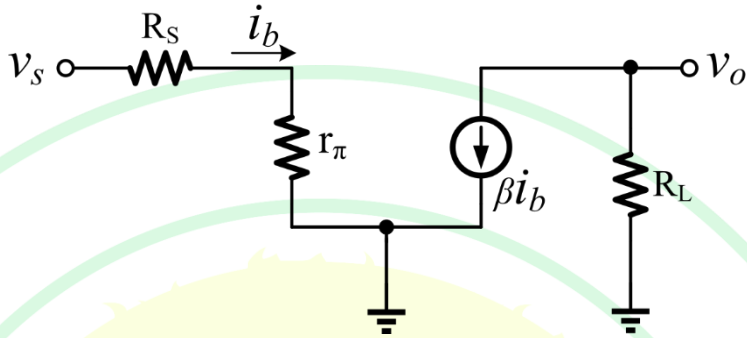
Q.18	Consider a continuous-time, real-valued signal $f(t)$ whose Fourier transform $F(\omega) = \int_{-\infty}^{\infty} f(t) \exp(-j\omega t) dt$ exists. Which one of the following statements is always TRUE?
(A)	$ F(\omega) \leq \int_{-\infty}^{\infty} f(t) dt$
(B)	$ F(\omega) > \int_{-\infty}^{\infty} f(t) dt$
(C)	$ F(\omega) \leq \int_{-\infty}^{\infty} f(t) dt$
(D)	$ F(\omega) \geq \int_{-\infty}^{\infty} f(t) dt$

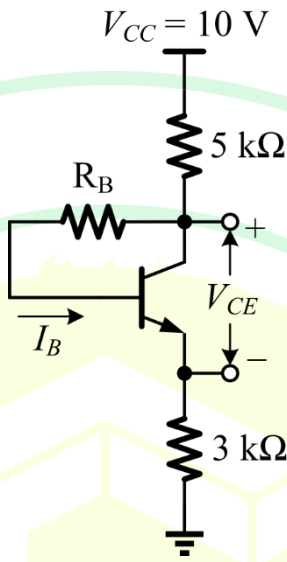
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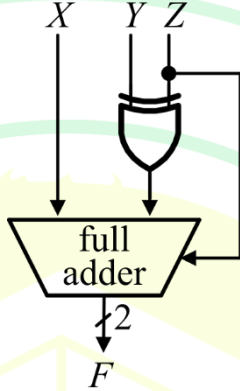
Q.19	Consider a part of an electrical network as shown below. Some node voltages, and the current flowing through the 3Ω resistor are as indicated. The voltage (in Volts) at node X is _____.
	
(A)	$20/3$
(B)	$32/3$
(C)	$22/3$
(D)	$2/3$
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Q.20	Let i_C, i_L, and i_R be the currents flowing through the capacitor, inductor, and resistor, respectively, in the circuit given below. The AC admittances are given in Siemens (S). Which one of the following is TRUE?
	
(A)	$i_C = 0.25\angle 180^\circ \text{ A}$, $i_L = 0.1\angle 0^\circ \text{ A}$, $i_R = 0.2\angle 90^\circ \text{ A}$
(B)	$i_C = 4\angle 180^\circ \text{ A}$, $i_L = 10\angle 0^\circ \text{ A}$, $i_R = 5\angle 90^\circ \text{ A}$
(C)	$i_C = 0.25\angle 270^\circ \text{ A}$, $i_L = 0.1\angle 90^\circ \text{ A}$, $i_R = 0.2\angle 90^\circ \text{ A}$
(D)	$i_C = 4\angle 90^\circ \text{ A}$, $i_L = 10\angle 270^\circ \text{ A}$, $i_R = 5\angle 0^\circ \text{ A}$

Q.21	A simplified small-signal equivalent circuit of a BJT-based amplifier is given below. The small-signal voltage gain V_o/V_s (in V/V) is _____.
	
(A)	$\frac{-\beta R_L}{R_s + r_\pi}$
(B)	$\frac{+\beta R_L}{R_s}$
(C)	$\frac{-\beta R_L}{R_s}$
(D)	$\frac{+\beta R_L}{R_s + r_\pi}$

Q.22	The ideal BJT in the circuit given below is biased in the active region with a β of 100. If I_B is $10\ \mu\text{A}$, then V_{CE} (in Volts, rounded off to two decimal places) is _____.
	
(A)	4.95
(B)	3.03
(C)	1.92
(D)	3.73

Q.23	A 3-input majority logic gate has inputs X, Y, and Z. The output F of the gate is logic '1' if two or more of the inputs are logic '1'. The output F is logic '0' if two or more of the inputs are logic '0'. Which one of the following options is a Boolean expression of the output F?
(A)	$XY + YZ + ZX$
(B)	$X \oplus Y \oplus Z$
(C)	$X + Y + Z$
(D)	XYZ

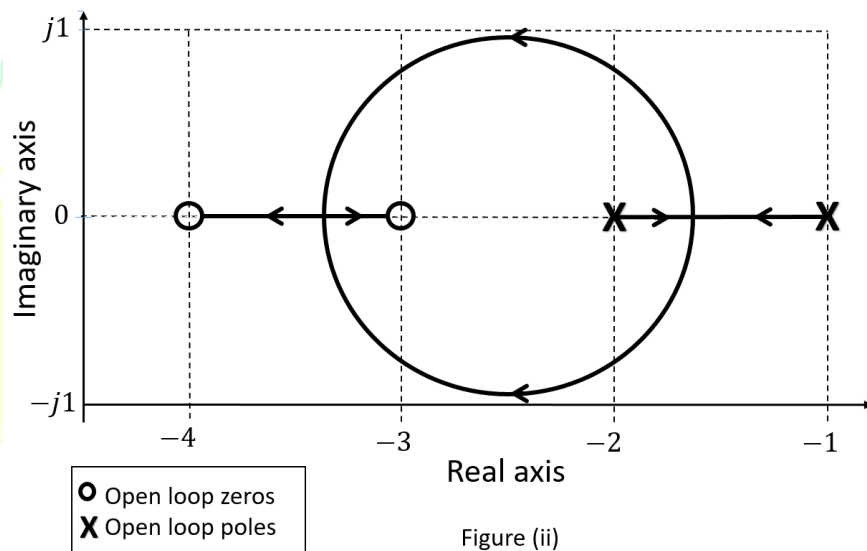
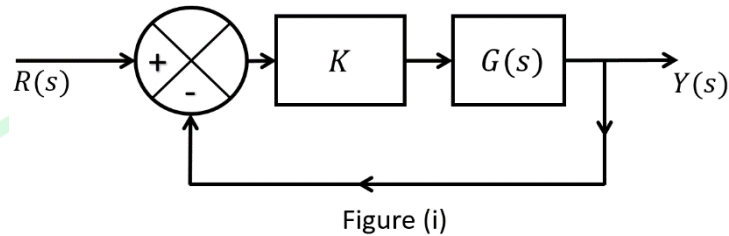
Q.24	A full adder and an XOR gate are used to design a digital circuit with inputs X, Y, and Z, and output F, as shown below. The input Z is connected to the carry-in input of the full adder. If the input Z is set to logic '1', then the circuit functions as _____ with X and Y as inputs.
	
(A)	an adder
(B)	a subtractor
(C)	a multiplier
(D)	a binary to Gray code converter

Q.25	Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$, defined as $f(x) = 2x^3 - 3x^2 - 12x + 1.$ Which of the following statements is/are correct? (Here, $\mathbb{R}$ is the set of real numbers.)
(A)	f has no global maximizer
(B)	f has no global minimizer
(C)	$x = -1$ is a local minimizer of f
(D)	$x = 2$ is a local maximizer of f

Q.26

Consider the unity-negative-feedback system shown in Figure (i) below, where gain $K \geq 0$. The root locus of this system is shown in Figure (ii) below.

For what value(s) of K will the system in Figure (i) have a pole at $-1 + j1$?



(A)

$$K = 5$$

(B)

$$K = 1/5$$

(C)

For no positive value of K

(D)

For all positive values of K

Q.27	Let $x[n]$ be a discrete-time signal whose Z-transform is $X(z)$. Which of the following statements is/are TRUE?
(A)	The discrete-time Fourier transform (DTFT) of $x[n]$ always exists
(B)	The region of convergence (RoC) of $X(z)$ contains neither poles nor zeros
(C)	The discrete-time Fourier transform (DTFT) exists if the region of convergence (RoC) contains the unit circle
(D)	If $x[n] = \alpha\delta[n]$, where $\delta[n]$ is the unit impulse and α is a scalar, then the region of convergence (RoC) is the entire Z-plane

Q.28	Consider a message signal $m(t)$ which is bandlimited to $[-W, W]$, where W is in Hz. Consider the following two modulation schemes for the message signal: • Double sideband-suppressed carrier (DSB-SC): $f_{DSB}(t) = A_c m(t) \cos(2\pi f_c t)$ • Amplitude modulation (AM): $f_{AM}(t) = A_c (1 + \mu m(t)) \cos(2\pi f_c t)$ Here, A_c and f_c are the amplitude and frequency (in Hz) of the carrier, respectively. In the case of AM, μ denotes the modulation index. Consider the following statements: - An envelope detector can be used for demodulation in the DSB-SC scheme if $m(t) > 0$ for all t. - An envelope detector can be used for demodulation in the AM scheme <u>only</u> if $m(t) > 0$ for all t. Which of the following options is/are correct?
(A)	(i) is TRUE
(B)	(i) is FALSE
(C)	(ii) is TRUE
(D)	(ii) is FALSE

Q.29	Which of the following statements is/are TRUE with respect to an ideal opamp?
(A)	It has an infinite input resistance
(B)	It has an infinite output resistance
(C)	It has an infinite open-loop differential gain
(D)	It has an infinite open-loop common-mode gain

Q.30	Which of the following statements is/are TRUE with respect to ideal MOSFET-based DC-coupled single-stage amplifiers having finite load resistors?
(A)	The common-gate amplifier has an infinite input resistance
(B)	The common-source amplifier has an infinite input resistance
(C)	The input and output voltages of the common-source amplifier are in phase
(D)	The input and output voltages of the common-drain amplifier are in phase

Q.31	Which of the following can be used as an n-type dopant for silicon? Select the correct option(s).
(A)	Arsenic
(B)	Boron
(C)	Gallium
(D)	Phosphorous

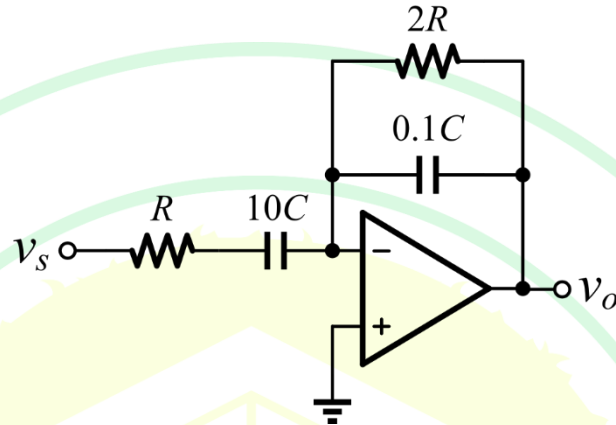
Q.32	The function $y(t)$ satisfies $t^2 y''(t) - 2ty'(t) + 2y(t) = 0,$ where $y'(t)$ and $y''(t)$ denote the first and second derivatives of $y(t)$, respectively. Given $y'(0) = 1$ and $y'(1) = -1$, the maximum value of $y(t)$ over $[0,1]$ is _____ (rounded off to two decimal places).
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Q.33	The generator matrix of a (6,3) binary linear block code is given by $G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}.$ The minimum Hamming distance d_{min} between codewords equals _____ (answer in integer).
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Q.34

All the components in the bandpass filter given below are ideal. The lower -3 dB frequency of the filter is 1 MHz.

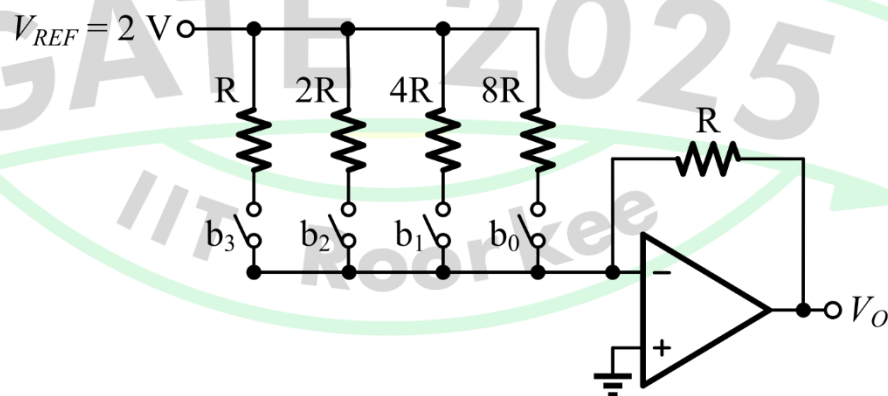
The upper -3 dB frequency (in MHz, rounded off to the nearest integer) is _____.



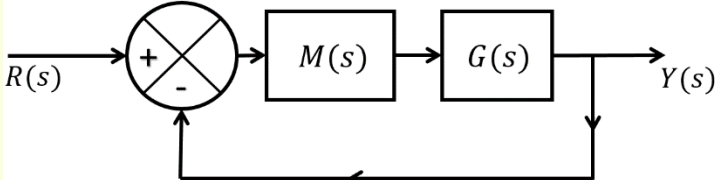
Q.35

A 4-bit weighted-resistor DAC with inputs b_3 , b_2 , b_1 , and b_0 (MSB to LSB) is designed using an ideal opamp, as shown below. The switches are closed when the corresponding input bits are logic '1' and open otherwise.

When the input $b_3b_2b_1b_0$ changes from 1110 to 1101, the magnitude of the change in the output voltage V_O (in mV, rounded off to the nearest integer) is _____.

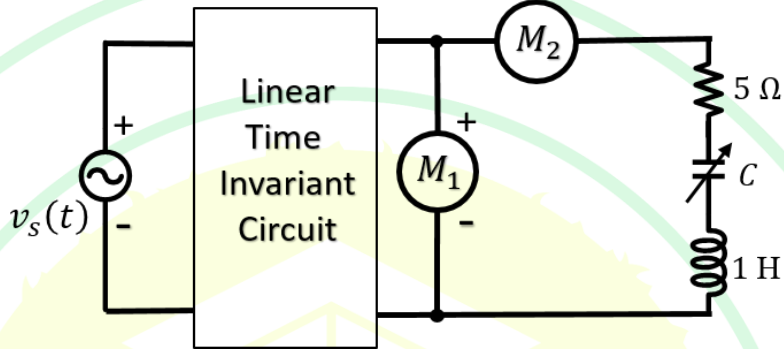


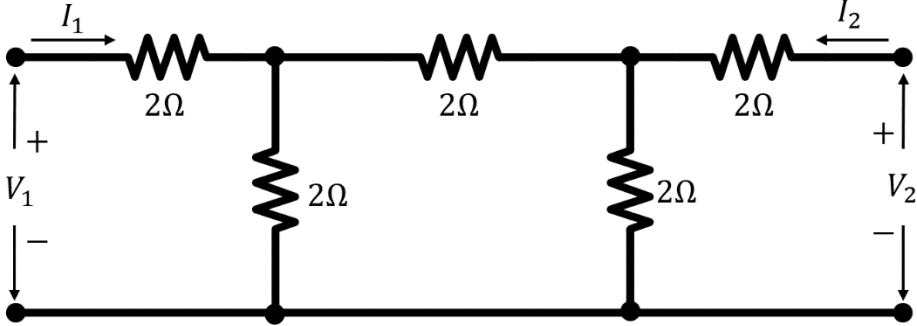
Q.36 – Q.65 Carry TWO marks Each

Q.36	Let $G(s) = \frac{1}{10s^2}$ be the transfer function of a second-order system. A controller $M(s)$ is connected to the system $G(s)$ in the configuration shown below. Consider the following statements. (i) There exists no controller of the form $M(s) = \frac{K_I}{s}$, where K_I is a positive real number, such that the closed loop system is stable. (ii) There exists at least one controller of the form $M(s) = K_P + sK_D$, where K_P and K_D are positive real numbers, such that the closed loop system is stable. Which one of the following options is correct?
	
(A)	(i) is TRUE and (ii) is FALSE
(B)	(i) is FALSE and (ii) is TRUE
(C)	Both (i) and (ii) are FALSE
(D)	Both (i) and (ii) are TRUE

Q.37	Consider the polynomial $p(s) = s^5 + 7s^4 + 3s^3 - 33s^2 + 2s - 40$. Let (L, I, R) be defined as follows. L is the number of roots of $p(s)$ with negative real parts. I is the number of roots of $p(s)$ that are purely imaginary. R is the number of roots of $p(s)$ with positive real parts. Which one of the following options is correct?
(A)	$L = 2, I = 2$, and $R = 1$
(B)	$L = 3, I = 2$, and $R = 0$
(C)	$L = 1, I = 2$, and $R = 2$
(D)	$L = 0, I = 4$, and $R = 1$

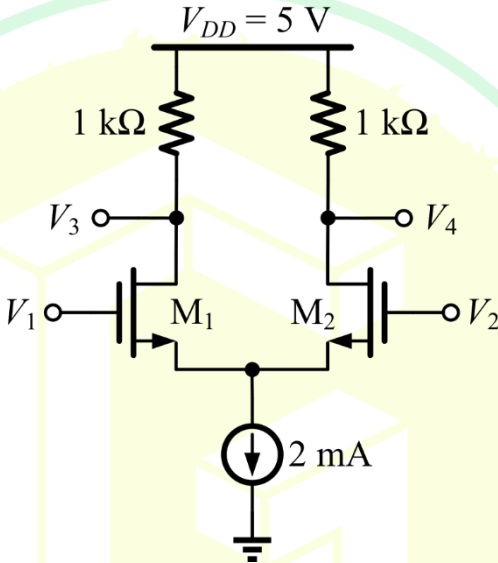
Q.38	Consider a continuous-time finite-energy signal $f(t)$ whose Fourier transform vanishes outside the frequency interval $[-\omega_c, \omega_c]$, where ω_c is in rad/sec. The signal $f(t)$ is uniformly sampled to obtain $y(t) = f(t)p(t)$. Here, $p(t) = \sum_{n=-\infty}^{\infty} \delta(t - \tau - nT_s),$ with $\delta(t)$ being the Dirac impulse, $T_s > 0$, and $\tau > 0$. The sampled signal $y(t)$ is passed through an ideal lowpass filter $h(t) = \omega_c T_s \frac{\sin(\omega_c t)}{\pi \omega_c t}$ with cutoff frequency ω_c and passband gain T_s. The output of the filter is given by _____.
(A)	$f(t)$ if $T_s < \pi/\omega_c$
(B)	$f(t - \tau)$ if $T_s < \pi/\omega_c$
(C)	$f(t - \tau)$ if $T_s < 2\pi/\omega_c$
(D)	$T_s f(t)$ if $T_s < 2\pi/\omega_c$

Q.39	In the circuit below, M_1 is an ideal AC voltmeter and M_2 is an ideal AC ammeter. The source voltage (in Volts) is $v_s(t) = 100 \cos(200t)$. What should be the value of the variable capacitor C such that the RMS readings on M_1 and M_2 are 25 V and 5 A, respectively?
	
(A)	25 μF
(B)	4 μF
(C)	0.25 μF
(D)	Insufficient information to find C

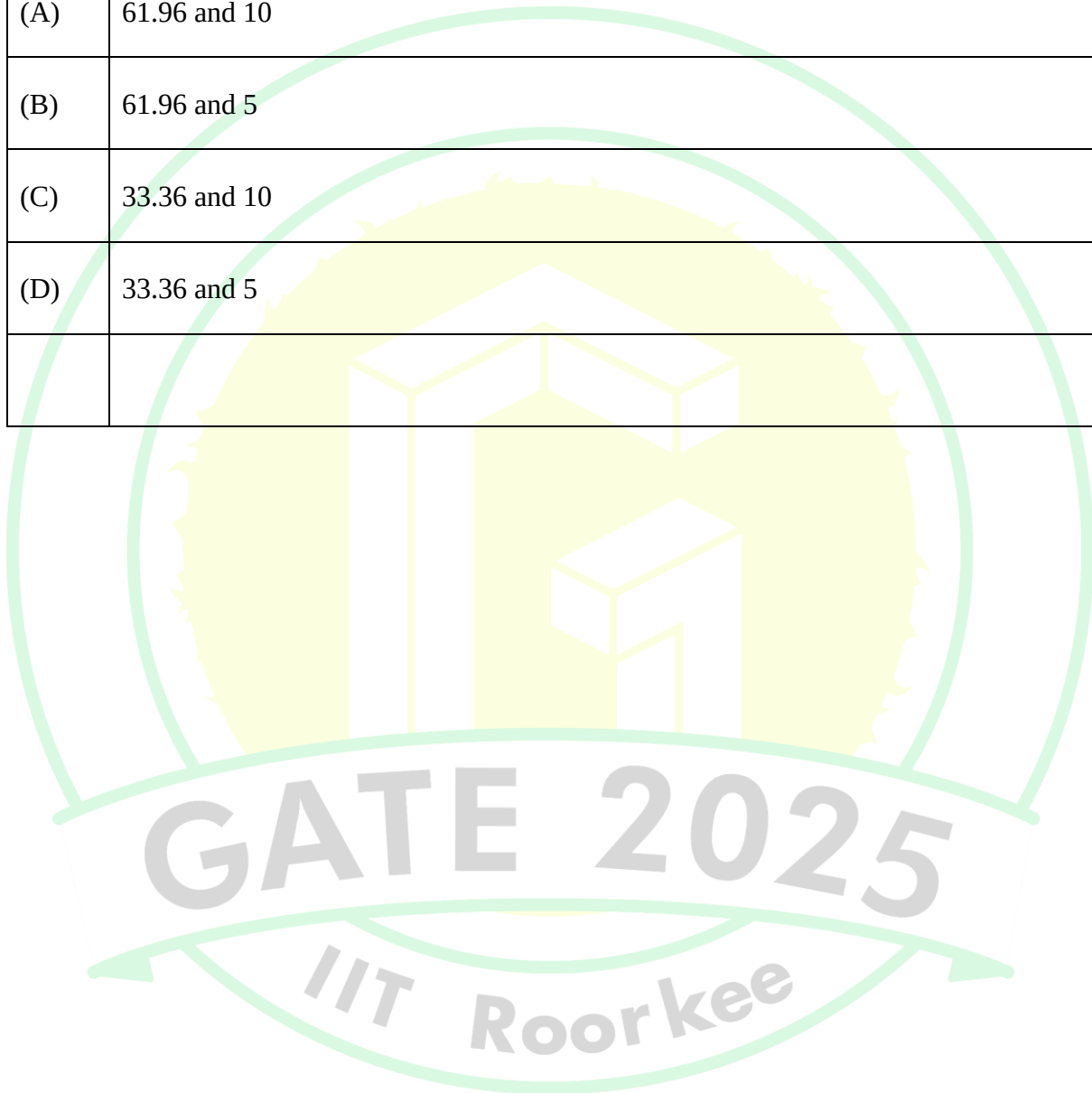
Q.40	The Z-parameter matrix of a two port network relates the port voltages and port currents as follows: $\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = Z \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$ The Z-parameter matrix (with each entry in Ohms) of the network shown below is _____.
	
(A)	$\begin{bmatrix} 10/3 & 2/3 \\ 2/3 & 10/3 \end{bmatrix}$
(B)	$\begin{bmatrix} 2/3 & 10/3 \\ 10/3 & 2/3 \end{bmatrix}$
(C)	$\begin{bmatrix} 10 & 2 \\ 2 & 10 \end{bmatrix}$
(D)	$\begin{bmatrix} 10/3 & 1/3 \\ 1/3 & 10/3 \end{bmatrix}$

Q.41	A source transmits symbol S that takes values uniformly at random from the set $\{-2, 0, 2\}$. The receiver obtains $Y = S + N$, where N is a zero-mean Gaussian random variable independent of S. The receiver uses the maximum likelihood decoder to estimate the transmitted symbol $\hat{S}$. Suppose the probability of symbol estimation error P_e is expressed as follows: $P_e = \alpha P(N > 1),$ where $P(N > 1)$ denotes the probability that N exceeds 1. What is the value of α ?
(A)	$1/3$
(B)	1
(C)	$2/3$
(D)	$4/3$

Q.42	Consider a real-valued random process $f(t) = \sum_{n=1}^N a_n p(t - nT),$ where $T > 0$ and N is a positive integer. Here, $p(t) = 1$ for $t \in [0, 0.5T]$ and 0 otherwise. The coefficients a_n are pairwise independent, zero-mean unit-variance random variables. Read the following statements about the random process and choose the correct option. (i) The mean of the process $f(t)$ is independent of time t. (ii) The autocorrelation function $E[f(t)f(t + \tau)]$ is independent of time t for all τ. (Here, $E[\cdot]$ is the expectation operation.)
(A)	(i) is TRUE and (ii) is FALSE
(B)	Both (i) and (ii) are TRUE
(C)	Both (i) and (ii) are FALSE
(D)	(i) is FALSE and (ii) is TRUE

Q.43	The identical MOSFETs M_1 and M_2 in the circuit given below are ideal and biased in the saturation region. M_1 and M_2 have a transconductance g_m of 5 mS. The input signals (in Volts) are: $V_1 = 2.5 + 0.01 \sin \omega t$ $V_2 = 2.5 - 0.01 \sin \omega t$ The output signal V_3 (in Volts) is _____.
	
(A)	$3 + 0.05 \sin \omega t$
(B)	$3 - 0.1 \sin \omega t$
(C)	$4 + 0.1 \sin \omega t$
(D)	$4 - 0.05 \sin \omega t$

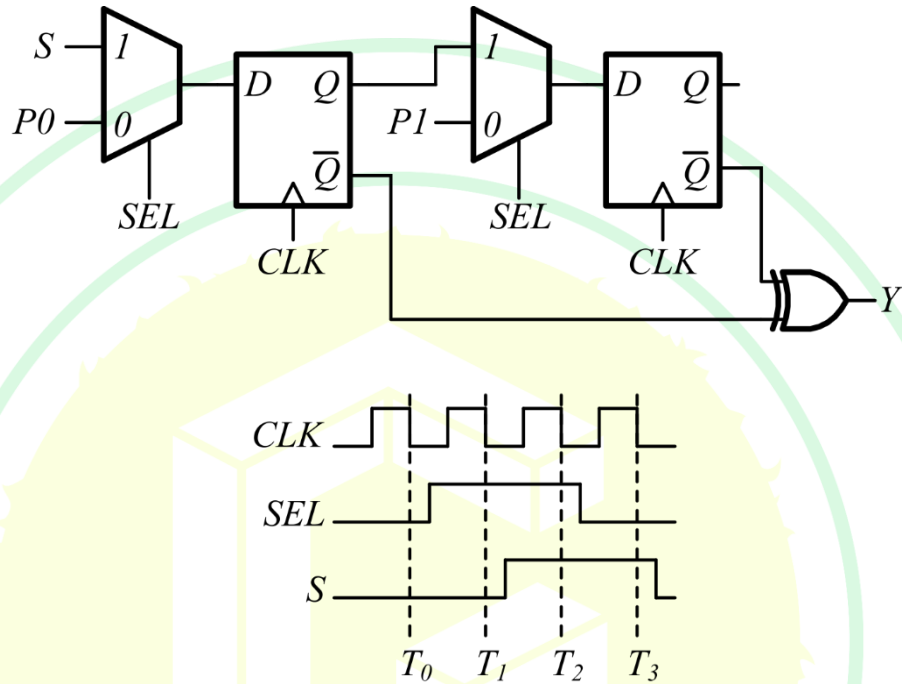
Q.44	A 10-bit analog-to-digital converter (ADC) has a sampling frequency of 1 MHz and a full scale voltage of 3.3 V. For an input sinusoidal signal with frequency 500 kHz, the maximum SNR (in dB, <i>rounded off to two decimal places</i>) and the data rate (in Mbps) at the output of the ADC are _____, respectively.
(A)	61.96 and 10
(B)	61.96 and 5
(C)	33.36 and 10
(D)	33.36 and 5



Q.45

A positive-edge-triggered sequential circuit is shown below. There are no timing violations in the circuit. Input $P0$ is set to logic '0' and $P1$ is set to logic '1' at all times. The timing diagram of the inputs SEL and S are also shown below.

The sequence of output Y from time T_0 to T_3 is _____.



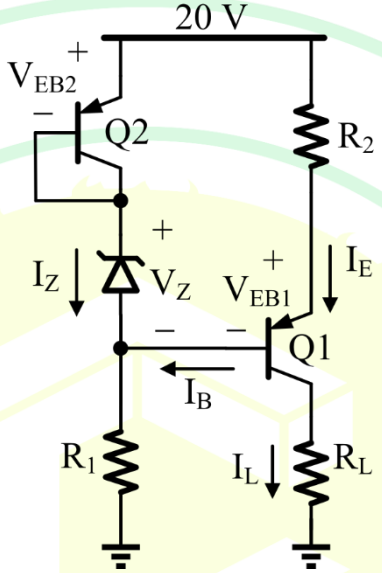
(A) 1011

(B) 0100

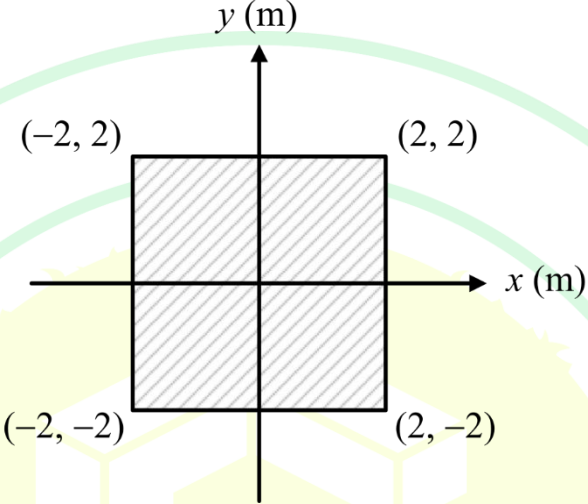
(C) 0010

(D) 1101

Q.46	The intrinsic carrier concentration of a semiconductor is $2.5 \times 10^{16} / \text{m}^3$ at 300 K. If the electron and hole mobilities are $0.15 \text{ m}^2/\text{Vs}$ and $0.05 \text{ m}^2/\text{Vs}$, respectively, then the intrinsic resistivity of the semiconductor (in $\text{k}\Omega\cdot\text{m}$) at 300 K is _____. (Charge of an electron $e = 1.6 \times 10^{-19} \text{ C}$.)
(A)	1.65
(B)	1.25
(C)	0.85
(D)	1.95

Q.47	In the circuit shown, the identical transistors Q1 and Q2 are biased in the active region with $\beta = 120$. The Zener diode is in the breakdown region with $V_Z = 5\text{ V}$ and $I_Z = 25\text{ mA}$. If $I_L = 12\text{ mA}$ and $V_{EB1} = V_{EB2} = 0.7\text{ V}$, then the values of R_1 and R_2 (in $k\Omega$, rounded off to one decimal place) are _____, respectively.
	
(A)	0.6 and 0.4
(B)	1.4 and 2.5
(C)	14.0 and 25.0
(D)	6.0 and 4.0

Q.48	The electron mobility μ_n in a non-degenerate germanium semiconductor at 300 K is $0.38 \text{ m}^2/\text{Vs}$. The electron diffusivity D_n at 300 K (in cm^2/s, rounded off to the nearest integer) is _____. (Consider the Boltzmann constant $k_B = 1.38 \times 10^{-23} \text{ J/K}$ and the charge of an electron $e = 1.6 \times 10^{-19} \text{ C}$.)
(A)	26
(B)	98
(C)	38
(D)	10

Q.49	A square metal sheet of $4\text{ m} \times 4\text{ m}$ is placed on the x-y plane as shown in the figure below. If the surface charge density (in $\mu\text{C}/\text{m}^2$) on the sheet is $\rho_s(x, y) = 4 y$, then the total charge (in μC, rounded off to the nearest integer) on the sheet is _____.
	
(A)	16
(B)	85
(C)	64
(D)	256

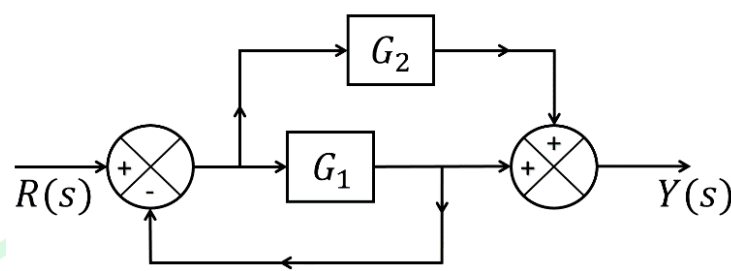
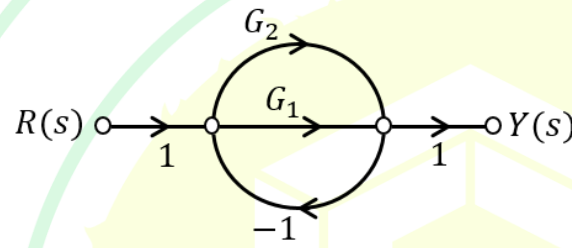
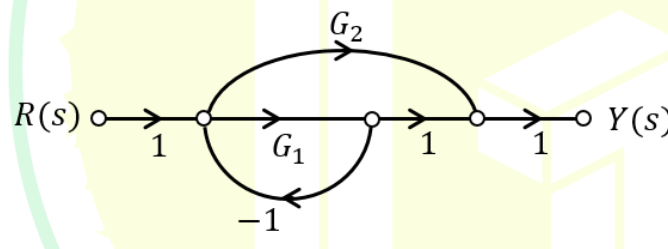
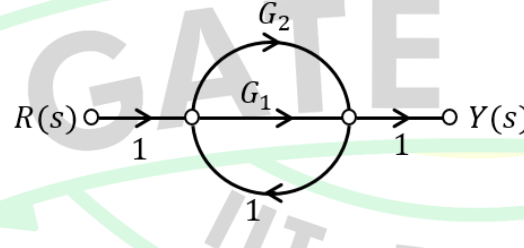
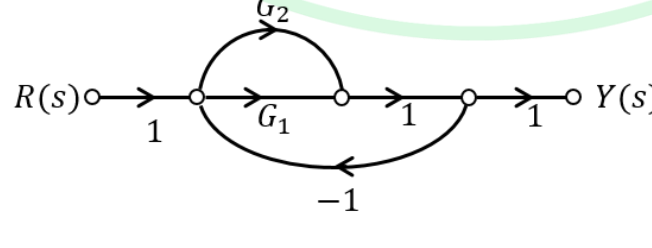
Q.50	An electric field of 0.01 V/m is applied along the length of a copper wire of circular cross-section with diameter 1 mm. Copper has a conductivity of 5.8×10^7 S/m. The current (in Amperes, <i>rounded off to two decimal places</i>) flowing through the wire is _____.
(A)	0.46
(B)	1.82
(C)	0.58
(D)	1.12

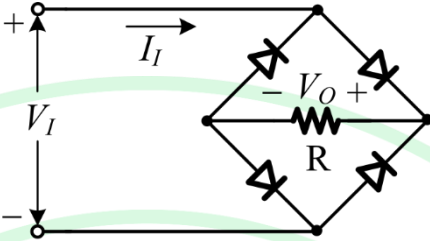
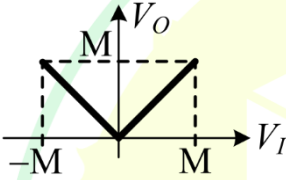
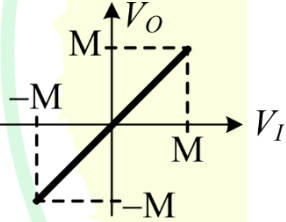
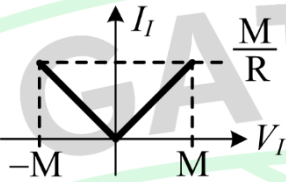
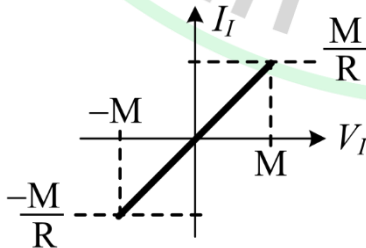
Q.51	Consider a non-negative function $f(x)$ which is continuous and bounded over the interval $[2, 8]$. Let M and m denote, respectively, the maximum and the minimum values of $f(x)$ over the interval. Among the combinations of α and β given below, choose the one(s) for which the inequality $\beta \leq \int_2^8 f(x) dx \leq \alpha$ is guaranteed to hold.
(A)	$\beta = 5m, \alpha = 7M$
(B)	$\beta = 6m, \alpha = 5M$
(C)	$\beta = 7m, \alpha = 6M$
(D)	$\beta = 7m, \alpha = 5M$

Q.52	Which of the following statements involving contour integrals (evaluated counter-clockwise) on the unit circle C in the complex plane is/are TRUE?
(A)	$\oint_C e^z dz = 0$
(B)	$\oint_C z^n dz = 0$, where n is an even integer
(C)	$\oint_C \cos z dz \neq 0$
(D)	$\oint_C \sec z dz \neq 0$

Q.53	Consider a system where $x_1(t)$, $x_2(t)$, and $x_3(t)$ are three internal state signals and $u(t)$ is the input signal. The differential equations governing the system are given by $\frac{d}{dt} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} u(t).$ Which of the following statements is/are TRUE?
(A)	The signals $x_1(t)$, $x_2(t)$, and $x_3(t)$ are bounded for all bounded inputs
(B)	There exists a bounded input such that at least one of the signals $x_1(t)$, $x_2(t)$, and $x_3(t)$ is unbounded
(C)	There exists a bounded input such that the signals $x_1(t)$, $x_2(t)$, and $x_3(t)$ are unbounded
(D)	The signals $x_1(t)$, $x_2(t)$, and $x_3(t)$ are unbounded for all bounded inputs

Q.54	The random variable X takes values in $\{-1, 0, 1\}$ with probabilities $P(X = -1) = P(X = 1) = \alpha$ and $P(X = 0) = 1 - 2\alpha$, where $0 < \alpha < 1/2$. Let $g(\alpha)$ denote the entropy of X (in bits), parameterized by α. Which of the following statements is/are TRUE?
(A)	$g(0.4) > g(0.3)$
(B)	$g(0.3) > g(0.4)$
(C)	$g(0.3) > g(0.25)$
(D)	$g(0.25) > g(0.3)$
Q.55	Let $f(t)$ be a periodic signal with fundamental period $T_0 > 0$. Consider the signal $y(t) = f(\alpha t)$, where $\alpha > 1$. The Fourier series expansions of $f(t)$ and $y(t)$ are given by $f(t) = \sum_{k=-\infty}^{\infty} c_k e^{j\frac{2\pi}{T_0}kt} \quad \text{and} \quad y(t) = \sum_{k=-\infty}^{\infty} d_k e^{j\frac{2\pi}{T_0}\alpha kt}.$ Which of the following statements is/are TRUE?
(A)	$c_k = d_k$ for all k
(B)	$y(t)$ is periodic with a fundamental period αT_0
(C)	$c_k = d_k/\alpha$ for all k
(D)	$y(t)$ is periodic with a fundamental period T_0/α

Q.56	Consider a system represented by the block diagram shown below. Which of the following signal flow graphs represent(s) this system? Choose the correct option(s).
	
(A)	
(B)	
(C)	
(D)	

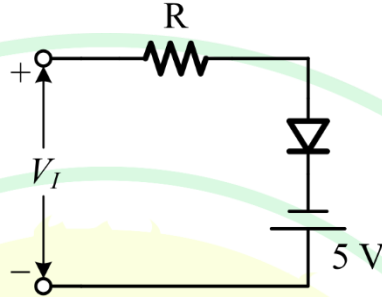
Q.57	All the diodes in the circuit given below are ideal. Which of the following plots is/are correct when V_I (in Volts) is swept from $-M$ to M ?
	
(A)	
(B)	
(C)	
(D)	

Q.58	Two fair dice (with faces labeled 1, 2, 3, 4, 5, and 6) are rolled. Let the random variable X denote the sum of the outcomes obtained. The expectation of X is _____ (rounded off to two decimal places).
Q.59	Consider the vectors $\mathbf{a} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 0 \\ 3\sqrt{2} \end{bmatrix}.$ For real-valued scalar variable x, the value of $\min_x \ \mathbf{a}x - \mathbf{b}\ _2$ is _____ (rounded off to two decimal places). $\ \cdot\ _2$ denotes the Euclidean norm, i.e., for $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$, $\ \mathbf{y}\ _2 = \sqrt{y_1^2 + y_2^2}$.
Q.60	X and Y are Bernoulli random variables taking values in $\{0,1\}$. The joint probability mass function of the random variables is given by: $\begin{aligned} P(X = 0, Y = 0) &= 0.06 \\ P(X = 0, Y = 1) &= 0.14 \\ P(X = 1, Y = 0) &= 0.24 \\ P(X = 1, Y = 1) &= 0.56 \end{aligned}$ The mutual information $I(X; Y)$ is _____ (rounded off to two decimal places).

Q.61

The diode in the circuit shown below is ideal. The input voltage (in Volts) is given by $V_I = 10 \sin 100\pi t$, where time t is in seconds.

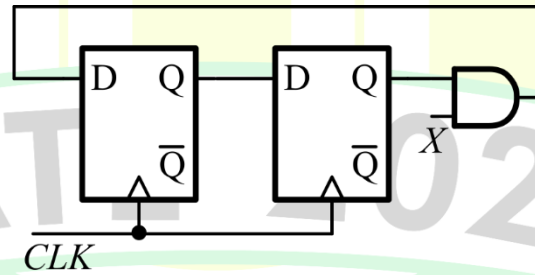
The time duration (in ms, rounded off to two decimal places) for which the diode is forward biased during one period of the input is _____.



Q.62

In the circuit shown below, the AND gate has a propagation delay of 1 ns. The edge-triggered flip-flops have a set-up time of 2 ns, a hold-time of 0 ns, and a clock-to-Q delay of 2 ns.

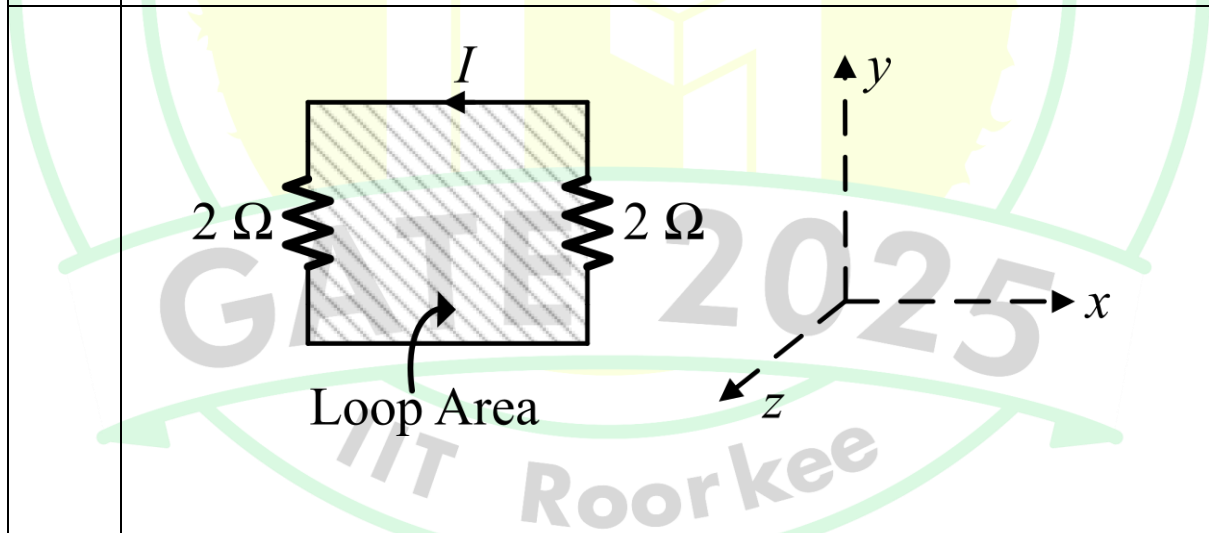
The maximum clock frequency (in MHz, rounded off to the nearest integer) such that there are no setup violations is _____.



Q.63	An ideal p-n junction germanium diode has a reverse saturation current of $10 \mu\text{A}$ at 300 K. The voltage (in Volts, <i>rounded off to two decimal places</i>) to be applied across the junction to get a forward bias current of 100 mA at 300 K is _____. (Consider the Boltzmann constant $k_B = 1.38 \times 10^{-23} \text{ J/K}$ and the charge of an electron $e = 1.6 \times 10^{-19} \text{ C}$.)
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Q.64	A 50Ω lossless transmission line is terminated with a load Z_L of $(50 - j75) \Omega$. If the average incident power on the line is 10 mW, then the average power delivered to the load (in mW, <i>rounded off to one decimal place</i>) is _____.
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Q.65	Two resistors are connected in a circuit loop of area 5 m^2, as shown in the figure below. The circuit loop is placed on the x-y plane. When a time-varying magnetic flux, with flux-density $B(t) = 0.5t$ (in Tesla), is applied along the positive z-axis, the magnitude of current I (in Amperes, <i>rounded off to two decimal places</i>) in the loop is _____.
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Graduate Aptitude Test in Engineering (GATE 2021)

Subject/Paper: Electronics and Communication Engineering (EC)

Q. No.	Session	Question Type MCQ/MSQ/NAT	Section Name	Answer Key/Range	Marks	Negative Marks
1	4	MCQ	GA	C	1	1/3
2	4	MCQ	GA	B	1	1/3
3	4	MCQ	GA	C	1	1/3
4	4	MCQ	GA	C	1	1/3
5	4	MCQ	GA	B	1	1/3
6	4	MCQ	GA	C	2	2/3
7	4	MCQ	GA	D	2	2/3
8	4	MCQ	GA	A	2	2/3
9	4	MCQ	GA	C	2	2/3
10	4	MCQ	GA	A	2	2/3
1	4	MCQ	EC	A	1	1/3
2	4	MCQ	EC	B	1	1/3
3	4	MCQ	EC	B	1	1/3
4	4	MCQ	EC	B	1	1/3
5	4	MCQ	EC	C	1	1/3
6	4	MCQ	EC	D	1	1/3
7	4	MCQ	EC	A	1	1/3
8	4	MCQ	EC	D	1	1/3
9	4	MCQ	EC	MTA	1	1/3
10	4	MCQ	EC	A	1	1/3

GATE 2021 Answer Key for Electronics and Communication Engineering (EC)

Q. No.	Session	Question Type MCQ/MSQ/NAT	Section Name	Answer Key/Range	Marks	Negative Marks
11	4	MCQ	EC	B	1	1/3
12	4	MCQ	EC	C	1	1/3
13	4	MCQ	EC	C	1	1/3
14	4	MCQ	EC	D	1	1/3
15	4	MCQ	EC	A	1	1/3
16	4	NAT	EC	8.0 to 8.0	1	0
17	4	NAT	EC	0 to 0	1	0
18	4	NAT	EC	0.5 to 0.5	1	0
19	4	NAT	EC	1.0 to 1.0	1	0
20	4	NAT	EC	4.5 to 4.5	1	0
21	4	NAT	EC	2.0 to 2.0	1	0
22	4	NAT	EC	94.2 to 94.6	1	0
23	4	NAT	EC	9.24 to 9.28	1	0
24	4	NAT	EC	2.80 to 3.20	1	0
25	4	NAT	EC	9.30 to 9.44	1	0
26	4	MCQ	EC	MTA	2	2/3
27	4	MCQ	EC	B	2	2/3
28	4	MCQ	EC	C	2	2/3
29	4	MCQ	EC	C	2	2/3
30	4	MCQ	EC	A	2	2/3
31	4	MCQ	EC	C	2	2/3
32	4	MCQ	EC	A	2	2/3
33	4	MCQ	EC	D	2	2/3



Q. No.	Session	Question Type	Subject Name	Key/Range	Mark
1	3	MCQ	GA	C	1
2	3	MCQ	GA	D	1
3	3	MCQ	GA	B	1
4	3	MCQ	GA	B	1
5	3	MCQ	GA	C	1
6	3	MCQ	GA	D	2
7	3	MCQ	GA	C	2
8	3	MCQ	GA	C	2
9	3	MCQ	GA	B	2
10	3	MCQ	GA	B	2
11	3	MCQ	EC	A	1
12	3	MCQ	EC	C	1
13	3	MCQ	EC	B	1
14	3	MCQ	EC	B	1
15	3	MCQ	EC	A	1
16	3	MCQ	EC	A	1
17	3	MCQ	EC	C	1
18	3	MCQ	EC	C	1
19	3	MCQ	EC	C	1
20	3	MCQ	EC	C	1
21	3	MCQ	EC	C	1
22	3	MCQ	EC	A	1
23	3	MCQ	EC	C	1
24	3	MCQ	EC	C	1
25	3	MSQ	EC	A,B	1
26	3	MSQ	EC	B,C	1
27	3	MSQ	EC	B,C	1
28	3	MSQ	EC	C	1
29	3	MSQ	EC	A,B,D	1
30	3	MSQ	EC	A,C	1
31	3	NAT	EC	4.0 to 4.0	1
32	3	NAT	EC	0.50 to 0.50	1
33	3	NAT	EC	0.0 to 0.0	1
34	3	NAT	EC	5.5 to 5.7	1
35	3	NAT	EC	2.99 to 3.01	1
36	3	MCQ	EC	B	2
37	3	MCQ	EC	A	2
38	3	MCQ	EC	A	2
39	3	MCQ	EC	B	2
40	3	MCQ	EC	C	2
41	3	MCQ	EC	C	2
42	3	MCQ	EC	A	2
43	3	MCQ	EC	A	2
44	3	MCQ	EC	A	2

45	3	MSQ	EC	B,D	2
46	3	MSQ	EC	C,D	2
47	3	MSQ	EC	A,C	2
48	3	MSQ	EC	A,C	2
49	3	MSQ	EC	A,C	2
50	3	MSQ	EC	A	2
51	3	MSQ	EC	A	2
52	3	MSQ	EC	MTA	2
53	3	NAT	EC	512 to 512	2
54	3	NAT	EC	3.9 to 4.1	2
55	3	NAT	EC	7.9 to 8.1	2
56	3	NAT	EC	0.57 to 0.61	2
57	3	NAT	EC	0.70 to 0.80	2
58	3	NAT	EC	-0.55 to -0.45	2
59	3	NAT	EC	-2.1 to -1.9	2
60	3	NAT	EC	-0.11 to -0.09	2
61	3	NAT	EC	1.00 to 1.00	2
62	3	NAT	EC	0.84 to 0.86	2
63	3	NAT	EC	0.22 to 0.25	2
64	3	NAT	EC	1.15 to 1.18	2
65	3	NAT	EC	0.48 to 0.52	2

Q. No.	Session	Question Type (QT) MCQ/MSQ/NAT	Subject Name (SN)	Key/Range (KY)	Mark (MK)
1	4	MCQ	GA	C	1
2	4	MCQ	GA	A	1
3	4	MCQ	GA	C	1
4	4	MCQ	GA	C	1
5	4	MCQ	GA	C	1
6	4	MCQ	GA	D	2
7	4	MCQ	GA	A	2
8	4	MCQ	GA	B	2
9	4	MCQ	GA	A	2
10	4	MCQ	GA	C	2
11	4	MCQ	EC	C	1
12	4	MCQ	EC	B	1
13	4	MCQ	EC	A	1
14	4	MCQ	EC	B	1
15	4	MCQ	EC	C	1
16	4	MCQ	EC	A	1
17	4	MCQ	EC	A	1
18	4	MCQ	EC	D	1
19	4	MCQ	EC	B	1
20	4	MCQ	EC	D	1
21	4	MCQ	EC	D	1
22	4	MCQ	EC	MTA	1
23	4	MCQ	EC	A	1
24	4	MCQ	EC	A	1
25	4	MCQ	EC	A	1
26	4	MCQ	EC	B	1
27	4	MCQ	EC	B	1
28	4	MCQ	EC	C	1
29	4	MCQ	EC	B	1
30	4	MSQ	EC	A,B	1
31	4	NAT	EC	10 to 10	1
32	4	NAT	EC	1.30 to 1.40	1
33	4	NAT	EC	80 to 80	1
34	4	NAT	EC	02 to 02	1
35	4	NAT	EC	04 to 04	1
36	4	MCQ	EC	B	2
37	4	MCQ	EC	B	2
38	4	MCQ	EC	A	2
39	4	MCQ	EC	A	2
40	4	MCQ	EC	A	2
41	4	MCQ	EC	A	2
42	4	MCQ	EC	A	2
43	4	MCQ	EC	A	2
44	4	MCQ	EC	B	2
45	4	MCQ	EC	B	2

46	4	MCQ	EC	MTA	2
47	4	MCQ	EC	A	2
48	4	MCQ	EC	C	2
49	4	MCQ	EC	A	2
50	4	MCQ	EC	D	2
51	4	MCQ	EC	A	2
52	4	MSQ	EC	B,C	2
53	4	MSQ	EC	A,B,D	2
54	4	MSQ	EC	B,C,D	2
55	4	NAT	EC	0 to 0	2
56	4	NAT	EC	2.20 to 2.30	2
57	4	NAT	EC	4 to 4	2
58	4	NAT	EC	15 to 15	2
59	4	NAT	EC	0.24 to 0.26	2
60	4	NAT	EC	0.12 to 0.14	2
61	4	NAT	EC	60.00 to 70.00	2
62	4	NAT	EC	8.30 to 8.34	2
63	4	NAT	EC	250 to 250 OR 500 to 500	2
64	4	NAT	EC	2.00 to 2.00	2
65	4	NAT	EC	1.97 to 1.99	2

MTA = Marks to ALL

**GRADUATE APTITUDE TEST IN ENGINEERING 2024**

अभियांत्रिकी स्नातक अभिक्षमता परीक्षा २०२४

ORGANISING INSTITUTE: INDIAN INSTITUTE OF SCIENCE, BENGALURU

**Electronics & Communication Engineering (EC)****Final Answer Key**

Q. No.	Session	Question Type	Section	Key/Range	Mark
1	7	MCQ	GA	A	1
2	7	MCQ	GA	A	1
3	7	MCQ	GA	B	1
4	7	MCQ	GA	C	1
5	7	MCQ	GA	A	1
6	7	MCQ	GA	C	2
7	7	MCQ	GA	D	2
8	7	MCQ	GA	C	2
9	7	MCQ	GA	A	2
10	7	MCQ	GA	A	2
11	7	MCQ	EC	A	1
12	7	MCQ	EC	A	1
13	7	MCQ	EC	D	1
14	7	MCQ	EC	MTA	1
15	7	MCQ	EC	A	1
16	7	MCQ	EC	B	1
17	7	MCQ	EC	A	1
18	7	MCQ	EC	A	1
19	7	MCQ	EC	A	1
20	7	MCQ	EC	A	1
21	7	MCQ	EC	A	1
22	7	MCQ	EC	A	1
23	7	MCQ	EC	A	1
24	7	MSQ	EC	A;C;D	1
25	7	MSQ	EC	A;C	1
26	7	MSQ	EC	A;B;C	1
27	7	NAT	EC	2047 to 2047	1
28	7	NAT	EC	2.49 to 2.51	1
29	7	NAT	EC	11 to 11	1

30	7	NAT	EC	5 to 5	1
31	7	NAT	EC	2 to 2	1
32	7	NAT	EC	0.75 to 0.75	1
33	7	NAT	EC	0.50 to 0.50	1
34	7	NAT	EC	4 to 4	1
35	7	NAT	EC	0.250 to 0.250	1
36	7	MCQ	EC	A	2
37	7	MCQ	EC	A	2
38	7	MCQ	EC	B	2
39	7	MCQ	EC	A	2
40	7	MCQ	EC	B	2
41	7	MCQ	EC	B	2
42	7	MCQ	EC	B	2
43	7	MCQ	EC	D	2
44	7	MCQ	EC	MTA	2
45	7	MCQ	EC	B	2
46	7	MCQ	EC	B	2
47	7	MCQ	EC	A	2
48	7	MCQ	EC	MTA	2
49	7	MCQ	EC	D	2
50	7	MCQ	EC	B	2
51	7	MCQ	EC	B	2
52	7	MSQ	EC	A;C;D	2
53	7	MSQ	EC	A;C	2
54	7	MSQ	EC	A;B;D	2
55	7	MSQ	EC	A;B	2
56	7	MSQ	EC	A;D OR A;C;D	2
57	7	NAT	EC	45.00 to 45.30	2
58	7	NAT	EC	200 to 200	2
59	7	NAT	EC	112 to 112	2
60	7	NAT	EC	1.5 to 1.5	2
61	7	NAT	EC	4.10 to 4.50	2
62	7	NAT	EC	2.5 to 2.5	2
63	7	NAT	EC	0.17 to 0.19	2
64	7	NAT	EC	52.40 to 52.60	2
65	7	NAT	EC	4.00 to 4.26	2



GRADUATE APTITUDE TEST IN ENGINEERING 2025

अभियांत्रिकी स्नातक अभिक्षमता परीक्षा २०२५

Organising Institute: INDIAN INSTITUTE OF TECHNOLOGY ROORKEE



Answer Key for Electronics and Communication Engineering (EC)

Q. No.	Session	Q. Type	Section	Key/Range	Marks
1	6	MCQ	GA	A	1
2	6	MCQ	GA	D	1
3	6	MCQ	GA	B	1
4	6	MCQ	GA	B	1
5	6	MCQ	GA	A	1
6	6	MCQ	GA	B	2
7	6	MCQ	GA	D	2
8	6	MCQ	GA	MTA*	2
9	6	MCQ	GA	C	2
10	6	MCQ	GA	B	2
11	6	MCQ	EC	A	1
12	6	MCQ	EC	B	1
13	6	MCQ	EC	A	1
14	6	MCQ	EC	A	1
15	6	MCQ	EC	A	1
16	6	MCQ	EC	D	1
17	6	MCQ	EC	D	1
18	6	MCQ	EC	A	1
19	6	MCQ	EC	A	1
20	6	MCQ	EC	A	1
21	6	MCQ	EC	A	1
22	6	MCQ	EC	C	1
23	6	MCQ	EC	A	1
24	6	MCQ	EC	B	1
25	6	MSQ	EC	A;B	1
26	6	MSQ	EC	C	1
27	6	MSQ	EC	C;D	1
28	6	MSQ	EC	A;D	1
29	6	MSQ	EC	A;C	1
30	6	MSQ	EC	B;D	1

31	6	MSQ	EC	A;D	1
32	6	NAT	EC	0.25 to 0.25	1
33	6	NAT	EC	3 to 3	1
34	6	NAT	EC	50 to 50	1
35	6	NAT	EC	250 to 250	1
36	6	MCQ	EC	D	2
37	6	MCQ	EC	A	2
38	6	MCQ	EC	A	2
39	6	MCQ	EC	A	2
40	6	MCQ	EC	A	2
41	6	MCQ	EC	D	2
42	6	MCQ	EC	A	2
43	6	MCQ	EC	D	2
44	6	MCQ	EC	A	2
45	6	MCQ	EC	A	2
46	6	MCQ	EC	B	2
47	6	MCQ	EC	A	2
48	6	MCQ	EC	B	2
49	6	MCQ	EC	C	2
50	6	MCQ	EC	A	2
51	6	MSQ	EC	A	2
52	6	MSQ	EC	A;B	2
53	6	MSQ	EC	B	2
54	6	MSQ	EC	B;C	2
55	6	MSQ	EC	A;D	2
56	6	MSQ	EC	B	2
57	6	MSQ	EC	A;D	2
58	6	NAT	EC	6.95 to 7.05	2
59	6	NAT	EC	2.95 to 3.05	2
60	6	NAT	EC	0.0 to 0.0	2
61	6	NAT	EC	13.32 to 13.34	2
62	6	NAT	EC	200 to 200	2
63	6	NAT	EC	0.23 to 0.24	2
64	6	NAT	EC	6.3 to 6.5	2
65	6	NAT	EC	0.62 to 0.63	2

*MTA= Marks To All